

Frames, Returns, and Port Power

Four central shafts, loaded reactions, and electrical counterparts

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2026-09-29

Abstract

A planet's rotating stub is brought to a separate central output through two correctly phased universal joints. Together with sun, carrier and ring, this gives four accessible shafts with two independent speeds. Contact constraints and attached free bodies determine the two-coordinate steady load family and every signed port work. One loaded-planet control delivers $7/3$ joule in one second and changes carrier work by the same amount, while a held ring retains a nonzero reaction. Two ideal transformer pairs and one four-tap common-flux winding independently reproduce the complete terminal relations, including the planet receiver. A specified initialized compliant extension maps inertial and elastic stores to capacitive and magnetic states. Coin and spoke controls distinguish counts from physical loads; switched returns and prepared capacitor states expose the additional laws needed for finite connections. Frame-dependent effective stores, compound ratios, supplied references and prepared limits delimit the correspondence. A companion develops joint, switching, material and observation questions, with independent uncertainties for work and endpoint stores. All worked results are exact or explicitly conditional within declared models.

PDF created: 2026-09-29 19:02:20 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-gear>

1 Four shafts and a loaded planet

Bring the rotation of a planet gear to the centre of a planetary train through two correctly phased universal joints. The sun, carrier, ring and planet output then have four separately accessible shafts. Which motions and loads can those shafts sustain, and which electrical connections reproduce their signed powers? This construction organizes the article. The two meshes leave two independent speeds; attached free bodies determine two independent steady load coordinates. An ideal four-tap winding and a two-transformer network realize those same terminal relations, including a loaded planet output.

Figure 1 fixes the physical meaning of the fourth shaft. The rotating planet body is keyed to a stub that turns in a bearing carried by the carrier. Its support pin and the stub are different parts. Two joint centres connect that orbiting stub to a separate central output, o ; the output shares the sun's axis without being connected to the sun. Throughout, p denotes the orbiting body. The equality $\omega_o = \omega_p$ follows from the specified joint geometry and phasing.

Four separate shafts; one rotating planet stub

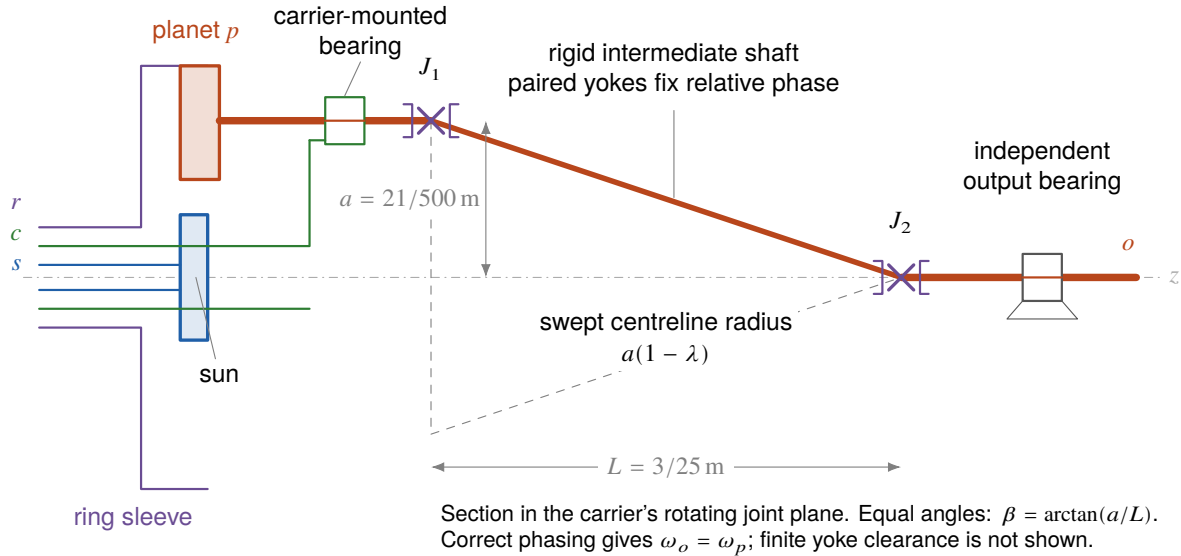


Figure 1. Schematic four-shaft construction, shown in a carrier-dependent axial plane. Separate sun, carrier and ring sleeves terminate before the illustrated joint sweep; their connections continue to their respective gear members. The planet stub rotates in a carrier-mounted bearing. Crosses and paired yoke marks show the two joints and the relative phasing fixed by the intermediate shaft. Dashed lines bound the swept centreline; real yokes, shaft radii, bearing loads and collision clearances require a separate design.

For the reference teeth $(Z_s, Z_p, Z_r) = (24, 18, 60)$ and module $m_0 = 1/500$ m, the pitch radii are $(3/125, 9/500, 3/50)$ m and the offset is $a = 21/500$ m. With carrier angle ϕ , take joint centres $J_1 = (a \cos \phi, a \sin \phi, z_1)$ and $J_2 = (0, 0, z_1 + L)$, where $L = 3/25$ m. The intermediate member has length $\sqrt{a^2 + L^2}$ and equal joint angles $\beta = \arctan(a/L)$. Its centreline sweeps radius $a(1 - \lambda)$ at fraction λ along the member. These dimensions specify a kinematic layout; the drawing does not establish a manufacturing clearance or a spatial joint-force solution. Equal operating angles and correct relative phasing are also requirements in ordinary double-joint practice [Belden](#). Here the joint plane turns with the carrier, and the rigid intermediate shaft fixes the yokes relative to that plane.

The central worked comparison holds the ring and uses shaft speeds $(7/2, 1, 0, -7/3)$ rad/s. Adding a $+1$ N m receiver torque at the negative-running planet output changes carrier work by $+7/3$ J over one second and delivers $7/3$ J to that receiver. The held-ring reaction changes as well, although its ground work remains zero. Both changes follow from the attached free bodies, before any work balance is formed.

The companion, *Finite Transfers and Open Energy Balances*, develops the corresponding physical questions: loaded joint forces, support and phase supplies, winding commutation, prepared cell states, and calibrated observations of each transfer. It includes a separate switched tapped-secondary apparatus motivated by SERPS. The main derivations below establish the four-shaft construction and its electrical realization; finite state models and exact controls state what extends beyond the ideal terminal relation. The later sections and appendix retain the detailed compound, reference, preparation and event arguments needed to delimit those claims.

1.1 Ports, states, and signed observations

All angular components are positive by the right-hand rule about the common $+z$ axis. We use angular rates ω in radians per second and $\nu = \omega / (2\pi)$ in revolutions per second. A frame rotating at Ω reports $\omega^f = \omega - \Omega$. A port power is positive into its stated system. For an interval with event times t_e , define

$$W_j(t_0, t_1) = \int_{t_0}^{t_1} P_j(t) dt, \quad P_j = e_j f_j, \quad (1)$$

$$r_E = E(x(t_1)) - E(x(t_0)) - \sum_j W_j - \sum_e W_e^{\text{imp}}. \quad (2)$$

Here $e_j f_j$ means torque times angular rate, force dotted with velocity, or voltage times current; an exported heat channel can equivalently be written $-T\dot{S}_{\text{out}}$ for a reservoir at fixed temperature T . Each port is integrated separately before the sum is formed. A constitutive endpoint energy does not specify the work of an unmodelled preparation path. Positive r_E means that the stored-energy increase exceeds the accounted net inward work; negative r_E means that it falls short. Both signs remain eligible outcomes. For a physical discrepancy, the interval, boundary, event states, signed magnitude, uncertainty, and completed checks must accompany the result. An absent measurement leaves the sign and magnitude unknown.

Table 1. Notation used across the mechanical and electrical models. Subscripts distinguish local states from these scalar quantities.

Symbol	Meaning
q	Number of planets
$\chi = F'(x)$	Instantaneous lead-out transmission factor
$\mathcal{T}$	Coupling torque in the lead-out
τ_*	Applied torque scale in the compound train
T	Duration of an integration interval
$p_j = V_j I_j$	Single-terminal power contribution
$W_j, W_j , \int P_j dt$	Signed work, its magnitude, and integrated absolute power

We distinguish a whole-system boundary from component boundaries. A mesh or pin has two sides; they are indispensable when examining a planet but internal to an enclosure containing both connected bodies. Similarly, winding powers and terminal contributions must not be added again to the source, receiver, and heat powers of a complete circuit. Ideal constraints below have no energy coordinates; their empty store is zero by definition. Finite bodies and windings have explicit nonempty stores evaluated at both endpoints.

Signed sums also differ from absolute observations. For oriented coordinates y_j and positive weights w_j , $\sum_j w_j |y_j|$ depends on the observation map, units, and weights. A linear constraint on the signed coordinates does not make their absolute sum conserved, nor make it energy. For example, if $E = x^T H x / 2$ with positive definite H and $y = O x$, Cauchy–Schwarz gives the exact instantaneous geometric bound

$$\sum_j w_j |y_j| \leq \sqrt{2E(x) \max_{\sigma_j = \pm 1} (w \odot \sigma)^T O H^{-1} O^T (w \odot \sigma)}. \quad (3)$$

It uses the current state energy, not a presumption that the initial energy persists. Extra conserved linear functionals restrict the admissible set only when actually derived. Neither a gear turn count nor a raw winding-linkage sum is automatically eligible for a conserved-sum growth claim.

1.2 Resolving a transfer and resolving a remainder

Let measured effort and flow be $\hat{e}, \hat{f}$, with independently established pointwise errors $|e - \hat{e}| \leq u_e$ and $|f - \hat{f}| \leq u_f$ on the declared interval. Expanding their product, then integrating its three error terms, gives the exact bound

$$|W - \widehat{W}| \leq \int_{t_0}^{t_1} (|\hat{e}|u_f + |\hat{f}|u_e + u_e u_f) dt, \quad \widehat{W} = \int_{t_0}^{t_1} \hat{e}\hat{f} dt. \quad (4)$$

The errors must refer to the actual conjugate port quantities at common times. Additional bounds cover channel timing, finite bandwidth, integration of recorded signals, and interval endpoints. For example, if a time displacement is at most u_t and the displaced flow satisfies $|\dot{f}| \leq L_f$, the additional product error with an aligned effort bounded by E_* is at most $E_* L_f u_t (t_1 - t_0)$. This follows by integrating $|f(t + \delta t) - f(t)| \leq L_f u_t$; it does not bound an unobserved fast transient without such a derivative bound. Endpoint-time errors also need bounds on the omitted terminal intervals. Count a timing contribution only once if it is already included in the pointwise errors.

For a complete measured balance, a conservative enclosure is

$$u_r = u_{E,0} + u_{E,1} + \sum_j u_{W,j} + \sum_e u_{\text{imp},e}, \quad r_E \in [\hat{r}_E - u_r, \hat{r}_E + u_r]. \quad (5)$$

Here each work bound includes its justified observation errors; resolved finite edges and ideal impulses describe disjoint transfers. Endpoint correlation can reduce this enclosure only when established independently. An unknown thermal state, magnetic state, or missing port is an undetermined physical contribution, not an uncertainty allowance. All evaluated model integrals in this article are exact, so their numerical integration residual is zero. This says nothing about the uncertainty of physical observations.

2 Rolling constraints and three kinds of count

2.1 A coin, a spoke, and an independently loaded output

A marked body carried once around a centre makes the reference distinction visible. Let ϕ be the orbital angle and θ its orientation relative to fixed axes. With $\Omega = \dot{\phi}$ and $\omega = \dot{\theta}$, integrating $\omega = (\omega - \Omega) + \Omega$ gives the fixed-axis count as the carrier-relative count plus the carrier count, including reversals.

Table 2. Three different constraints on a marked body. Counts are unwrapped signed angles, not final photographs.

Constraint during one positive orbit	Fixed-axis turns	Carrier-relative turns
Free pivot, initially zero spin and zero centre torque	0	-1
Rigid lock to a spoke	1	0
Equal circles, exterior rolling without slip	2	1

For exterior rolling around a fixed circle of radius R , a moving circle of radius r has centre radius $a = R + r$. Its contact velocity is $a\Omega - r\omega$, so

$$\omega = (1 + R/r)\Omega, \quad \omega - \Omega = (R/r)\Omega. \quad (6)$$

Interior rolling at $R > r$ instead gives $(R - r)\Omega + r\omega = 0$ and $\omega = (1 - R/r)\Omega$. The degenerate $R = r$ interior case has no orbit phase. A toothed exterior realization with $Z_s = Z_p$ also needs $Z_r = 3Z_s$ if a

ring is added: a held sun then requires $\omega_r = 4\Omega/3$. Holding both sun and ring would force $\Omega = 0$. Each contact contributes its own constraint.

The centre-of-mass identity $\int \rho \, dm = 0$ in the rigid velocity field $\mathbf{v} = \mathbf{V}_C + \omega \mathbf{e}_z \times \rho$ gives

$$K^0 = \frac{1}{2}Ma^2\Omega^2 + \frac{1}{2}I_C\omega^2, \quad H_{C,z} = I_C\omega. \quad (7)$$

A spoke-locked body has nonzero inertial spin when $\Omega \neq 0$. The free-pivot and clamped-body comparisons in Tesla's 1919 discussion [Tesla \(1919\)](#) make this distinction concrete. Writing a total energy using a radius of gyration includes its second moment; that effective radius does not change the actual centre velocity $a\Omega$. An impulse-free release preserves the body's instantaneous velocity field. Torque and work of any physical locking or release device are additional laws.

For a loaded exterior roller at constant $\Omega > 0$, apply a ground-referenced receiver couple $-\mathcal{T}$, $\mathcal{T} > 0$. With contact tangent force F_t and carrier drive force F_d , the independent spin and centre equations are $-rF_t - \mathcal{T} = 0$ and $F_d + F_t = 0$. Thus the drive torque is $(1 + R/r)\mathcal{T}$, and the held central support torque is $-R\mathcal{T}/r$. Separate integrals over one orbit give

$$W_d = 2\pi(1 + R/r)\mathcal{T}, \quad W_{\text{rec}} = -2\pi(1 + R/r)\mathcal{T}, \quad W_s = 0. \quad (8)$$

The contact point has zero ground velocity; its centre and spin powers $F_t a \Omega$ and $-rF_t \omega$ cancel after both are evaluated. Both endpoints of (7) agree for the stipulated steady motion. The model residual is therefore zero. The assumed ideal torque connection, adequate traction and lossless support do not specify an actual planet lead-out.

As a dimensioned control take $R = 2$ m, $r = 1$ m, $M = 1$ kg, $I_C = 2/5$ kg m², $\Omega = 1/2$ rad/s and $\mathcal{T} = 1$ N m. On $[0, 2]$ s, the separate works are $(3, -3, 0)$ J, and each kinetic endpoint is $63/40$ J. Unloaded preparation with $\Omega = t/2$ on the same time interval instead gives $F_t = -3/5$ N, $F_d = 21/10$ N and $W_d = 63/10$ J. Direct endpoint evaluation gives orbital $9/2$ J and spin $9/5$ J. Preparation and steady receipt are separate histories.

2.2 A parallel-axis train

Expressing mesh motion relative to the carrier is the standard kinematic construction illustrated by [Culpepper \(2002\)](#). Here the two contact constraints are retained separately before deriving the train relation.

Let Z_s, Z_p, Z_r be positive integer tooth counts of the sun, each planet, and the internal ring. For common module m_0 , the pitch radii are $r_j = m_0 Z_j / 2$. The planet centre is at radius a . The simple train requires

$$a = r_s + r_p = r_r - r_p, \quad Z_r = Z_s + 2Z_p. \quad (9)$$

Pure rolling at the two pitch points gives two independently defined planet rates:

$$\omega_p = \omega_c - \frac{Z_s}{Z_p}(\omega_s - \omega_c), \quad (10)$$

$$\omega_p = \omega_c + \frac{Z_r}{Z_p}(\omega_r - \omega_c). \quad (11)$$

Eliminating ω_p yields the division-free Willis relation

$$Z_s(\omega_s - \omega_c) + Z_r(\omega_r - \omega_c) = 0. \quad (12)$$

The ratio form is valid only when its denominator is nonzero. Equation (12) also covers the locked train.

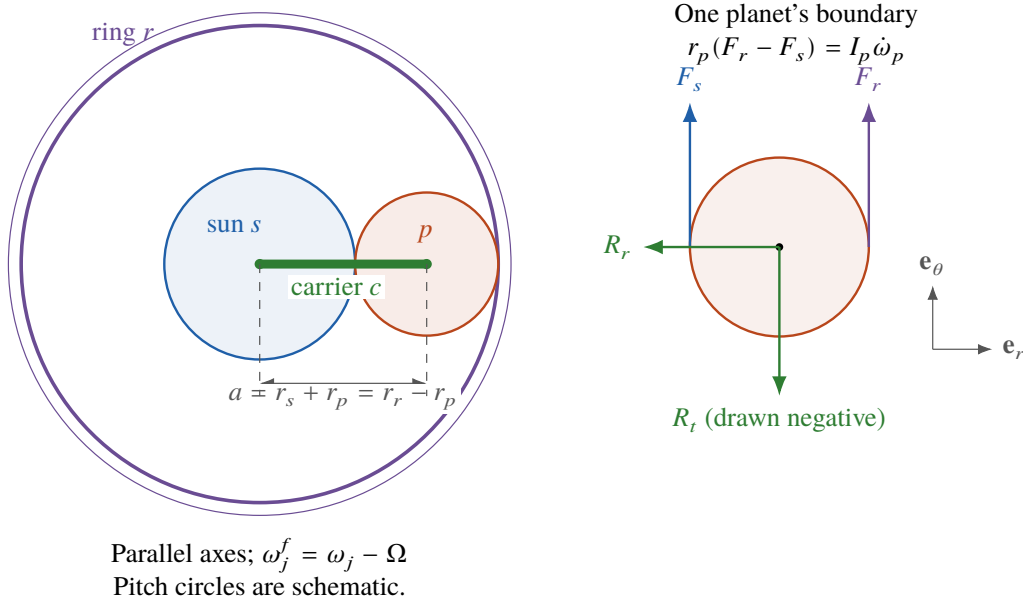


Figure 2. Schematic pitch geometry and forces on one planet. Blue identifies the sun, orange the planet, green the carrier, and purple the ring. Tangential mesh forces on the planet have signed components F_s, F_r ; its pin force is $R_t \mathbf{e}_\theta + R_r \mathbf{e}_r$. Only one of the three planets is drawn.

The geometry of Figure 2 permits scalar subtraction of rates because all rotation axes are parallel. Helical thrust, bevel geometry, and frames rotating about other axes require vector kinematics and additional force components. Positive pitch radii alone do not establish manufacturability: equally spaced multiple planets also require compatible tooth phasing and clearance. For the simple train with q equally spaced planets, a familiar phase condition follows by advancing adjacent mesh phases through $2\pi/q$: $(Z_s + Z_r)/q$ must be an integer. All algebra below can instead describe a single planet or a stipulated equal-load assembly; it is not proof that every integer pair admits three equally spaced, noninterfering planets.

2.3 Signed displacement, tooth passage, and a shaft readout

On the same explicit interval define

$$N_p^f = \frac{1}{2\pi} \int_{t_0}^{t_1} (\omega_p - \Omega) dt, \quad Q_p = \frac{Z_p}{2\pi} \int_{t_0}^{t_1} (\omega_p - \omega_c) dt. \quad (13)$$

Q_p is an oriented tooth-pitch passage at a named mesh; integer crossing counts additionally need an initial tooth phase. The sign at the other mesh can be chosen oppositely, but that choice must be retained. Absolute traffic is $Z_p \int |\omega_p - \omega_c| dt / (2\pi)$, which equals $|Q_p|$ only without a reversal. The magnitude of the net body count $|N_p^f|$ is another observable again.

Table 3. Three distinct meanings of a planet's accumulated rotation.

Observable	Definition	Dependence
Body displacement	N_p^f	Observation frame
Tooth-pitch passage	Q_p at a named mesh	Relative motion; invariant under a common frame change
Output displacement	$(\Delta\theta_o - \Delta\theta_\sigma)/(2\pi)$	Linkage and stator σ

Subtracting the integrands, without any dynamical assumption, proves

$$N_p^0 - N_p^c = N_c^0, \quad N_p^r = N_p^c + N_c^r. \quad (14)$$

A common frame subtraction leaves every relative rate unchanged and hence leaves Q_p unchanged. These identities apply to arbitrary integrable paths, including reversals; rational constant rates are only a convenient example.

For specificity, take the positive-integer family

$$Z_s \in \{12, 17, 18, 24, 30, 36, 48, 60, 101\}, \quad Z_p \in \{12, 13, 18, 24, 36\}.$$

It has $36 \leq Z_r \leq 173$ under (9). Over $[0, 7/2]$ s, eight distinct rate choices are described by the following complete definitions, with the other rates obtained from (11)–(12).

Table 4. Rate choices for ground, carrier, sun, and ring observations. The interval is not one second, so rates and accumulated counts must not be interchanged.

Choice	(ν_s, ν_c) in rev/s	Defining feature
1	$(7/3, 5/4)$	Forward motion
2	$(-2, 3)$	Counterrotation
3	$(1 + Z_r/Z_s, 1)$	Ring held
4	$(0, 1)$	Sun held
5	$(1, 0)$	Carrier held
6	$(1, 1)$	Locked relative motion
7	$(1 + Z_p/Z_s, 1)$	Planet stationary in ground
8	$(1, Z_s/(Z_s + Z_p))$	Same stationary condition with sun rate fixed

Write $d = \omega_s - \omega_c$ and $(k_s, k_c, k_r, k_p) = (1, 0, -Z_s/Z_r, -Z_s/Z_p)$. Every rate is $\omega_j = \omega_c + k_j d$, and every interbody relative rate is $(k_j - k_l)d$. The four k_j are distinct, so all relative rates vanish if and only if $d = 0$. In the full (ω_s, ω_c) plane, the four absolute-rate zeros and this relative-rate zero are **five lines** through the origin, not five rays. A transverse passage through a nontrivial linear zero reverses sign; a tangency or the identically zero path does not.

3 Forces, torques, and moving-frame work

3.1 Free bodies before power

Use $q = 3$ equally loaded planets, no pin couple, and inertias I_s, I_r, I_c, I_p ; m_p is one planet's mass. Mesh forces F_s, F_r are tangential components on one planet, with the signs in Figure 2. The following Newton–Euler equations determine the remaining forces and shaft torques from a prescribed τ_s and kinematically compatible accelerations:

$$F_s = \frac{\tau_s - I_s \dot{\omega}_s}{qr_s}, \quad F_r = F_s + \frac{I_p \dot{\omega}_p}{r_p}, \quad (15)$$

$$\tau_r = I_r \dot{\omega}_r + qr_r F_r, \quad R_t = m_p a \dot{\omega}_c - F_s - F_r, \quad (16)$$

$$R_r = -m_p a \dot{\omega}_c^2, \quad \tau_c = I_c \dot{\omega}_c + qaR_t. \quad (17)$$

The radial equation assumes fixed a and neglects other radial mesh forces; it isolates the declared tangential pitch-contact model. A pressure-angle model would change the radial reaction.

When all rates are constant, (17) gives

$$F_r = F_s = \frac{\tau_s}{qr_s}, \quad \tau_r = \frac{Z_r}{Z_s} \tau_s, \quad \tau_c = -\frac{Z_s + Z_r}{Z_s} \tau_s, \quad \tau_s + \tau_r + \tau_c = 0. \quad (18)$$

These are consequences of the free bodies. The planet's real axial couple $r_p(F_r - F_s) = I_p \dot{\omega}_p$ vanishes exactly when $\dot{\omega}_c = (Z_s/Z_p)(\dot{\omega}_s - \dot{\omega}_c)$ for $I_p > 0$. Thus a steady idler has zero net spin-couple work in either frame, although its body count differs. Its separate contact and pin powers need not vanish.

Let

$$J = I_s + I_r + I_c + q(I_p + m_p a^2), \quad (19)$$

$$H = I_s \omega_s + I_r \omega_r + (I_c + qm_p a^2) \omega_c + qI_p \omega_p = J \omega_c + Bd, \quad (20)$$

$$B = I_s - I_r Z_s/Z_r - qI_p Z_s/Z_p. \quad (21)$$

Summing the free bodies gives $\sum \tau = \dot{H}$ on (9). Before imposing that assembly, the surplus is $q[(r_s - a + r_p)F_s + (r_r - a - r_p)F_r]$. It need not vanish. This is a compatibility requirement of these equations, not a failure of the angular-momentum law for physical mechanisms.

The equation $\sum \tau = 0$ alone constrains accelerations to $J\dot{\omega}_c + B\dot{d} = 0$ and need not make them zero. For example, choose positive effective inertias $I_s = I_r = qI_p = I_c + qm_p a^2 = 1 \text{ kg m}^2$ and tooth ratios $Z_s/Z_r = 2/5$, $Z_s/Z_p = 4/3$. Then $J = 4$, $B = -11/15$ in the corresponding units. Taking $\dot{\omega}_c = 1 \text{ rad/s}^2$ and $\dot{d} = 60/11 \text{ rad/s}^2$ accelerates every body while $\dot{H} = 0$. The torque ratios in (18) cannot be inferred in this case.

3.2 The four-shaft operating family

For the correctly phased axial model, write $u = \omega_c$ and $d = \omega_s - \omega_c$. The two pitch constraints give

$$\omega = \begin{pmatrix} \omega_s \\ \omega_c \\ \omega_r \\ \omega_o \end{pmatrix} = B \begin{pmatrix} u \\ d \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -Z_s/Z_r \\ 1 & -Z_s/Z_p \end{pmatrix}. \quad (22)$$

The sun/carrier minor has determinant -1 , so these four accessible shafts have exactly two independent speeds. This division-free form includes every held shaft. The four absolute-speed zero lines are $u+d=0$, $u=0$, $Z_r u - Z_s d = 0$, and $Z_p u - Z_s d = 0$. All six interbody relative rates vanish on the single line $d=0$. Let F_{si}, F_{ri} be tangential sun and ring forces on planet i , and R_{ti} its carrier-pin force. Let τ_{oi} be the external torque transmitted from its output; the equal-angle massless linkage transmits this axial torque without a baseline carrier-support torque. The body equations are

$$r_s \sum_i F_{si} = \tau_s - I_s \dot{\omega}_s, \quad r_p (F_{ri} - F_{si}) + \tau_{oi} = I_{pi} \dot{\omega}_p, \quad (23)$$

$$R_{ti} + F_{si} + F_{ri} = m_{pi} a \dot{\omega}_c, \quad \tau_r = I_r \dot{\omega}_r + r_r \sum_i F_{ri}, \quad (24)$$

$$\tau_c = I_c \dot{\omega}_c + a \sum_i R_{ti}. \quad (25)$$

I_c excludes planet orbit; added output inertia belongs in the relevant spin demand. The pin equation includes orbital acceleration once. For one steady loaded planet, these equations give $F_s = \tau_s/r_s$, $F_r = F_s - \tau_o/r_p$ and $R_t = -F_s - F_r$, hence

$$\tau_r = \frac{Z_r}{Z_s} \tau_s - \frac{Z_r}{Z_p} \tau_o, \quad (26)$$

$$\tau_c = -\left(1 + \frac{Z_r}{Z_s}\right) \tau_s + \left(\frac{Z_r}{Z_p} - 1\right) \tau_o, \quad B^T \tau = 0. \quad (27)$$

The orthogonality follows from forces and moment arms. Only afterward does $\tau^T \omega = 0$ follow. The operating family has coordinates (u, d, τ_s, τ_o) before source and load laws select a subset. A passive receiver must satisfy $\tau_o \omega_o \leq 0$ at its actual reference.

For (24, 18, 60) teeth, the following exact controls use $[0, 1]$ s. Each displayed work is its own constant torque–rate integral.

Table 5. Four complete external ports. Positive work enters the assembly; a held port can carry a nonzero reaction.

State	$(\omega_s, \omega_c, \omega_r, \omega_o)$, rad/s	$(\tau_s, \tau_c, \tau_r, \tau_o)$, N m	(W_s, W_c, W_r, W_o) , J
Carrier held	$(1, 0, -2/5, -4/3)$	$(2, -14/3, 5/3, 1)$	$(2, 0, -2/3, -4/3)$
Ring held, $\tau_o = 0$	$(7/2, 1, 0, -7/3)$	$(2, -7, 5, 0)$	$(7, -7, 0, 0)$
Ring held, $\tau_o = 1$	$(7/2, 1, 0, -7/3)$	$(2, -14/3, 5/3, 1)$	$(7, -14/3, 0, -7/3)$
Ring held, $\tau_o = 3$	$(7/2, 1, 0, -7/3)$	$(2, 0, -5, 3)$	$(7, 0, 0, -7)$
Sun held	$(0, 1, 7/5, 7/3)$	$(2, -14/3, 5/3, 1)$	$(0, -14/3, 7/3, 7/3)$
Output held	$(7/4, 1, 7/10, 0)$	$(2, -14/3, 5/3, 1)$	$(7/2, -14/3, 7/6, 0)$
Co-rotation	$(1, 1, 1, 1)$	$(2, -14/3, 5/3, 1)$	$(2, -14/3, 5/3, 1)$
Opposed rates	$(-2, 1, 11/5, 5)$	$(2, -14/3, 5/3, 1)$	$(-4, -14/3, 11/3, 5)$
Planet supplies work	$(7/2, 1, 0, -7/3)$	$(2, -28/3, 25/3, -1)$	$(7, -28/3, 0, 7/3)$

The zero-torque control at the ring-held speeds has four zero works. Reversing every speed reverses every work while preserving the torque solution. Zero speeds with the displayed reaction set give an

all-held zero-work control. With $\tau_o = 3\tau_s/4$, the ring torque vanishes; with $\tau_o = 3\tau_s/2$, the carrier torque vanishes. These are ordinary load boundaries. A throughput ratio with $D = \sum_j \max(\tau_j \omega_j, 0) = 0$ is undefined, including moving zero-torque states.

Choose four effective inertias of 1 kg m^2 , with the carrier entry including planet orbit and the last entry combining planet and output spin. At each endpoint of the ring-held controls, $K^0 = \frac{1}{2} \sum_j \omega_j^2 = 673/72 \text{ J}$. For example a 1 kg planet permits bare carrier inertia $1 - a^2 > 0$ in SI units, and the combined spin can be split into two $1/2 \text{ kg m}^2$ inertias. No preparation work is assigned to these initialized states. The separately integrated works and independently evaluated endpoints give $r_E = 0$ with no event or impulse.

For q planets the steady equations determine $\sum_i F_{si}$ and each $F_{ri} - F_{si}$, leaving $q - 1$ force-sharing freedoms. A declared equal bilateral stiffness and equal total contact deflection give

$$F_{si} = \frac{\tau_s}{qr_s} + \frac{\tau_{oi} - \bar{\tau}_o}{2r_p}, \quad F_{ri} = F_{si} - \frac{\tau_{oi}}{r_p}, \quad \bar{\tau}_o = \frac{1}{q} \sum_i \tau_{oi}. \quad (28)$$

Indeed $F_{si} + F_{ri}$ is then common, and summing the first equation recovers the sun force. This is a bilateral or preloaded-contact selection, with zero tooth error. At $q = 3$, $\tau_s = 2$, loads $(3, 0, 0) \text{ N m}$ give $F_s = (250/3, 0, 0) \text{ N}$ and $F_r = -F_s$. Loads $(1, 1, 1)$ give $F_s = (250/9, 250/9, 250/9) \text{ N}$. Both have $\tau_c = 0$, $\tau_r = -5 \text{ N m}$; their individual forces and takeoffs differ. A common output combining three takeoffs needs its own phase offsets, load sharing and support model.

For acceleration the equation is $B^T \tau = B^T M B(\dot{u}, \dot{d})^T$. With $M = I$, $u = 1 + t$, $d = 5/2 + 60t/11$, and zero constraint force, $\tau = \dot{\omega} = (71/11, 1, -13/11, -69/11)$ in SI units on $[0, 1] \text{ s}$. Their sum vanishes although all four accelerations are nonzero. Each work is $\tau_j \omega_j(0) + \tau_j \dot{\omega}_j/2$; each endpoint is $\sum_j \omega_j(t)^2/2$. Substitution gives the same total increase. The steady torque law must therefore remain attached to its zero-acceleration assumption.

3.3 The actual port sides

For the unloaded three-shaft subcase, the boundary encloses all gears, carrier and pins, and its physical inputs are the sun, carrier and ring shafts. A loaded planet adds its fourth port as derived above. Internal transfers on one planet's boundary are listed separately:

Table 6. Force–velocity and torque–rate products. Here $\dot{a} = 0$. Rolling equates the two contact-point velocities; it does not set each contact-side power to zero.

Port or side	Signed instantaneous power in frame f
Shaft $j = s, r, c$ into complete train	$\tau_j(\omega_j - \Omega)$
Sun mesh into one planet	$F_s r_s(\omega_s - \Omega)$
Same mesh into sun	$-F_s r_s(\omega_s - \Omega)$
Ring mesh into one planet	$F_r r_r(\omega_r - \Omega)$
Same mesh into ring	$-F_r r_r(\omega_r - \Omega)$
Pin into one planet	$R_t a(\omega_c - \Omega) + R_r \dot{a}$
Same pin into carrier	Negative of planet-side pin power

A sliding variant must replace ideal no-slip contact by a declared relative velocity v_{slip} and opposing friction force. With Coulomb coefficient $\mu_f \geq 0$ and normal force $F_N \geq 0$, the pair's mechanical power sum is $-D$, where $D = \mu_f F_N |v_{\text{slip}}|$. Both contact points receive the same subtraction $\Omega \times \mathbf{r}$, so v_{slip}

and D are frame invariant. The exported heat work is $-\int D dt$, or $-\int T \dot{S}_{\text{out}} dt$ under immediate thermal export. The pure-rolling idealization has $D = 0$; it does not bound real mesh losses, even for an illustrative $0 \leq \mu_f \leq 1/2$.

3.4 Relative kinetic energy and effective centrifugal potential

Define the constitutive ground kinetic energy

$$K^0 = \frac{1}{2} I_s \omega_s^2 + \frac{1}{2} I_r \omega_r^2 + \frac{1}{2} (I_c + q m_p a^2) \omega_c^2 + \frac{1}{2} q I_p \omega_p^2. \quad (29)$$

Replacing every angular rate by its frame-relative value gives $K^f = K^0 - \Omega H + J \Omega^2/2$. The effective centrifugal potential of all the mass, including spin contributions, is $U^f = -J \Omega^2/2$. Hence

$$E^f = K^f + U^f = K^0 - \Omega H, \quad E^0 - E^f = \Omega H. \quad (30)$$

For one planet the centrifugal part is $-\Omega^2 (I_p + m_p a^2)/2$, of which the orbital part is $-m_p \Omega^2 a^2/2$. This is a coordinate-dependent effective potential, not a newly created physical reservoir. It can make E^f negative. For fixed J , the Euler-force power summed over the bodies is $-\dot{\Omega} (H - J \Omega)$: for a spin it is the fictitious torque $-I \dot{\Omega}$ times $\omega - \Omega$, and for translation it is $(-m \dot{\Omega} \times \mathbf{r}) \cdot \mathbf{v}^f$. The explicit potential-time term is $-J \Omega \dot{\Omega}$. They must be retained separately when using E^f . The second is a parameter term in the effective store, not a physical shaft or a second heat loss. Multiplying (17) by the corresponding velocities gives

$$\dot{K}^0 = \sum_{j=s,r,c} \tau_j \omega_j - D, \quad (31)$$

$$\dot{E}^f = \sum_{j=s,r,c} \tau_j (\omega_j - \Omega) - D - \dot{\Omega} (H - J \Omega) - J \Omega \dot{\Omega}. \quad (32)$$

Thus a complete effective-frame balance includes $-\dot{\Omega} H$ in addition to physical port products. No physical frame-driving supply work is inferred from this mathematical term.

A useful correction concerns Coriolis force. The rotating-velocity and Coriolis formulas are standard; see MIT (2022), Section 31.4. At fixed a the planet-centre velocity is $a(\omega_c - \Omega) \mathbf{e}_\theta$ in frame f , so

$$\mathbf{F}_{\text{Cor}} = 2m_p \Omega a (\omega_c - \Omega) \mathbf{e}_r, \quad \mathbf{F}_{\text{Cor}} \cdot \mathbf{v}^f = 0. \quad (33)$$

It is zero in ground and carrier frames, but generally nonzero in a sun or ring frame. Constant radius guarantees zero Coriolis **power**, not zero Coriolis force in every frame. Radial motion adds a tangential component; Coriolis force still does no total instantaneous work.

The real axial torque component is unchanged by rotating coordinates about z , even during acceleration. Real force vectors are also unchanged as geometrical objects, but their coordinate components are not. If R_r, R_t are constant in the carrier basis, then $R_x = R_r \cos \phi - R_t \sin \phi$ and $R_y = R_r \sin \phi + R_t \cos \phi$, with $\dot{\phi} = \omega_c - \Omega$. Each fixed-axis component has full-cycle peak-to-peak excursion $2\sqrt{R_r^2 + R_t^2}$; a shorter interval need not reach those extrema. Stationarity requires zero relative rotation or zero force, with additional possibilities only for specially varying components.

3.5 An exact steady ledger and its preparation

Choose $(Z_s, Z_p, Z_r) = (24, 18, 60)$, $m_0 = 1/500$ m, $\tau_s = 47/4$ N m, and held-ring rates $(\nu_s, \nu_c, \nu_r, \nu_p) = (7/2, 1, 0, -7/3)$ rev/s. The operation interval is $[2/5, 19/10]$ s, of duration $3/2$ s. Each work in the next

table is its constant effort–flow product times that duration, integrated before combining.

Table 7. Exact operation works. Opposite internal sides have opposite signs and cancel on the whole-train boundary. The sun and ring mesh sides are different; there is no single work common to every mesh side.

Signed work	Ground, J	Carrier, J
Sun shaft	$987\pi/8$	$705\pi/8$
Held-ring shaft	0	$-705\pi/8$
Carrier shaft	$-987\pi/8$	0
One planet-side pin	$-329\pi/8$	0
One sun-mesh planet side	$329\pi/8$	$235\pi/8$
One ring-mesh planet side	0	$-235\pi/8$

For endpoint evaluation choose $m = \mu r^2$, $I = \mu r^4/2$ for each disc, with $\mu = 490 \text{ kg/m}^2$; μ includes the circular area factor. The carrier disc has radius $a = 21/500 \text{ m}$. The ring has outer radius $r_o = r_r + 1/250 \text{ m}$ and $I_r = \mu(r_o^4 - r_r^4)/2$. Planet orbital inertia is additional. Direct substitution gives

$$J = \frac{16851149}{6250000000} \text{ kg m}^2, \quad H_* = \frac{166698\pi}{48828125} \text{ kg m}^2/\text{s}, \quad (34)$$

$$K_*^0 = \frac{18864657\pi^2}{3125000000} \text{ J}, \quad E_*^c = -\frac{2472687\pi^2}{3125000000} \text{ J}. \quad (35)$$

Both endpoints of operation have these respective values because their states are explicitly the same. The separately summed external shaft works are zero in either frame, and $r_E = 0$. A held ring has nonzero carrier-frame work; deleting its reaction port would leave $705\pi/8 \text{ J}$ unbalanced. This difference is present in shaft and mesh sides as well as pins.

The same endpoint can be reached by a specified smooth preparation. Put $F(z) = 3z^2 - 2z^3$; on $[0, 2/5] \text{ s}$ take $\omega_j = \omega_{j*}F(t/(2/5))$ and $\tau_s = (47/4)F$. On $[19/10, 5/2] \text{ s}$ replace F by $1 - F((t - 19/10)/(3/5))$. Accelerations and applied torque meet the steady values continuously. There are no impulses. The free bodies yield $\tau_j = a_j F + b_j \dot{F}$, where

$$(a_s, a_r, a_c) = \left(\frac{47}{4}, \frac{235}{8}, -\frac{329}{8} \right), \quad (b_s, b_r, b_c) = \left(0, -\frac{1639197\pi}{625000000}, \frac{18864657\pi}{3125000000} \right) \quad (36)$$

in torque and angular-momentum units respectively. Each shaft integral on a ramp of duration T is exactly

$$W_j^f = (\omega_{j*} - \Omega_*) \left(\frac{13}{35} T a_j \pm \frac{1}{2} b_j \right), \quad (37)$$

with $+$ for preparation and $-$ for relaxation. This follows from $\int_0^1 F^2 dz = 13/35$ and $\int F dF = \pm 1/2$. For the internal sides write

$$F_s = f_s F + g_s \dot{F}, \quad F_r = f_r F + g_r \dot{F}, \quad R_t = -2f_s F + g_t \dot{F},$$

where $f_s = (47/4)/(qr_s)$, $g_s = -I_s \omega_{s*}/(qr_s)$, $g_r = g_s + I_p \omega_{p*}/r_p$, and $g_t = m_p a \omega_{c*} - g_s - g_r$. The separately integrated planet-side works are therefore

$$W_{sp}^f = r_s (\omega_{s*} - \Omega_*) \left(\frac{13T}{35} f_s \pm \frac{g_s}{2} \right), \quad (38)$$

$$W_{rp}^f = r_r(\omega_{r*} - \Omega_*) \left(\frac{13T}{35} f_s \pm \frac{g_r}{2} \right), \quad (39)$$

$$W_{pin,p}^f = a(\omega_{c*} - \Omega_*) \left(-\frac{26T}{35} f_s \pm \frac{g_t}{2} \right). \quad (40)$$

Each opposite side has the negative of its corresponding expression; no internal sum is added to the whole-train shaft works. The ground endpoints are 0, K_*^0 during preparation and K_*^0 , 0 during relaxation. Carrier endpoints are 0, E_*^c and E_*^c , 0.

The separately integrated Euler and potential-time terms on preparation are $-\Omega_*(H_* - J\Omega_*)/2$ and $-J\Omega_*^2/2$; both reverse on relaxation. The ground-minus-carrier shaft-work sum is $\Omega_*H_*/2 = 166698\pi^2/48828125$ J, and the corresponding effective-field difference is the same. Their total is $\Omega_*H_* = 333396\pi^2/48828125$ J. For the distinct forward rates $(\nu_s, \nu_c) = (7/3, 5/4)$ the same formula applies with its own H_* ; a carrier-held path has $\Omega_* = 0$ and no frame difference. These statements follow from the path, not constant stores.

In general the shaft difference is $\int \Omega dH$ and the effective-field difference is $\int H d\Omega$. Integration by parts proves

$$\int_{t_0}^{t_1} \Omega \dot{H} dt + \int_{t_0}^{t_1} H \dot{\Omega} dt = [\Omega H]_{t_0}^{t_1}. \quad (41)$$

The equal split requires a proportional history. For the explicit path $H = H_*z$, $\Omega = \Omega_*[z + bz(1 - z)]$, $0 \leq z \leq 1$, its two shares are $1/2 + b/6$ and $1/2 - b/6$. They can lie outside $[0, 1]$ when one term opposes the total. When $B \neq 0$ in (21), compatible sun and carrier rates can be reconstructed from H and $\Omega = \omega_c$. Prescribing only endpoint rates leaves these individual works undetermined.

3.6 The locked control and the two limits

Set every rate to 2π rad/s and set the drive torque to zero. Equation (17) leaves only $R_r = -83349\pi^2/3125000$ N, with zero radial power. Over $[0, 3/2]$ s, $N_p^0 = N_c^0 = 3/2$, $N_p^c = 0$, and no teeth pass either mesh. Every physical port work is exactly zero. Independent stores are

$$K^0(t_0) = K^0(t_1) = \frac{16851149\pi^2}{3125000000} \text{ J}, \quad E^c(t_0) = E^c(t_1) = -\frac{16851149\pi^2}{3125000000} \text{ J}. \quad (42)$$

In carrier coordinates $K^c = 0$, whereas U^c is negative. Assigning the positive ground value to both effective stores would contradict (30). Both balances nonetheless give $r_E = 0$.

Locking the rates while retaining $\tau_s = 47/4$ N m is a different control. It transmits the self-equilibrated torque set (18); for example the ground pin power per planet is $-329\pi/12$ W and its carrier-frame power is zero. On the approach $\omega_c = 2\pi$, $\omega_s = \omega_c + \varepsilon$, fixed drive gives a nonzero limiting maximum shaft work $987\pi/8$ J over $3/2$ s; that maximum is constant on the portion where the carrier port dominates. Scaling the drive to zero with ε makes every port work approach zero. Both paths are continuous, but only the latter reaches the zero-drive null. The varying counts alone imply neither result about work.

4 What a physical lead-out reports

4.1 Stator, phase reference, and finite-interval ripple

A lead-out connects the planet axle to a parallel output axis offset by a ; it does not connect that output to the sun. Denote the planet and carrier angles by θ_p, ϕ , and the output and stator angles by θ_o, θ_σ . The

reported count is $N_o^\sigma = \Delta(\theta_o - \theta_\sigma)/(2\pi)$. For the **same output motion**, subtraction gives

$$N_o^0 - N_o^c = \Delta\phi/(2\pi). \quad (43)$$

With finite loading, physically remounting the stator may change that motion; then (43) must be applied separately to each trajectory.

For a Hooke joint of angle $\beta < \pi/2$, let $c_\beta = \cos \beta > 0$ and let $H_\beta(x)$ be the continuous lift of $\text{atan2}(\sin x, c_\beta \cos x)$, with $H_\beta(0) = 0$. Then $H_\beta(x + \pi) = H_\beta(x) + \pi$ and $\tan H_\beta(x) = \tan x/c_\beta$. The joint plane in a parallel-offset Z arrangement is carried by the carrier. Writing $x = \theta_p - \phi$, a declared two-joint kinematic map is

$$\theta_o = \phi + H_\beta\left(H_\beta(x) + \gamma + \frac{\pi}{2}\right) + \frac{\pi}{2}, \quad (44)$$

where constant additive angles have no effect on displacement. With correct carrier-fixed phasing $\gamma = 0$, this reduces to $\theta_o = \theta_p$ up to a constant. A rigid intermediate shaft supplies that fixed relative phasing. With carrier-fixed quarter-turn error $\gamma = \pi/2$, write the output as $\phi + F_\beta(x)$ up to a constant; then

$$\tan F_\beta(x) = \frac{\tan x}{c_\beta^2}, \quad \chi_\beta(x) = F'_\beta(x) = \frac{c_\beta^2}{c_\beta^4 \cos^2 x + \sin^2 x}, \quad c_\beta^2 \leq \chi_\beta \leq c_\beta^{-2}. \quad (45)$$

This is a twice-per-relative-revolution ripple. Since $F_\beta(x) - x$ is bounded and π -periodic, the mean count agrees with the planet count, but an arbitrary finite interval has the endpoint correction $\delta = [F_\beta(x) - x]_{t_0}^{t_1}/(2\pi)$. It is exactly zero over complete ripple periods. It must not be dropped merely because the mean ratio is one.

Holding the phase in ground instead prescribes $\gamma = -\phi$. Equation (44) then gives $\theta_o = \theta_p - \phi +$ a bounded phase-dependent term. A grounded stator has the carrier-relative **mean** count, with an endpoint ripple correction δ_g determined by that equation. An exact claim for all finite intervals would be false. The device that enforces this drifting phase has not been specified dynamically; its required effort and supply work remain unknown.

Table 8. Eleven mountings, with finite-interval phase corrections retained. Coincidence with a particular body count in a special state is not an identity for all states.

Coupling and phase	Ground stator count	Carrier stator count
Correct double Hooke, carrier phase	N_p^0	N_p^c
Quarter-turn double Hooke, carrier phase	$N_p^0 + \delta$	$N_p^c + \delta$
Double Hooke, ground phase	$N_p^c + \delta_g$	Not assigned a physical mounting here
Ideal Oldham or equivalent constant-ratio offset path	N_p^0	N_p^c
Bevel path, $\kappa = 1$	N_p^0	N_p^c
Bevel path, $\kappa = 2$	$N_c^0 + 2N_p^c$	$2N_p^c$

The ideal Oldham constraint is $\theta_o = \theta_p$ even as the offset precesses. Orthogonal sliding coordinates are $a \cos x$ and $a \sin x$; each has excursion amplitude a (full stroke $2a$), and their maximum absolute value

never exceeds a . This kinematic statement can also represent an ideal constant-velocity Schmidt path; it does not make their detailed force systems identical. The bevel path obeys $\theta_o = \phi + \kappa(\theta_p - \phi)$, which proves the last two cases.

Whether an actual orbiting lead-out follows these phase and stator predictions under a small load remains a measurement question [OP-EPI-05]. A carrier encoder and an output encoder can resolve the predicted full-turn count and the twice-per-relative-revolution ripple at low speed; changing the phase datum distinguishes a carrier-fixed joint plane from a ground-referenced actuator. Their finite-interval differences must include δ or δ_g . These are ordinary angle observations, without exceptional energy sensitivity. Equation (43) rules out a single unchanged rotor angle having equal readings against two differently rotating stators. It does not prove that no differential device can encode relative tooth passage, or that no device of any kind can produce an invariant output.

4.2 Load, support, and cycle work

For the carrier-fixed map $\theta_o = \phi + F(x)$, put $d = \omega_p - \omega_c$, $\chi = F'(x)$, and $\omega_o = \omega_c + \chi d$. A linkage boundary includes a rotor of spin inertia I_R and mass m_R orbiting at a , an output inertia I_o , and massless coupling members. Its ground kinetic store is $K_L^0 = I_R \omega_p^2/2 + m_R a^2 \omega_c^2/2 + I_o \omega_o^2/2$; its other-frame stores follow (30) with their own J, H . The stator, imposed output torque τ_L , and drive are external.

Let the drag magnitude be $d_0 \geq 0$, with $s_\sigma = \text{sgn}(\omega_o - \omega_\sigma)$ away from a zero. The output equation gives coupling torque $\mathcal{T} = I_o \dot{\omega}_o - \tau_L + d_0 s_\sigma$. Virtual angular displacements in $\theta_o(\theta_p, \phi)$ give coupling reactions $\mathcal{T} \chi$ and $\mathcal{T}(1 - \chi)$. The external drive must also accelerate the input rotor, so $\tau_d = I_R \dot{\omega}_p + \mathcal{T} \chi$ and $\tau_u = \mathcal{T}(1 - \chi)$ at the drive and carrier support. Their work integrals and those of the load, drag, and rotor pin are

$$W_d^f = \int (I_R \dot{\omega}_p + \mathcal{T} \chi)(\omega_p - \Omega) dt, \quad W_u^f = \int \mathcal{T}(1 - \chi)(\omega_c - \Omega) dt, \quad (46)$$

$$W_L^f = \int \tau_L(\omega_o - \Omega) dt, \quad W_{\text{drag}}^f = \int (-d_0 s_\sigma)(\omega_o - \Omega) dt, \quad (47)$$

$$W_{\text{pin}}^f = \int m_R a \dot{\omega}_c a(\omega_c - \Omega) dt. \quad (48)$$

All integrals in this section have the explicitly stated readout or cycle interval. At constant ω_c the rotor-pin power is zero, whereas the support port can be nonzero. The orbital acceleration is counted only at the pin, not again in τ_u . Summing all five instantaneous products gives $I_R \dot{\omega}_p(\omega_p - \Omega) + I_o \dot{\omega}_o(\omega_o - \Omega) + m_R a^2 \dot{\omega}_c(\omega_c - \Omega)$. For K_L^f add the Euler power $-\dot{\Omega}(H_L - J_L \Omega)$; for $E_L^f = K_L^f - J_L \Omega^2/2$ also add $-J_L \Omega \dot{\Omega}$. These terms reproduce the derivative of the independently evaluated store, with $J_L = I_R + I_o + m_R a^2$ and $H_L = I_R \omega_p + I_o \omega_o + m_R a^2 \omega_c$. Constant input rates remove the input-spin and orbital acceleration powers and give the cycle controls below. This torque-path model does not resolve transverse Cardan joint forces or bending-couple ports; two joints alone do not determine that system. Attaching it to a train with the readout load changes the planet and carrier equations, as derived below.

The physical dissipation is $D_\sigma = d_0 |\omega_o - \omega_\sigma|$. It equals minus the rotor drag power only in the stator frame. For the reference train's ideal Oldham path, $d_0 = 1/50 \text{ N m}$ and $[0, 7/2] \text{ s}$ give

$$\int D_0 dt = \frac{49\pi}{150} \text{ J}, \quad \int D_c dt = \frac{7\pi}{15} \text{ J}. \quad (49)$$

These are different physical mountings. A change of coordinates on either fixed mounting leaves its relative drag dissipation unchanged. Separately integrating the two drag-torque and relative-rate products

gives the finite comparison

$$Q_c - Q_0 = \frac{7\pi}{15} - \frac{49\pi}{150} = \frac{7\pi}{50} \text{ J.} \quad (50)$$

Hold the stated motion in both mountings and observe the actual drag torque, relative angle, and the drive and support works. A changed trajectory calls for its own integrals. Dissipation is internal conversion when the frictional bodies lie inside the apparatus; heat exported across that larger boundary and their thermal endpoint stores are determined separately.

One conditional measurement budget makes the scale explicit. Suppose each recorded drag torque is bounded by $21/1000 \text{ N m}$, each relative rate by 21 rad/s , and calibrated pointwise errors are at most $1/1000 \text{ N m}$ and $1/100 \text{ rad/s}$. These envelopes contain both prescribed rates. Equation (4) gives the following bound for the difference of the two $7/2 \text{ s}$ integrals:

$$u_{Q_c - Q_0}^{\text{product}} \leq 7 \left(\frac{21}{1000} \frac{1}{100} + 21 \frac{1}{1000} + \frac{1}{1000} \frac{1}{100} \right) \text{ J} = \frac{7427}{50000} \text{ J.} \quad (51)$$

The proposed total bound $1/5 \text{ J}$ leaves $2573/50000 \text{ J}$ for timing and other observation errors. If independently met, it separates the predicted $7\pi/50 \text{ J}$ from zero using torque and encoder observations. These are calibration requirements, not specifications of an unnamed instrument. Resolving the full apparatus remainder additionally needs all other ports and endpoint stores in (5).

A prescribed $\tau_L = -47/40 \text{ N m}$ is not necessarily a passive load: when $\omega_o < 0$ its ground-frame power is positive. Describing a torque as a brake therefore requires its relative velocity as well as its sign.

At constant ω_p, ω_c , the quarter-turn Hooke path has cycle duration $T_b = \pi/|d|$. Assume τ_L and s_σ remain constant through that cycle. Set $\mathcal{T}_0 = -\tau_L + d_0 s_\sigma$. Since $F(x + \pi) - F(x) = \pi$, $\int (1 - \chi) dt = 0$. Also $\dot{\omega}_o = d\dot{\chi}$. The ground support work integrates exactly to

$$W_u^0 = \omega_c \left\{ \mathcal{T}_0 \int (1 - \chi) dt + I_o d \left[\chi - \frac{\chi^2}{2} \right]_{t_0}^{t_0 + T_b} \right\} = 0. \quad (52)$$

The output rate and all stores return to their initial values. This proves zero cycle support work for this torque path; it makes no claim for changing drag sign, compliance, backlash, joint friction, or the unresolved transverse force system. It is stronger within that model than merely observing a small cycle residual, and narrower than a claim about real joints.

For an axial lead-out length $L = 3/25 \text{ m}$, the geometry gives $c_\beta^2 = L^2/(L^2 + a^2)$. Choosing $a = 137/1000 \text{ m}$ gives $c_\beta^2 = 14400/33169$, the exact illustrative value in Figure 3. Its large articulation is a kinematic example only.

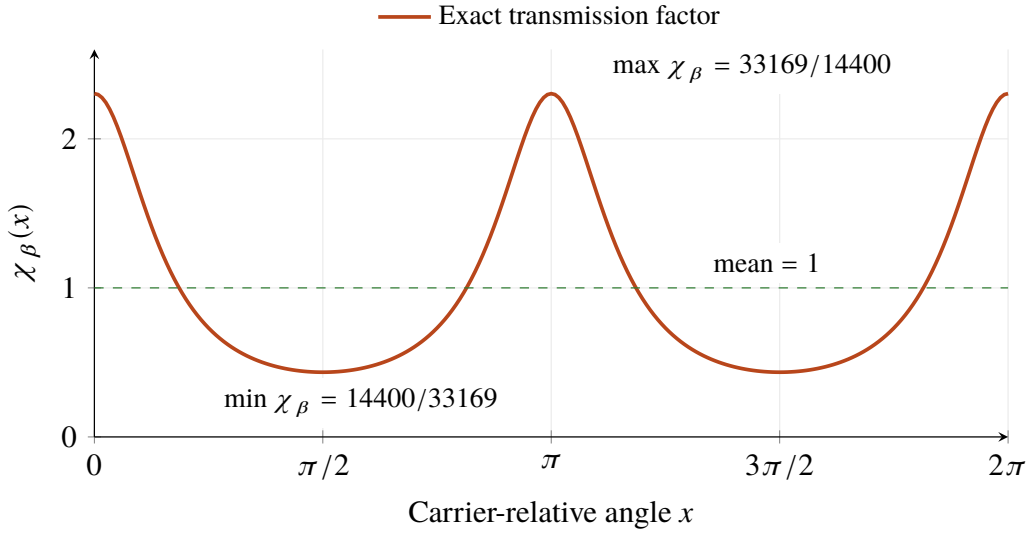


Figure 3. Exact quarter-turn Hooke transmission factor from equation (45), with $\cos^2 \beta = 14400/33169$. The ripple is periodic in the carrier-relative input angle. Its mean is one even though the instantaneous factor ranges between the annotated exact bounds. The illustrative articulation is not asserted to be practical.

For the bevel path $\chi = \kappa$, the support work is instead $\mathcal{T}_0(1 - \kappa)(\omega_c - \Omega)T_b$. As an exact example choose $Z_s = Z_p = 12$, $Z_r = 36$, $(\nu_s, \nu_c) = (7/3, 5/4)$, $\kappa = 2$, $\tau_L = -47/40$, $d_0 = 1/50$ in SI units. Then $\nu_p = 1/6$, $\nu_o = -11/12$, $T_b = 6/13$ s, and $\mathcal{T}_0 = 231/200$ N m for either listed stator. Thus

$$W_u^0 = -\frac{693\pi}{520} \text{ J}, \quad W_u^c = 0. \quad (53)$$

The other integrals in (48) balance it with unchanged endpoint stores. This signed support transfer is the comparison for a loaded apparatus; the combined physical energy remainder is a separate quantity.

4.3 The loaded train and its unresolved reaction work

Does the train with its attached lead-out retain an unexplained energy change after its drive, receiver, moving supports, and endpoint states have all been included [OP-EPI-33]? The separate torque-path balances do not yet determine that physical result. Its signed magnitude is unknown. In particular, the transverse joint forces, bending couples, and the actuator that holds a ground-referenced phase require their own mechanical constraints and ports.

An exact axial extension makes the change caused by attachment explicit. Take q identical, equally phased lead-outs, one on each planet, with stators fixed in ground. Each input rotor is an additional body, with inertia I_R and mass m_R , rather than a second counting of the planet gear. Enclose the train and all lead-outs; keep the three shaft drives, output receivers, and thermal reservoirs outside. With the same kinematic constraints, the loaded free bodies give

$$F_s = \frac{\tau_s - I_s \dot{\omega}_s}{qr_s}, \quad F_r = F_s + \frac{I_p \dot{\omega}_p + \tau_d}{r_p}, \quad (54)$$

$$R_t = m_p a \dot{\omega}_c - F_s - F_r, \quad \tau_r = I_r \dot{\omega}_r + qr_r F_r, \quad (55)$$

$$\tau_c = I_c \dot{\omega}_c + qaR_t + q(\tau_u + m_R a^2 \dot{\omega}_c). \quad (56)$$

Here $\tau_d = I_R \dot{\omega}_p + \mathcal{T} \chi$ and $\tau_u = \mathcal{T}(1 - \chi)$ retain their earlier meanings. The planet now supplies τ_d ; the carrier supplies both the linkage support torque and the extra orbital acceleration. A carrier-mounted stator would additionally return its drag reaction to the carrier and requires that changed free body.

On the same declared interval $[t_0, t_1]$, the external works in ground are the three separate $\int \tau_j \omega_j dt$, the receiver work $q \int \tau_L \omega_o dt$, and heat export $-q \int d_0 |\omega_o| dt$ for this ground stator. On separate subboundaries the common axle works are $W_{p \rightarrow L} = \int \tau_d \omega_p dt$ and $W_{L \rightarrow p} = -\int \tau_d \omega_p dt$. They cancel only after integration on that shared trajectory. The support and orbital sides cancel in the same way. Multiplying (56) and the output equation by their own velocities gives

$$\frac{d}{dt}(K^0 + qK_L^0) = \tau_s \omega_s + \tau_r \omega_r + \tau_c \omega_c + q\tau_L \omega_o - qd_0 |\omega_o|. \quad (57)$$

Both endpoint stores are evaluated from all the spin and orbital rates. For any compatible smooth path the separate exact integrals give zero residual in this axial model. At a finite force change, retain both power sides; an impact requires an additional stated event law. Unspecified physical transverse or phase-drive works are not supplied by (57).

For a complete finite axial control, use the reference train $(Z_s, Z_p, Z_r) = (24, 18, 60)$, $q = 3$, module $1/500$ m, and hence $(r_s, r_p, r_r, a) = (3/125, 9/500, 3/50, 21/500)$ m. Prescribe $(\omega_s, \omega_c, \omega_r, \omega_p) = (7\pi, 2\pi, 0, -14\pi/3)$ rad/s throughout $[t_0, t_1] = [2/5, 19/10]$ s. Attach three ideal Oldham paths with ground stators and $d_0 = 1/50$ N m. Choose the new receiver torque $\tau_L = +47/40$ N m on each negative-speed output, making each receiver passive on this interval. With $\chi = 1$ and $\tau_s = 47/4$ N m, the output and loaded free bodies give independently

$$\tau_d = -\frac{239}{200}, \quad \tau_u = 0, \quad \tau_r = \frac{697}{40}, \quad \tau_c = -\frac{819}{25} \text{ N m}. \quad (58)$$

The boundary is the train and all three lead-outs, with the receiver and heat reservoirs external. Under the immediate-export thermal idealization, each signed effort–flow product integrates separately over $3/2$ s:

Table 9. Separately integrated transfers for the loaded constant-rate control. The receiver and heat entries each occur three times.

Port into the combined axial system	Power product	Work, J
Sun	$\tau_s \omega_s$	$987\pi/8$
Held ring	$\tau_r \omega_r$	0
Carrier	$\tau_c \omega_c$	$-2457\pi/25$
Receiver j , each $j = 1, 2, 3$	$\tau_L \omega_o$	$-329\pi/40$
Drag heat j , each $j = 1, 2, 3$	$-T_j \dot{S}_j = -d_0 \omega_o $	$-7\pi/50$

For finite added-body parameters I_R, I_o, m_R , evaluation from each specified endpoint state gives

$$E(t_0) = E(t_1) = K_*^0 + 3 \left[\frac{I_R + I_o}{2} \left(\frac{14\pi}{3} \right)^2 + \frac{m_R a^2}{2} (2\pi)^2 \right]. \quad (59)$$

Here K_*^0 is the reference train's ground kinetic store at these rates; the added rotor masses and inertias are counted only in the bracket. The external works independently sum to $987\pi/8 - 2457\pi/25 - 987\pi/40 - 21\pi/50 = 0$ J, so the residual is zero. The common axle transfers $\int_{t_0}^{t_1} \tau_d \omega_p dt = 1673\pi/200$ J into each

lead-out and its negative into its planet. Each support and orbital-pin integral is zero at these constant rates. These internal sides cancel after integration. There are no events within this operation interval; preparation and reset are additional intervals. Numerical integration error is zero, while the physical thermal and unresolved reaction contributions remain undetermined.

The unloaded reference carrier work is $-987\pi/8$ J. Thus attachment changes that signed carrier work by $5019\pi/200$ J while the three passive receivers collect $987\pi/40$ J and the drag dissipates $21\pi/50$ J. This is a substantial torque-and-angle target at the stated speeds. For example, conditional measured envelopes $|\widehat{\tau}_c| \leq 45$ N m, $|\widehat{\omega}_c| \leq 7$ rad/s and pointwise errors $u_\tau = 1/10$ N m, $u_\omega = 1/100$ rad/s bound the two carrier-work product errors together by $3453/1000$ J. A total comparison bound of 5 J leaves $1547/1000$ J for timing and other observation errors, well below the predicted difference. Actual calibration determines whether this target is met; it does not close the full physical balance.

For such a reaction at its stated application point, use

$$W_{\text{reaction}} = \int_{t_0}^{t_1} (\mathbf{F} \cdot \mathbf{v} + \mathbf{M} \cdot \boldsymbol{\omega}) dt. \quad (60)$$

The point velocity and body's angular velocity are conjugate to the force and couple. A fixed support has zero flow; a moving support or a bending degree of freedom needs its own determination. The current massless-joint model supplies too few constraints for all transverse reactions, leaving their physical works open.

A low-speed experiment with synchronized shaft torque and encoder readings at the input, receiver, and moving support can investigate these finite transfers under the stated uncertainty requirements. Record the actual trajectory and both endpoint speeds. Determining the remaining joint and phase-actuator contributions requires explicit support constraints and their corresponding force or supply observations. The gross axial comparison is accessible even while that fuller reconstruction remains unfinished. A nonzero support-cycle integral and a nonzero whole-boundary residual are reported separately. If a signed remainder survives the completed measurements and their uncertainty, it remains open with its magnitude and conditions. No missing reaction is assigned that remainder. A rotating-frame account then adds its separately integrated effective terms to its own evaluated stores; it does not add them to the ground physical balance.

5 Windings, returns, and the electrical observation

5.1 Four electrical connections and a loaded planet terminal

Assign (S, C, R, P) to the mechanical (s, c, r, o) . P is the accessible planet connection; O is a separate physical external reference conductor. With α in V/(rad/s), define

$$V_j - V_O = \alpha \omega_j, \quad I_j = \tau_j / \alpha, \quad (V_j - V_O)I_j = \tau_j \omega_j. \quad (61)$$

Currents enter the winding assembly at their named terminals; every external branch uses its actual return to O . Translating both potentials preserves a port voltage; physically moving a return changes the branch.

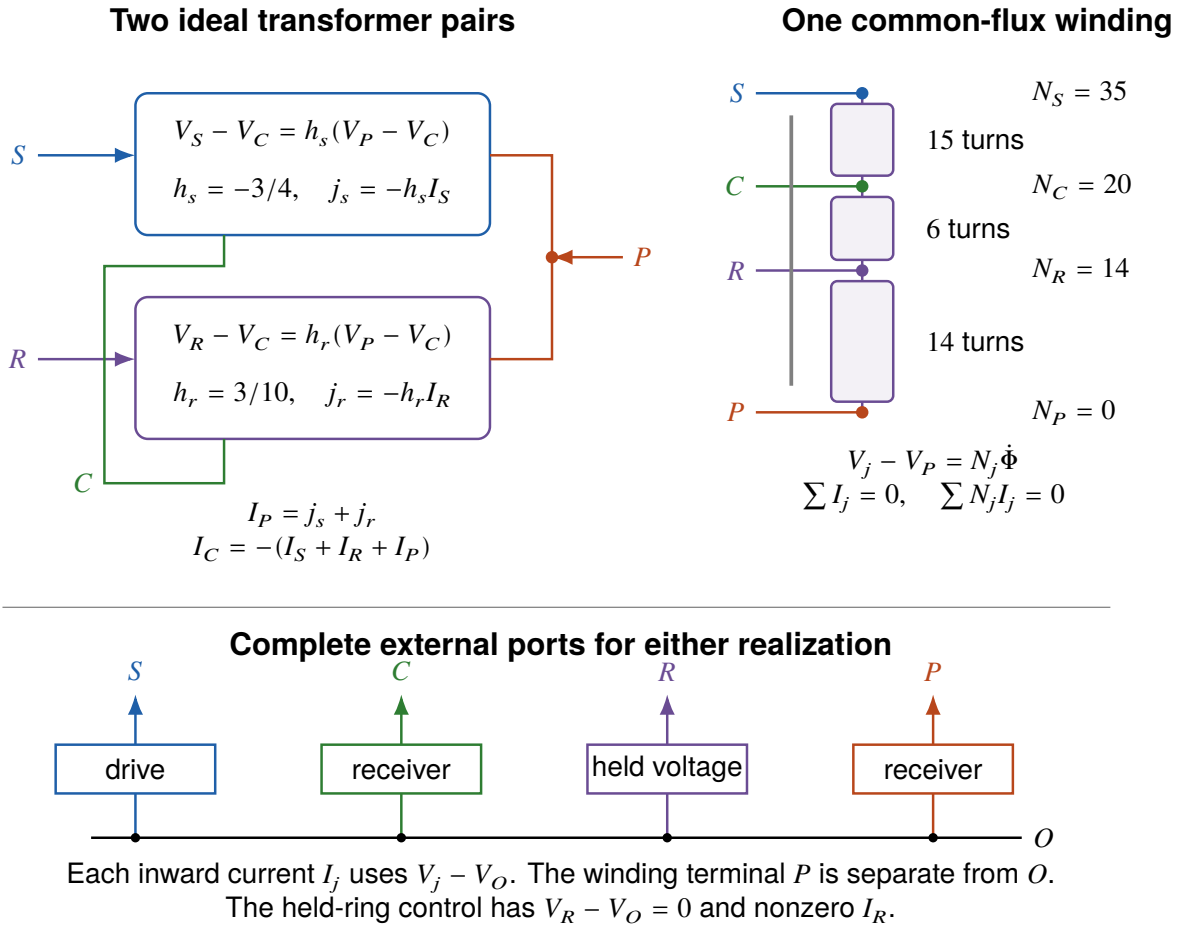


Figure 4. Two ideal realizations of the same four-terminal relation. Left: signed sun and ring transformer ratios with their secondary currents entering the planet node. Right: four taps on one common-flux winding; numbers are cumulative turns from P. Each external port is a terminal-to-O connection. P is loaded, and O is an external physical return separate from the winding.

Set $h_s = -Z_p/Z_s$ and $h_r = Z_p/Z_r$. The two ideal transformers impose

$$V_S - V_C = h_s(V_P - V_C), \quad V_R - V_C = h_r(V_P - V_C), \quad (62)$$

$$j_s = -h_s I_S, \quad j_r = -h_r I_R, \quad (63)$$

$$I_P = j_s + j_r = \frac{Z_p}{Z_s} I_S - \frac{Z_p}{Z_r} I_R, \quad I_C = -(I_S + I_R + I_P). \quad (64)$$

The signed current laws come from ideal ampere-turn cancellation; KCL retains the external P branch. Secondary cancellation $j_s + j_r = 0$ would apply only to an unloaded planet node. Multiplying each winding voltage by its own current then verifies each pair's zero total power. Solving these equations independently gives the same voltage and current families as (22) and (27).

A single common flux supplies a second construction. For general positive teeth with $Z_r = Z_s + 2Z_p$, choose cumulative tap positions

$$(N_S, N_C, N_R, N_P) = (Z_r(Z_s + Z_p), Z_s Z_r, Z_s(Z_r - Z_p), 0)/g, \quad (65)$$

where g is their greatest common divisor. They are ordered $P < R < C < S$: the three positive section counts before reduction are $Z_s(Z_s + Z_p)$, $Z_s Z_p$, and $Z_r Z_p$. Multiplying all counts by a common positive integer preserves ideal ratios. For (24, 18, 60) the taps are $(S, C, R, P) = (35, 20, 14, 0)$. With $e = \dot{\Phi}$, the three voltages to P are $(35, 20, 14)e$. Section currents directed from S towards P are I_S , $I_S + I_C$ and $I_S + I_C + I_R = -I_P$. The independent winding and node equations are therefore

$$15I_S + 6(I_S + I_C) + 14(I_S + I_C + I_R) = 0, \quad I_S + I_C + I_R + I_P = 0. \quad (66)$$

Subtracting tap voltages gives both ratios in (64); substituting KCL into the general ampere-turn sum gives its I_P law. Conversely those equations give both tap constraints. Their coefficient matrices have rank two, including zero-voltage states by the multiplied, division-free equations. This proves ideal terminal equivalence.

At $\alpha = 1$ the recurring one-second control is

	S	C	R	P
$V_j - V_O$ (V)	7/2	1	0	-7/3
I_j (A)	2	-14/3	5/3	1
W_j (J)	7	-14/3	0	-7/3

(67)

Here $j_s = 3/2$ A, $j_r = -1/2$ A, so $P - O$ is a receiver. The zero-voltage ring carries a nonzero reaction current. Unit capacitors to O give 673/72 J at both endpoints, matching the effective unit inertias. Every work follows its own integral; ideal winding constraints add no magnetic state. Removing the planet load gives (7, -7, 0, 0) J; setting $I_P = 3$ A gives (7, 0, 0, -7) J. All operating controls in Section 3.2 map in the same way.

The finite turn count, leakage, copper resistance and core law are additional parameters. Neither ideal realization establishes their equality, or sustained DC operation within a finite core's flux range. A phase-dependent joint ratio χ requires a specified variable electrical coupling and its control work. A voltage coordinate shift does not implement that coupling.

5.2 An initialized finite receiver at P

A specified finite model retains three nodes (S, C, P) relative to a ring clamped at $O = 0$, three unit capacitors, and four winding currents. In order sun primary/secondary, ring primary/secondary, use

$$B_f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad (68)$$

$$L_f = \text{diag} \left[\begin{pmatrix} 9/16 & -3/8 \\ -3/8 & 1 \end{pmatrix}, \begin{pmatrix} 9/100 & 3/20 \\ 3/20 & 1 \end{pmatrix} \right] H, \quad (69)$$

$$R_f = \text{diag}(9/160, 1/10, 9/1000, 1/10) \Omega. \quad (70)$$

Both inductance blocks are positive definite, with coupling magnitude $1/2$. A $1\ \Omega$ source resistance connects $U(t)$ to S ; receivers $G_C = 1/2\ \text{S}$ and $G_P(t) > 0$ connect C, P to O . KCL and the winding laws give

$$C_f \dot{v} = b - Gv - B_f i, \quad L_f \dot{i} = B_f^T v - R_f i, \quad b = (U, 0, 0)^T, \quad G = \text{diag}(1, 1/2, G_P). \quad (71)$$

Here and below matrix coefficients carry the indicated SI units. Start with $v = i = 0$. On successive intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, and $[3, 5]$ s, take $(U, G_P) = (1, 1/4), (1, 1/2), (1, 1), (0, 1)$ in volts and siemens. Finite resistance preserves every current path; all constitutive states are continuous at the load and source steps. Power can jump, while store jumps and model impulses are zero.

For the reactive boundary, separately integrate source $U(U - V_S)$, source-resistor export $-(U - V_S)^2$, each copper export $-R_{f,j} i_j^2$, carrier receipt $-V_C^2/2$, and planet receipt $-G_P V_P^2$. Their quadratic-form integrals are the exact matrix functions developed below. Evaluate $E = (v^T C_f v + i^T L_f i)/2$ directly at every endpoint. The winding/node terms cancel only after both sides are calculated, giving $r_E = 0$. A thermally insulated boundary instead adds $\dot{H} = (U - V_S)^2 + i^T R_f i$ and includes H at the endpoints. It omits those two heat-export terms. The retained source resistance during relaxation permits asymptotic decay; finite relaxation is not an exact reset.

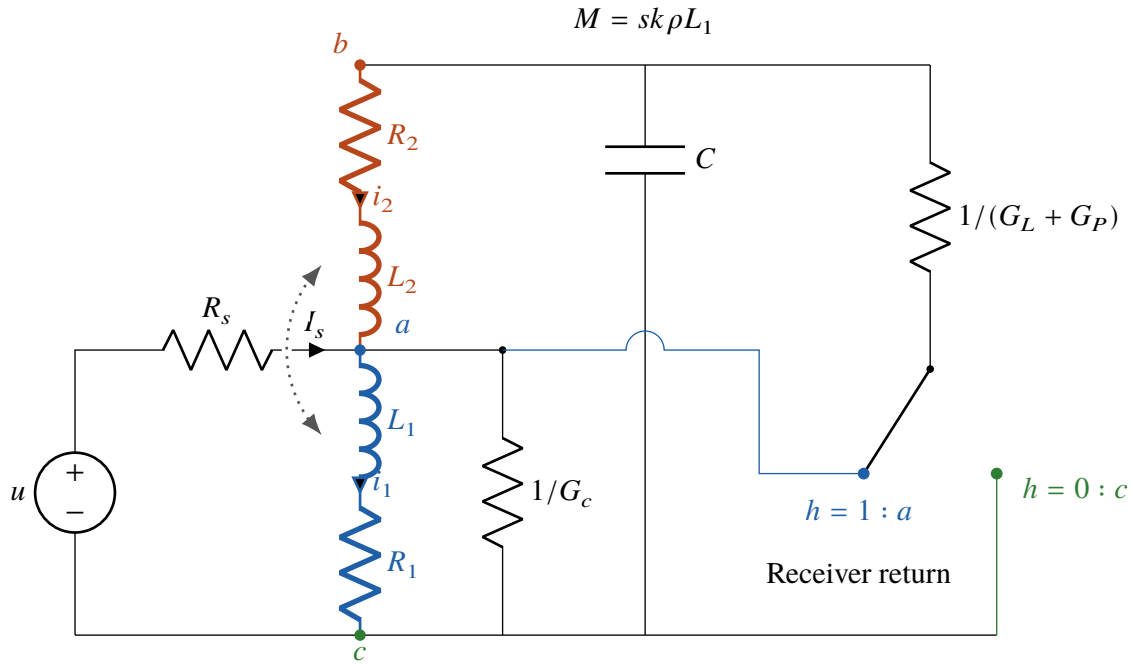
A compliant mechanical counterpart has $J = \alpha^2 C_f$, $z = L_f i / \alpha$ and $K = \alpha^2 L_f^{-1}$. At $\alpha = 1$, its independent laws are $J \dot{\omega} = b - G\omega - B_f K z$ and $\dot{z} = B_f^T \omega - R_f K z$. For each signed ratio $h = -3/4, 3/10$, the positive elastic energy $(z_1/h + z_2)^2/6 + (z_1/h - z_2)^2/2$ has the corresponding inverse-inductance Hessian. A speed source connected through a unit viscous element supplies $\Omega_d(\Omega_d - \omega_s)$ and exports $-(\Omega_d - \omega_s)^2$; its source rotor is outside the boundary. The carrier and planet dashpots are separate receivers. This construction maps actual elastic and inertial states as well as port powers. It adds compliant coordinates to the rigid four-shaft relation. The companion develops finite winding, material and support observations needed to identify a physical realization [Nilre and Herlin \(2026\)](#).

5.3 The tapped and isolated connections

Winding 1 is oriented from terminal a to common return c ; winding 2 is oriented from b to a . Current enters each winding at its first terminal. Define

$$v_1 = V_a - V_c, \quad v_2 = V_b - V_a, \quad v_o = v_1 + v_2, \quad (I_a, I_b, I_c) = (i_1 - i_2, i_2, -i_1). \quad (72)$$

The winding-assembly currents obey $I_a + I_b + I_c = 0$. A source supplying an extra shunt resistor has another current I_s , not simply I_a . Figure 5 specifies the conductive connections independently of winding polarity and magnetic coupling.



Capacitor return stays at c ; bridge crossing is unconnected.

Figure 5. Schematic tapped circuit with winding resistances R_1, R_2 . The source reaches a through R_s ; the coupled winding sections are a – c and b – a . The capacitor remains across b – c . The receiver selector is shown at a and can move to c . Resistors marked $1/G_c$ and $1/(G_L + G_P)$ represent the specified conductances; zero conductance means an open branch. Dotted coupling is magnetic, and the wire bridge has no junction.

An isolated control keeps winding 1 across its source and places winding 2 across its own capacitor and receiver. Its two returns are physically distinct; assigning coordinate potential zero to each does not connect them. When terminal contributions are combined, both returns must be retained.

The standard distinction between a finite coupled-inductance model and an ideal transformer is explicit in [Haus and Melcher \(1989\)](#), Section 9.7. Here we use it as a constitutive distinction and derive the required power relations directly. Put $N_1, N_2 > 0$, polarity $s = \pm 1$, and $n = sN_2/N_1$. A shared-flux idealization has $\lambda_1 = N_1 \Phi$, $\lambda_2 = sN_2 \Phi$. Its voltage and ampere-turn constraints are

$$v_2 = nv_1, \quad i_1 = -ni_2, \quad v_o = (1+n)v_1, \quad I_a = -(1+n)i_2. \quad (73)$$

The first follows by differentiation of the linkage relations, and the second from $N_1 i_1 + sN_2 i_2 = 0$. They imply zero total ideal winding power without using power balance to define the currents. Three terminals give two independent voltages, not three independent two-terminal energy ports.

The weighted linkage constraint is $\lambda_2 - n\lambda_1 = 0$, whereas

$$|\lambda_1| + |\lambda_2| = (1 + |n|)|\lambda_1|, \quad \lambda_1 + \lambda_2 = (1 + n)\lambda_1. \quad (74)$$

The absolute sum changes with turns ratio, and the unweighted signed sum usually changes with excitation. Finite leakage further prevents reducing both linkages to one core flux. These observations do not establish an unweighted conserved-state functional or an absolute-sum growth envelope.

5.4 Which terminal represents which body

For the restricted ring-mesh correspondence choose planet $\leftrightarrow b$, carrier $\leftrightarrow a$, and ring $\leftrightarrow c$. Then

$$\omega_p - \omega_c = -\frac{Z_r}{Z_p}(\omega_c - \omega_r) \quad \leftrightarrow \quad v_2 = nv_1, \quad n = -Z_r/Z_p < -2. \quad (75)$$

The last inequality uses positive Z_s, Z_p and (9). Positive ratios and the opposed null $n = -1$ are useful electrical controls outside this simple planetary assignment.

Define a signed voltage integral, rather than a copper-turn count,

$$\Psi_{xy}(t_0, t_1) = \int_{t_0}^{t_1} (V_x - V_y)dt, \quad \Psi_{bc} = \Psi_{ba} + \Psi_{ac}. \quad (76)$$

It has volt-second units, not joules. For $(Z_s, Z_p, Z_r) = (24, 12, 48)$ with ring held and carrier at 1 rev/s over $[0, 1]$ s, $N_p^c = -4$ and $N_p^r = -3$. Under the kinematic scale 1 V/(rev/s), $v_1 = 1$ V, $n = -4$, $\Psi_{ba} = -4$ V s, and $\Psi_{bc} = -3$ V s. Equation (75) proves the corresponding identity for every rate choice in the earlier family, not only this example.

The choice of velocity as voltage and force as current is the mobility analogy developed by [Firestone \(1933\)](#). Its rotational version needs a dimensional scale α using radians:

$$V = \alpha \omega, \quad I = \tau / \alpha, \quad VI = \tau \omega, \quad C = J / \alpha^2. \quad (77)$$

Thus angular speed corresponds here to voltage, torque to current, and inertia to capacitance. The finite magnetic store below is not automatically an inertia store. A voltage integral also needs its initial linkage and copper correction:

$$\Delta \lambda_j = \int_{t_0}^{t_1} (v_j - R_j i_j) dt. \quad (78)$$

For the series readout both linkage changes and both copper drops contribute. No single node-potential integral is identified with core flux. An independent sun terminal is supplied by Section 5.1; the restricted three-terminal map here does not supply it. Accelerating reference supplies and independently phased linkages require the additional constructions below.

5.5 A common zero and a missing return

For a fixed physical circuit, changing every potential to $V_j + g(t)$ leaves all complete port voltages unchanged. Individual terminal contributions $p_j = V_j I_j$ instead obey

$$p_j^g - p_j = g I_j, \quad \sum_j (p_j^g - p_j) = g \sum_j I_j = 0, \quad W_j^g - W_j = \int_{t_0}^{t_1} g I_j dt. \quad (79)$$

The identity includes the return, even when its original potential is zero. It holds for time-dependent g as well as a constant. Load power is $(V_b - V_c)i_L$, not one terminal contribution $V_b i_L$.

Take the ideal winding assembly as boundary, with $n = 1$, $G_L = 1$ S, and $v_1 = (t - 2)$ V for time t in seconds on $[0, 7/2]$ s. There are no topology changes or impulses; the voltage passes continuously through zero at 2 s. Writing $v = t - 2$ numerically in SI units, the currents are $(I_a, I_b, I_c) = (4v, -2v, -2v)$ and baseline potentials $(v, 2v, 0)$. The two primitives are $\int v^2 dt = 91/24$ and $\int v dt = -7/8$. Integrating each signed terminal product gives:

Table 10. Exact missing-return control on the ideal winding boundary. The physical receiver obtains $91/6$ J in either coordinate system.

Signed work or balance	$V_c = 0, \text{ J}$	$g = 3 \text{ V, J}$
W_a	$91/6$	$14/3$
W_b	$-91/6$	$-119/12$
W_c	0	$21/4$
Sum of complete terminal works	0	0
Initial and final ideal stores	$0, 0$	$0, 0$
r_E	0	0

Omitting c after the shift yields $r_E = +21/4 \text{ J}$. It is exactly the missing return work, determined by its own integral rather than assigned from the residual. The same polynomial proof covers $n \in \{-4, -2, -101/100, -1, -99/100, -1/2, -1/4, 1/4, 1/2, 1, 2, 4\}$ and $g = g_0 + g_1 t$, including $g_0 \in \{-3, 0, 3\} \text{ V}$, $g_1 \in \{-1, 0, 1\} \text{ V/s}$. It also covers nonpolynomial common offsets such as $g = 3 + (7/10) \cos(5t) \text{ V}$ by (79). The ideal endpoint store is an empty sum, not a claim that a real core contains no energy.

Physical grounding remains different: a real bond, probe common-mode branch, or chassis capacitance adds a path and potentially a store [OP-TRF-01]. A low-voltage isolated winding, known shunt conductance G_b , and ordinary differential voltage and current readings can distinguish an added dissipative return from a mere coordinate shift. If the attachment is held at constant voltage difference V_b for duration T , its separate heat delivery is $G_b V_b^2 T$ exactly; without that held-voltage assumption the measured instantaneous product must be integrated on its actual path. Floating and bonded arrangements can therefore be distinguished at ordinary voltage and current levels. Stray capacitance adds $C_b V_b^2 / 2$ endpoint storage; its unknown magnitude is not determined by the gauge identity.

6 Loaded readouts and finite stores

6.1 Two exact receiving experiments

Moving a negligible-loading probe return from a to c selects a different voltage on one trajectory. Moving a finite load changes the circuit; the two trajectories must be established independently. Equation (76) holds within each trajectory, not across two different loaded experiments.

Use the ideal algebraic boundary with $n = -4$, a source fixing $v_1 = 1 \text{ V}$, and a 1 S load on $[0, 1] \text{ s}$. For return a , the load current is $j_L = -4 \text{ A}$ and source current is 16 A ; for return c they are -3 A and 9 A . The load-side currents into the transformer are $-j_L$ at b and $+j_L$ at the return. Direct integration gives:

Table 11. Separate exact experiments, with no internal event or impulse. The two load contributions decompose the load work and are not added again.

Quantity	Return <i>a</i>	Return <i>c</i>
Loaded voltage integral, V s	−4	−3
Source work into boundary, J	16	9
Complete load work into boundary, J	−16	−9
Load contribution at <i>b</i> , J	−12	−9
Load contribution at its return, J	−4	0
Initial/final ideal store, J	0/0	0/0
r_E , J	0	0

The extra 7 J of load delivery accompanies an independently integrated extra 7 J of source input. If instead $v_1 = (t - 2)$ V on $[0, 4]$ s, all three signed voltage integrals are zero, while

$$W_{L,a} = -16 \int_0^4 (t - 2)^2 dt = -\frac{256}{3} \text{ J}, \quad W_{L,c} = -9 \int_0^4 (t - 2)^2 dt = -48 \text{ J}. \quad (80)$$

The source products integrate separately to the opposite values. A continuous sign reversal changes neither the ideal empty endpoints nor the balance. Zero net readout is not zero transferred work, as Figure 6 illustrates.

The complete curves on $0 \leq t \leq 4$ s, in the stated SI units, follow from the polynomial primitives:

$$\Psi_{ba}(0, t) = -4(t^2/2 - 2t), \quad \Psi_{bc}(0, t) = -3(t^2/2 - 2t), \quad (81)$$

$$-W_{L,a}(0, t) = 16(t^3/3 - 2t^2 + 4t), \quad -W_{L,c}(0, t) = 9(t^3/3 - 2t^2 + 4t). \quad (82)$$

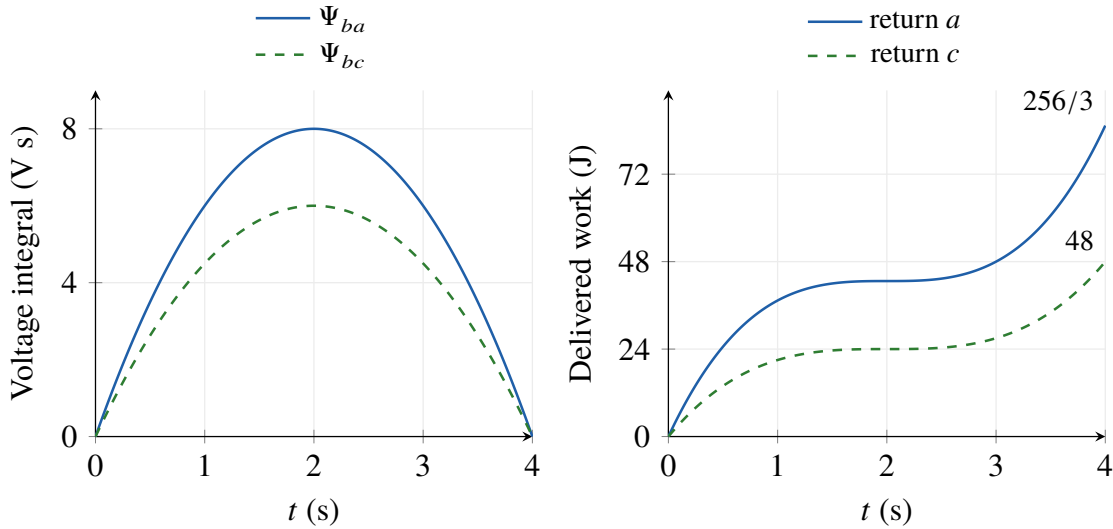


Figure 6. Exact reversing-drive control: the voltage integrals return to zero, while delivered load works increase as the integral of squared voltage. Time is in seconds; the two load connections are distinct ideal experiments driven by the same prescribed $v_1 = t - 2$.

For a general ideal ratio and a b – c resistive load, $i_2 = -G_L(1+n)v_1$, $i_1 = nG_L(1+n)v_1$ and $P_L = -G_L(1+n)^2v_1^2$. At $n = -1$ the output is identically zero; without imposed circulating current all terminal currents vanish while both winding voltages can be nonzero. For $n = -1 + \epsilon$, load work is $-G_L\epsilon^2 \int v_1^2 dt$ and approaches zero continuously, while winding-power direction reverses across the null. The values $-101/100, -1, -99/100$ are explicit two-sided controls. The b – a load instead sees nv_1 and need not vanish there. A finite transformer does not automatically inherit this ideal null.

6.2 The unresolved complete-cycle comparison

What remains of the receiving-work difference when both circuits are prepared, operated, physically switched, relaxed, and reset with every controller supply included is an open energy question [OP-TRF-05]. The exact 7 J difference in the one-second ideal control already accompanies an independently integrated 7 J source-input difference. The complete finite-cycle difference, and any unexplained energy remainder within either physical apparatus, have unknown magnitudes and signs.

For each return choice $\ell \in \{a, c\}$, take the boundary to contain the windings, capacitor, resistors, and the switching and controller stores. Electrical supplies, receiver, probe, and fixed-temperature heat reservoirs are external. Every supply has its own vi product; receiver and probe powers are negative into this boundary, and each exported heat is $-T\dot{S}$. Choose ordered endpoints $t_0 < \dots < t_5$ so the five disjoint stages are preparation, operation, switching, relaxation, and reset. Define

$$W_{j,\text{cycle}}^{(\ell)} = \sum_{m=0}^4 \int_{t_m}^{t_{m+1}} e_j^{(\ell)} f_j^{(\ell)} dt + \sum_e W_{j,e}^{(\ell),\text{imp}}, \quad (83)$$

$$r_{E,\text{cycle}}^{(\ell)} = E^{(\ell)}(t_5) - E^{(\ell)}(t_0) - \sum_j W_{j,\text{cycle}}^{(\ell)}. \quad (84)$$

The continuous integrals and ideal impulse terms represent disjoint transfers. A resolved finite switching path needs no duplicate impulse. Both endpoint energies include magnetic, capacitive, controller, and material thermal states evaluated from independently established constitutive laws and observations. The resistors and windings lie inside this physical boundary: their Joule conversion is internal, and only heat actually crossing the boundary enters as $-T\dot{S}$. If a supply rail is brought inside an enlarged boundary, add its endpoint store and cancel its two already integrated port sides; do not retain its output as an additional external input.

The finite models below specify some of these paths exactly. A physical experiment can record source and controller voltage–current products, both winding currents, and capacitor voltages throughout all five stages. The low-voltage drive and a deliberately slow preparation and reset permit ordinary differential voltage and current observations. Independently calibrated capacitances and a constant inductance matrix determine the corresponding linear-model stores. The physical comparison additionally needs temperatures, a caloric law for the enclosed dissipative bodies, actual external heat transfer, and the magnetic-state qualification developed next. The resistor-reset control below gives a resolvable half-voltage endpoint. It is a part of this comparison, not a preparation of the winding states.

The independently integrated works may account for the store change within the established uncertainty, or a signed remainder may survive. Retain either outcome with its actual interval and states. An unspecified winding preparation, switch supply, thermal store, or reset path remains an undetermined contribution; none is assigned the remainder by definition. The delivered-work difference $-W_{L,\text{cycle}}^{(a)} + W_{L,\text{cycle}}^{(c)}$ is reported separately from the two residuals. Return of the visible capacitor voltage alone does not establish return of every state.

6.3 Thermal endpoints and magnetic material states

How much of a finite departure from the linear winding prediction is resolved by observed thermal and magnetic material states [OP-TRF-03]? A useful first comparison repeats a specified low-voltage preparation at two independently recorded initial temperatures, recording winding voltages and currents, capacitor voltages, and temperature changes on each path. The finite winding preparation below supplies a 93/1600 J linear magnetic target. Compare its separately measured input work, actual heat export, and independent endpoint observations with that target. A departure may remain after the thermal contribution is determined; it retains its signed value and uncertainty. Neither the current readings nor that remainder determine an unobserved magnetic internal state.

For the first whole-apparatus comparison, enclose the windings, resistors, capacitor, and controller; keep supplies, receiver, and heat reservoirs external. Define a nonoverlapping physical endpoint model

$$E_{\text{phys}} = E_C + E_{\text{mag}}(i, \zeta, \vartheta) + E_{\text{controller}} + U_{\text{thermal}} + E_{\text{other}}. \quad (85)$$

The calibrated capacitor contribution is E_C ; ϑ denotes the observed material temperatures and ζ the declared magnetic internal state. The constitutive partition assigns the caloric material energy to U_{thermal} and any remaining magnetic state dependence to E_{mag} . It must specify their coupling without counting the same energy twice. Controller electrical stores are separate from the controller body's contribution to U_{thermal} . Mechanical or other stores, if present, enter E_{other} with their own states. A nonuniform temperature field requires its own caloric evaluation or a justified enclosure.

The reciprocal expression $i^T \mathbf{L} i / 2$ determines the magnetic store of the declared constant linear model. Magnetic linearity is also the stated condition for the inductance-matrix terminal relation in [Haus and Melcher \(1989\)](#). Calibrating that matrix alone does not determine $E_{\text{mag}}(i, \zeta, \vartheta)$ for a core with remanence or hysteresis. Record which material states are independently observed and which remain undetermined. With the same measured ports define separately

$$\hat{r}_{E,\text{lin}} = \Delta \left(\hat{E}_C + \frac{1}{2} \hat{i}^T \mathbf{L} \hat{i} + \hat{E}_{\text{controller}} + \hat{U}_{\text{thermal}} + \hat{E}_{\text{other}} \right) - \sum_j \hat{W}_j - \sum_e \hat{W}_e^{\text{imp}}. \quad (86)$$

This is an observed discrepancy against a specified model, with calibration uncertainty included. A complete physical residual instead needs both values of (85). An unknown magnetic contribution stays unknown; it is not set equal to (86).

An exact caloric control shows why thermal endpoints matter even without a magnetic core. Enclose a resistor with $R = 1/10 \, \Omega$ and prescribe $i = 1 \, \text{A}$ over $[0, 1] \, \text{s}$, with an adiabatic boundary on that interval. Declare a constant $C_{\text{th}} > 0$ and initial $\vartheta(0) = \vartheta_{\text{ref}}$. The independent caloric law and its endpoint evaluation are

$$C_{\text{th}} \dot{\vartheta} = Ri^2, \quad U_{\text{th}} = C_{\text{th}}(\vartheta - \vartheta_{\text{ref}}), \quad (87)$$

$$\vartheta(1 \, \text{s}) = \vartheta_{\text{ref}} + \frac{1 \, \text{J}}{10C_{\text{th}}}, \quad (U_{\text{th}}(0), U_{\text{th}}(1 \, \text{s})) = (0, 1/10) \, \text{J}. \quad (88)$$

The external ports integrate independently as

$$W_{\text{el}} = \int_0^{1 \, \text{s}} (Ri)i \, dt = \frac{1}{10} \, \text{J}, \quad W_h = - \int_0^{1 \, \text{s}} T \dot{S}_{\text{out}} \, dt = 0. \quad (89)$$

There are no electrical energy coordinates in this ideal resistor and no impulsive transfers. Thus $r_E = 0$

with the independently evaluated thermal store and exact integrals. Omitting U_{th} gives $r_E = -1/10$ J. Entering another $-1/10$ J as heat export would artificially close that incomplete account while changing the stipulated adiabatic boundary. An electrical subboundary may export Ri^2 into a separate thermal subsystem; its receiving side then integrates to $+1/10$ J. Those two sides cancel only when the subsystem boundaries are combined.

For illustrative $C_{\text{th}} = 1$ J/K, the predicted temperature rise is $1/10$ K. Temperature observations with each endpoint error at most $1/100$ K, a calibrated heat-capacity error at most $1/100$ J/K, and observed rise at most $11/100$ K would bound the inferred thermal change error by

$$u_{\Delta U} \leq u_{C_{\text{th}}} |\Delta \hat{\vartheta}| + (\hat{C}_{\text{th}} + u_{C_{\text{th}}})(u_{\vartheta,0} + u_{\vartheta,1}) \leq \frac{213}{10000} \text{ J}. \quad (90)$$

To resolve the omitted $1/10$ J at total uncertainty $1/20$ J, the measured electrical work, actual external heat, timing, and remaining caloric errors must together contribute at most $287/10000$ J. In a physical apparatus the heat transfer is observed or bounded, rather than declared zero by insulation alone. These finite requirements give voltage, current, and temperature observations a specific task. The core comparison needs its additional magnetic-state information before it can claim a complete physical remainder. The numerical integration residual of the caloric control is zero; no value is assigned to an unmeasured material discrepancy.

6.4 The regular coupled-winding model

Let $\rho = N_2/N_1 > 0$, $s = \pm 1$, and $0 \leq k < 1$. Define

$$L_2 = \rho^2 L_1, \quad M = sk\rho L_1, \quad \mathbf{L} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix}, \quad E_m = \frac{1}{2} i^T \mathbf{L} i, \quad E_C = \frac{1}{2} C v_o^2. \quad (91)$$

Here $L_1, C > 0$. The determinant $L_1 L_2 (1 - k^2) > 0$ makes the magnetic store positive definite, including the mutual term once. The exact identity

$$E_m = \frac{1}{2} k L_1 (i_1 + s \rho i_2)^2 + \frac{1}{2} (1 - k) L_1 i_1^2 + \frac{1}{2} (1 - k) L_2 i_2^2 \quad (92)$$

separates the shared magnetizing mode and leakage terms. Taking k to one removes leakage but retains finite magnetizing inductance. At exactly $k = 1$ the differential equations need a constrained formulation; replacing them with (73) is a different model, not that limit alone.

Let $\eta = 1$ for the tapped connection and $\eta = 0$ for the isolated one, using each circuit's local return coordinates. The source voltage u acts through $R_s > 0$, and the effective core-loss conductance $G_c \geq 0$ is across winding 1. Put $G = G_L + G_P$ for load and resistive probe. Then

$$v_a = \frac{u - R_s(i_1 - \eta i_2)}{1 + R_s G_c}, \quad I_s = i_1 - \eta i_2 + G_c v_a, \quad (93)$$

$$\mathbf{L} \dot{i} = \begin{pmatrix} v_a \\ v_o - \eta v_a \end{pmatrix} - \text{diag}(R_1, R_2) i, \quad C \dot{v}_o = -i_2 - G v_o. \quad (94)$$

These circuit equations define the state $x = (i_1, i_2, v_o)^T$ before any work is evaluated. The core resistor is a declared loss path, not a hysteresis law. Resistor heat is exported immediately; no thermal storage is included.

The whole boundary contains the windings, capacitor, source resistor, and core-loss resistor. Its seven powers and their effort–flow meanings are:

Table 12. Complete finite-circuit boundary. The source, receiver, probe, and thermal reservoirs are external. The thermal equalities define immediate export to fixed-temperature reservoirs.

Port	Effort times flow, positive into the boundary
Ideal source	uI_s
Source-resistor heat export	$-(R_s I_s)I_s = -T_s \dot{S}_s$
Winding 1 copper heat export	$-(R_1 i_1)i_1 = -T_1 \dot{S}_1$
Winding 2 copper heat export	$-(R_2 i_2)i_2 = -T_2 \dot{S}_2$
Effective core-resistor heat export	$-v_a(G_c v_a) = -T_c \dot{S}_c$
Receiver	$-v_o(G_L v_o)$
Probe	$-v_o(G_P v_o)$

Multiply the magnetic equation by i^T , the capacitor equation by v_o , and use $u = v_a + R_s I_s$. This independently derives

$$\frac{d}{dt}(E_m + E_C) = uI_s - R_s I_s^2 - R_1 i_1^2 - R_2 i_2^2 - G_c v_a^2 - G_L v_o^2 - G_P v_o^2. \quad (95)$$

The winding-only subboundary instead uses E_m , all winding terminal contributions, and the two copper losses. The shunt core resistor is outside it. Omitting a shifted return in that subboundary gives exactly its missing $\int gI_c dt$ term, with either sign possible; the complete sum obeys (79) at all times.

At $k = 0$, an isolated zero-state secondary satisfies a homogeneous system, so $i_2 = v_o = 0$ exactly. A tapped circuit retains the conductive path through winding 2. Its initial forcing $v_o - v_a$ is generally nonzero, so it can respond without magnetic coupling. This is a topological difference, not a change of voltage label.

6.5 Event intervals and exact finite-state integrals

Use the illustrative baseline $L_1 = L_2 = 1/25$ H, $k = 19/20$, $s = +1$, $C = 1/500$ F, $R_s = 1/5$ Ω , $R_1 = R_2 = 1/10$ Ω , $G_c = 1/100$ S, $G_L = 1/10$ S, and $G_P = 0$. All initial states are zero unless explicitly changed. With t in seconds, the drive is $u = 2 \cos(14\pi t + 3/10)$ V, multiplied by $[1 - \cos(10\pi t)]/2$ during preparation.

Table 13. Four separate integration intervals. Winding currents and capacitor voltage are continuous; powers may jump at the load and source events.

Interval, s	Physical path	Endpoint and event condition
[0, 1/10]	Smooth preparation	Starts at $x = 0$; envelope reaches one
[1/10, 1/5]	Operation	Same circuit and full drive
[1/5, 3/10]	Changed load	G_L increases to 2/5 S
[3/10, 1/2]	Finite relaxation	$u = 0$; source resistance retains a path

Setting the source voltage to zero does not open its conductor. Removing only the load leaves the capacitor. For these finite-path events, integrating the state equations across a vanishing interval gives $x^+ = x^-$ and hence no store jump or model impulse. Source and load powers use their own event-side values. A real switch-control supply is absent from this ideal selector model; its unknown work is not claimed zero on apparatus.

The energy transferred when an energized readout return is physically moved from a to c , or back, remains an open question [OP-TRF-02]. Keep the capacitor across b – c for this experiment and distinguish it from a second experiment that reconnects a charged capacitor. Enclose the windings, capacitor, switch and controller stores, and any specified clamp. External source, receiver, probe, controller supplies, and heat reservoirs each retain their complete effort–flow products. For a resolved finite edge $[t_e, t_e + \delta]$,

$$r_{E,\text{event}} = E(t_e + \delta) - E(t_e) - \sum_j \int_{t_e}^{t_e + \delta} e_j f_j dt. \quad (96)$$

The physical remainder has unknown sign and magnitude. Winding currents, capacitor voltages, clamp state, controller state, and material thermal and magnetic states in (85) are required at both endpoints; a change of tap must retain the individual winding branches rather than merely change a ratio in an energy formula.

The explicit two-branch path in (110) below selects a finite overlap and an energized preparation, with a separate open-gap control. Its voltage–current envelopes and 1/50 J proposed whole-event uncertainty give ordinary synchronized observations a definite resolution target. Record each supply and receiver product and the event-side states; faster opening introduces an additional bandwidth requirement. The prepared-clamp calculation below supplies an exact single-capacitor comparison with separately integrated source and heat works. It does not assign those same works to an energized coupled-winding switch. A resolved finite path and an ideal impulse are alternative representations of a transfer, never two additions for it. The observation may agree with the independent store change within uncertainty, or retain a signed energy remainder. Keep the latter with its conditions and completed checks; no arc, switch loss, or stray capacitance receives it by definition. Preparation and reset of this event belong to (84).

A closed form retains every finite work without a numerical trajectory. On one constant-topology interval write (94) as $\dot{x} = Ax + bu(t)$. Represent the sinusoidal forcing by real oscillator coordinates z satisfying $\dot{z} = Dz$, $u = cz$. A cosine–sine pair at frequency ω has $D_\omega = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$. The preparation waveform is exactly $\cos(14\pi t + 3/10) - \frac{1}{2}\cos(24\pi t + 3/10) - \frac{1}{2}\cos(4\pi t + 3/10)$ in volts, so three such pairs suffice. The full-drive interval needs one pair; constant drive uses a zero-frequency coordinate. With $y = (x, z)$,

$$\dot{y} = Ay, \quad A = \begin{pmatrix} A & bc \\ 0 & D \end{pmatrix}, \quad y(t_0 + T) = e^{TA}y_0. \quad (97)$$

Every effort and flow is a linear functional of y . If they are $e^T y$ and $f^T y$, define $Q = (ef^T + fe^T)/2$, so its signed power is $y^T Q y$. The minus signs of exports belong in e or f .

Theorem 1 (Separate closed-form port works). *For a constant matrix A , define $B = A \otimes I + I \otimes A$ and the entire matrix function $\varphi_1(Z) = \sum_{m=0}^{\infty} Z^m / (m+1)!$. Column vectorization is denoted by vec . Each signed port integral is exactly*

$$\mathcal{W}(Q; A, y_0, T) = \text{vec}(Q)^T T \varphi_1(TB) \text{vec}(y_0 y_0^T). \quad (98)$$

It is valid even when B is singular.

Proof. $Y = yy^T$ obeys $\dot{Y} = AY + YA^T$, whence $\text{vec } Y(t_0 + t) = e^{tB} \text{vec } Y(t_0)$. Integrating its convergent exponential series gives $T \varphi_1(TB)$. Contracting with Q integrates that port's effort–flow product. No endpoint store or balance residual enters this calculation. $\square$

The infinite series defines a closed matrix function, without truncation or small-parameter expansion. An equivalent exact definition is the upper-right block of $\exp \left[T \begin{pmatrix} B & I \\ 0 & 0 \end{pmatrix} \right]$. Linear voltage integrals are likewise $\ell^T T \varphi_1(TA) y_0$. At a physical event, propagate the physical x continuously, initialize the forcing coordinates at the event's actual phase, and apply (98) on the next interval.

For clarity, let $\mathcal{W}_j(\eta, p)$ denote the sum of the four separate closed forms (98) for port j , with probe conductance p . Let $X_{\eta,p}(t)$ be the physical part of the successive exponentials and $\mathcal{E}(X) = i^T Li/2 + Cv_o^2/2$. The complete three-circuit comparison, with every transfer explicitly defined, is:

Table 14. Exact finite expressions, in joules, for $[0, 1/2]$ s. The matrices are fixed by (94), (97), the seven effort–flow pairs, and the interval table. No work is assigned as the remainder of another quantity.

Work or endpoint	Tapped	Isolated	Tapped, 10 Ω probe
Source	$\mathcal{W}_s(1, 0)$	$\mathcal{W}_s(0, 0)$	$\mathcal{W}_s(1, 1/10)$
Source heat	$\mathcal{W}_{R_s}(1, 0)$	$\mathcal{W}_{R_s}(0, 0)$	$\mathcal{W}_{R_s}(1, 1/10)$
Copper 1	$\mathcal{W}_{R_1}(1, 0)$	$\mathcal{W}_{R_1}(0, 0)$	$\mathcal{W}_{R_1}(1, 1/10)$
Copper 2	$\mathcal{W}_{R_2}(1, 0)$	$\mathcal{W}_{R_2}(0, 0)$	$\mathcal{W}_{R_2}(1, 1/10)$
Core heat	$\mathcal{W}_c(1, 0)$	$\mathcal{W}_c(0, 0)$	$\mathcal{W}_c(1, 1/10)$
Load	$\mathcal{W}_L(1, 0)$	$\mathcal{W}_L(0, 0)$	$\mathcal{W}_L(1, 1/10)$
Probe	0	0	$\mathcal{W}_P(1, 1/10)$
Initial store	0	0	0
Final store	$\mathcal{E}(X_{1,0}(1/2))$	$\mathcal{E}(X_{0,0}(1/2))$	$\mathcal{E}(X_{1,1/10}(1/2))$
Exact r_E	0	0	0

This table is not an efficiency ranking: equal source waveform and winding parameters do not match terminal impedance or output voltage. A resistive probe has its own negative work and changes A , hence all other works and endpoints. It can raise source input or reduce original-load delivery under particular conditions, but neither ordering is asserted as a theorem for all finite transients. Even $G_P = 0$ retains the output capacitor and therefore is not a probe on an otherwise unchanged bare winding.

The finite waveform, store, and signed-work functions are respectively the linear projection of (97), $\mathcal{E}(X(t))$, and (98) with its variable upper time. These exact expressions replace any inference from a plotted numerical trajectory: stores may rise, fall, and remain nonzero after relaxation. In fact a source-

free regular interval has $x(t_1) = e^{A(t_1-t_0)}x(t_0)$; the exponential is invertible. A nonzero state therefore cannot reach the exact zero state at finite time through that passive linear relaxation alone.

The complete-cycle question in (84) includes the still-open winding preparation and controller works. One accessible part has an exact comparison. For an explicit capacitor-only reset, a fixed resistor R across an initially charged C gives $v(t) = V_0 e^{-t/(RC)}$. On $[0, T]$ its own heat-port integral is

$$W_R = - \int_0^T \frac{V_0^2}{R} e^{-2t/(RC)} dt = -\frac{CV_0^2}{2} (1 - e^{-2T/(RC)}), \quad E(T) = \frac{CV_0^2}{2} e^{-2T/(RC)}. \quad (99)$$

A voltmeter and resistor-current measurement at ordinary voltage levels can resolve a nonzero remaining charge and the finite reset work. For example, $T = RC \ln 2$ gives $v(T) = V_0/2$, $E(T) = CV_0^2/8$, and $W_R = -3CV_0^2/8$ by (99). A measured voltage uncertainty interval excluding zero rules out exact reset at that endpoint. An interval containing zero does not prove exact reset; the finite-time nonzero result is a statement of the ideal exponential model. An active reset may instead reach zero in finite time, but its supply and switching ports must be included. The elementary reset does not specify how nonzero winding currents were prepared; their separate source and loss paths remain necessary.

6.6 Physical return loading in the finite circuit

For the tapped circuit keep the capacitor across b - c and set $v_b = V_b - V_c$. Let $h = 1$ select a receiver return at a and $h = 0$ at c , with $v_L = v_b - hv_a$ and $G = G_L + G_P$. Current conservation and the source resistor now give

$$I_s = i_1 - i_2 + G_c v_a - hGv_L, \quad (100)$$

$$v_a = \frac{u - R_s(i_1 - i_2) + R_s hGv_b}{1 + R_s(G_c + hG)}, \quad (101)$$

$$\mathbb{L}i = \begin{pmatrix} v_a \\ v_b - v_a \end{pmatrix} - \text{diag}(R_1, R_2)i, \quad C\dot{v}_b = -i_2 - Gv_L. \quad (102)$$

The expression for v_a uses $h^2 = h$; h denotes a discrete physical connection, not a continuously interpolated fractional return. Omitting $-hGv_L$ changes the source circuit. The seven powers retain their definitions except for load and probe, which are $-G_L v_L^2$ and $-G_P v_L^2$. With $j_L = G_L v_L$, their load-terminal decomposition is

$$P_{L,b} = -v_b j_L, \quad P_{L,\text{ret}} = hv_a j_L, \quad P_{L,b} + P_{L,\text{ret}} = -G_L v_L^2. \quad (103)$$

Integrating each of these before combining verifies the decomposition. The same multiplication proof as for (95) gives zero exact energy residual with stores $E_m + Cv_b^2/2$ for either h .

The nominal paired example has $n = -4$, $k = 19/20$, and $G_L = 1/10$ S. Its operation interval $[1/10, 1/5]$ s has the exact observations, with $T = 1/10$ s,

$$\Psi_L^{(h)} = \ell_h^T T \varphi_1(TA_h)y_h, \quad W_j^{(h)} = \mathcal{W}(Q_{jh}; A_h, y_h, T), \quad (104)$$

$$E_0^{(h)} = \mathcal{E}(x_h), \quad E_1^{(h)} = \mathcal{E}(X_h(1/5)). \quad (105)$$

Here $x_h = X_h(1/10)$ is independently prepared by its own first-interval matrix, y_h augments that state with the forcing phase, and $\ell_h^T y = v_L$. The two initial operation stores need not agree even though both preparations start from zero. Subtracting only already integrated works gives the exact paired balance

$$\Delta E^{(1)} - \Delta E^{(0)} = \sum_j (W_j^{(1)} - W_j^{(0)}).$$

There is no general ordering of the load deliveries. Two exact initial-state arguments already show opposite signs. At $x = (0, 0, 0)$ with $u \neq 0$, return a has $v_L = -u/[1 + R_s(G_c + G)]$, whereas return c has $v_L = 0$. At $x = (0, 0, V_0)$ with $u = 0$, return a has $v_L = V_0(1 + R_s G_c)/[1 + R_s(G_c + G)]$, smaller in magnitude than the return- c value V_0 when $R_s G > 0$. Continuity preserves these strict inequalities on some nonzero following interval when $G_L > 0$. At $G_L = 0$ both receiver works are zero. The prepared-state example specifies observation endpoints, not the missing preparation work.

On every one of these finite trajectories the additive readout relation is exact. The ideal winding ratio generally is not. Define $n = s\rho$ and $D_\Psi = \Psi_{ba} - n\Psi_{ac}$. Integrating both constitutive winding equations independently yields

$$D_\Psi = \Delta(\lambda_2 - n\lambda_1) + \int_{t_0}^{t_1} (R_2 i_2 - nR_1 i_1) dt, \quad \lambda = \mathbb{L}i. \quad (106)$$

The linkage endpoints are obtained from the state, while each copper integral has its own linear primitive. A nonzero D_Ψ is a finite-model voltage observation, not an energy residual. Its sign depends on state and interval.

6.7 An energized return change with two receiver branches

Keep the capacitor at b – c . Connect a separately controlled receiver conductance $G_a(t)$ between b and a , and another $G_o(t)$ between b and c , as in Figure 7. The winding and source connections remain those of the tapped circuit. Core-loss conductance retains the symbol G_c . Direct current conservation now gives

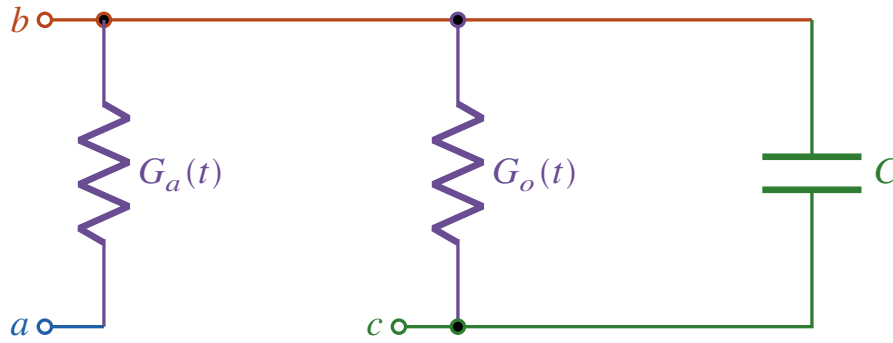
$$I_s = i_1 - i_2 + G_c v_a - G_a(v_b - v_a), \quad (107)$$

$$v_a = \frac{u - R_s(i_1 - i_2) + R_s G_a v_b}{1 + R_s(G_c + G_a)}, \quad (108)$$

$$\mathbb{L}i = \begin{pmatrix} v_a \\ v_b - v_a \end{pmatrix} - \text{diag}(R_1, R_2)i, \quad (109)$$

$$C\dot{v}_b = -i_2 - G_a(v_b - v_a) - G_o v_b. \quad (110)$$

These equations retain both actual returns during overlap. The old binary h is not interpolated. An interval with $G_a = G_o = 0$ is an open receiver gap with the declared source and winding paths still connected.



Separate returns; capacitor remains at $b-c$

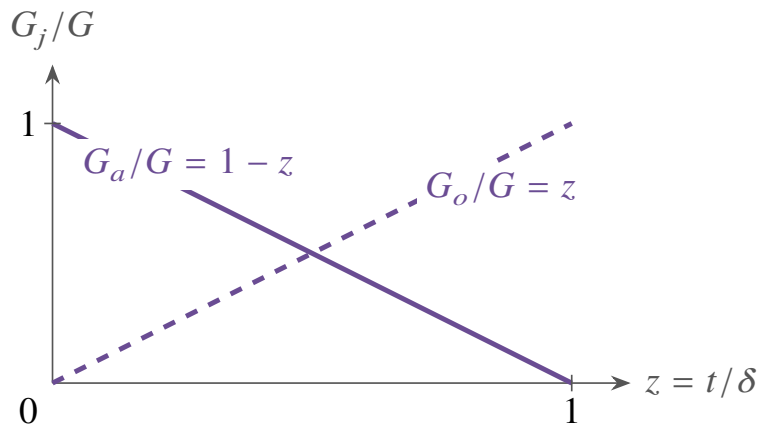


Figure 7. Schematic receiver connections during commutation. Both controlled branches remain distinct; G_a returns to a , while G_o and the fixed capacitor return to c . The source and tapped windings connect to these same terminals as in Figure 5. The lower plot gives the exact bounded overlap history in equation (111).

Choose $t_e = 0$, $\delta = 1/10$ s, $G = 1/10$ S, and the bounded history

$$z = \frac{t}{\delta}, \quad G_a = G(1 - z), \quad G_o = Gz, \quad 0 \leq t \leq \delta. \quad (111)$$

Before the edge take $(G_a, G_o) = (G, 0)$ and afterwards $(0, G)$. Both receivers conduct in the interior. Equal endpoint conductances isolate the return change from the simultaneous conductance increase in the later compound example. A distinct gap control is $G_a = G \max(1 - 3z, 0)$, $G_o = G \max(3z - 2, 0)$: the middle third has both branches open. It needs its own trajectory and work integrals.

For a finite energized initial state set $u = 1$ V during the edge, $R_s = 1/5 \Omega$, $R_1 = R_2 = 1/10 \Omega$, $G_c = 1/100$ S, $C = 1/100$ F, and

$$\mathbf{L} = \begin{pmatrix} 1/25 & -19/125 \\ -19/125 & 16/25 \end{pmatrix} \text{H}, \quad i(0) = \begin{pmatrix} 1/5 \\ 0 \end{pmatrix} \text{A}, \quad v_b(0) = 4 \text{V}, \quad E(0) = \frac{101}{1250} \text{J}. \quad (112)$$

This is the finite $n = -4$, $k = 19/20$ winding pair. Its preparation can be specified independently: on $[-1/10, 0]$ s, isolate the two windings and capacitor from the operating connections, and let $f =$

$(t + 1/10 \text{ s})/(1/10 \text{ s})$, $i = f(1/5, 0)^T \text{ A}$, and $v_b = 4f \text{ V}$. Two winding sources impose $u_1 = 2/25 + f/50 \text{ V}$ and $u_2 = -38/125 \text{ V}$; a capacitor source supplies $(4f \text{ V}, 2/5 \text{ A})$. For the winding/capacitor boundary with immediate resistive heat export, the separately integrated products are

$$W_{s,1}^{\text{prep}} = \int_{-1/10 \text{ s}}^0 u_1 i_1 dt = \frac{7}{7500} \text{ J}, \quad W_{s,2}^{\text{prep}} = \int_{-1/10 \text{ s}}^0 u_2 i_2 dt = 0, \quad (113)$$

$$W_{h,1}^{\text{prep}} = - \int_{-1/10 \text{ s}}^0 R_1 i_1^2 dt = -\frac{1}{7500} \text{ J}, \quad W_{h,2}^{\text{prep}} = - \int_{-1/10 \text{ s}}^0 R_2 i_2^2 dt = 0, \quad (114)$$

$$W_C^{\text{prep}} = \int_{-1/10 \text{ s}}^0 v_b (C \dot{v}_b) dt = \frac{2}{25} \text{ J}. \quad (115)$$

The independent initial store is zero; the final magnetic store is $1/1250 \text{ J}$ and capacitor store is $2/25 \text{ J}$. Their sum agrees with the five integrated works. Reconnection to the declared operating circuit retains both winding currents and capacitor voltage; the algebraic v_a may change finitely. There is no impulse for this ideal event. Physical preparation and selector supplies require their own observed works.

On the edge, enclose the windings, capacitor, source resistor and copper/core loss elements, keeping the source, two receivers, and heat reservoirs outside. For this immediate-export model the seven works are individually

$$W_s = \int_0^\delta u_s ds, \quad W_{h,s} = - \int_0^\delta R_s I_s^2 dt, \quad (116)$$

$$W_{h,1} = - \int_0^\delta R_1 i_1^2 dt, \quad W_{h,2} = - \int_0^\delta R_2 i_2^2 dt, \quad (117)$$

$$W_{h,c} = - \int_0^\delta G_c v_a^2 dt, \quad W_{L,a} = - \int_0^\delta (v_b - v_a) G_a (v_b - v_a) dt, \quad (118)$$

$$W_{L,o} = - \int_0^\delta v_b (G_o v_b) dt. \quad (119)$$

Each heat integrand is $-T_j \dot{S}_j$ under the declared thermal law. Evaluate the endpoint stores from $E(t) = i(t)^T L i(t)/2 + C v_b(t)^2/2$ at 0 and δ . Multiplying the independently specified current and capacitor equations gives $\dot{E} = u I_s - R_s I_s^2 - R_1 i_1^2 - R_2 i_2^2 - G_c v_a^2 - G_a (v_b - v_a)^2 - G_o v_b^2$. Consequently the exact model residual from these integrals and endpoint evaluations is zero, without an additional impulse. The matrix in (110) varies with time; the constant-matrix work formula cannot be substituted here. The path integrals and $E(\delta)$ are retained as exact conditional expressions, not assigned unevaluated numerical values. No numerical integration has been performed.

For the physical selector enlarge the boundary to include its stores and material thermal states. Its control supply has the separately measured work $\int_0^\delta v_{\text{ctrl}} i_{\text{ctrl}} dt$; any mechanical actuation has its conjugate work as well. Specifying G_a, G_o does not determine those transfers. Use actual heat crossing this enlarged boundary and (85), rather than exporting its internal Joule conversion again. Reactive reattachment and winding interruption require their own connection and event laws.

This 1 V source, initially 4 V capacitor, and 1/10 s edge give a concrete voltage–current investigation. Conditional observed envelopes $|\hat{v}| \leq 5 \text{ V}$, $|\hat{i}| \leq 1 \text{ A}$ and calibrated errors $u_v = 1/100 \text{ V}$, $u_i = 1/1000 \text{ A}$ bound one port's product error by $1501/1000000 \text{ J}$ on the edge. A proposed whole-event bound of $1/50 \text{ J}$ must include every channel, timing, endpoint, thermal, and controller contribution in (5); it is not guaranteed by the single-port bound. If a waveform exceeds these envelopes, recalculate its bound. Report both receiver works and the physical signed remainder. Agreement or a surviving discrepancy becomes a result of that specified commutation, with preparation and reset still recorded separately.

6.8 Parameter distinctions and constrained limits

The regular derivations cover both connections and both polarities. In particular, they include

$$\rho \in \{1/2, 1, 2\}, \quad k \in \{0, 1/2, 19/20, 99/100\}$$

without any interpolation assumption. The additional individually specified controls are important because they alter different parts of the equations:

Table 15. Individual regular controls, with other baseline parameters retained. Their joint physical realization is not established by the algebraic balance alone.

Parameter or condition	Alternatives to the baseline
Turns magnitude	1/4, 4
Initial load conductance, S	0, 1/1000, 1
Probe conductance, S	1/1000, 1/10
Drive frequency, Hz	0, 2, 20
Capacitance, F	1/2000, 1/100
Both copper resistances, Ω	0, 1/2
Core conductance, S	0, 1/10
Load multiplier at 1/5 s	0 (removal), 1 (no change)
Drive phase	$-\pi/2, 0, \pi/2$
Imposed (i_1, i_2, v_o) , in A, A, V	$(1/5, 0, 0); (0, -1/5, 0); (0, 0, 1)$
Null	Zero source and zero initial state

The paired-return family additionally uses opposed $\rho \in \{3, 4, 6\}$, the same four k values, $G_L \in \{0, 1/10, 1\}$ S, and both returns. Its separate $n = -4$ controls are zero and 20 Hz drive, initial $(1/5, 0, 0)$ or $(0, 0, 1)$, $G_P = 1/10$ S, reversed polarity, zero drive, and equal opposed sections $n = -1$. All four physical intervals remain separate; neither fixed-return experiment itself commutes the return.

Exactly $k = 1$, $C = 0$, $R_s = 0$, a shorted output, and zero turns ratio are not obtained by substituting into a formula that divides by the vanishing quantity. They need their own constrained equations and compatible initial states; a prepared capacitor short may require an impulse or a dissipative finite path. Saturation, remanence, hysteresis, frequency-dependent core loss, distributed capacitance, chassis returns, and thermal evolution are absent from (94). Their model residual has unknown sign and magnitude. No equality of ideal balances estimates that discrepancy.

7 Compound circulation and a transformer network

7.1 Two meshes and a common stepped planet

Let coaxial members A, B mesh with two rigidly connected planet sections P_A, P_B on carrier C . Set $s_X = -1$ for an external mesh and $s_X = +1$ for an internal mesh, and define

$$a = \frac{m_0}{2} (Z_A - s_A Z_{pA}) = \frac{m_0}{2} (Z_B - s_B Z_{pB}) > 0, \quad h_X = s_X \frac{Z_{pX}}{Z_X}, \quad R = \frac{h_A}{h_B}. \quad (120)$$

Rolling gives $\omega_X - \omega_C = h_X(\omega_P - \omega_C)$ and hence $\omega_A - \omega_C = R(\omega_B - \omega_C)$. With carrier rate $\omega_C = w$ and one held member, the other rates are

$$\begin{array}{c|cc} & \omega_A & \omega_B \\ \hline A \text{ held} & 0 & w(1 - 1/R) \\ B \text{ held} & w(1 - R) & 0 \end{array}, \quad \omega_P = w + \frac{\omega_A - w}{h_A}. \quad (121)$$

For q equally loaded massless planets, choose $\tau_A = \tau_*$. Their axial free-body equation is $s_A r_{pA} F_A + s_B r_{pB} F_B = 0$, with $q r_A F_A = \tau_*$. The pin reaction is $R_t = -(F_A + F_B)$, giving

$$(\tau_A, \tau_B, \tau_C) = \tau_*(1, -R, R - 1). \quad (122)$$

The carrier torque follows from $q a R_t$ using (120). These constraints, not an energy premise, imply $\sum \tau = 0$ and $\sum \tau \omega = 0$.

The nine named sides are three shaft powers $p_X^f = \tau_X(\omega_X - \Omega)$, two opposite sides of each mesh, and two opposite pin sides. When all planets are aggregated, the mesh sides are $\pm p_A^f, \pm p_B^f$ and the pin sides are $\pm p_C^f$. They are component diagnostics, not nine independent external supplies. Fix the physical ground throughput denominator $D_0 = \max_{X=A,B,C} |\tau_X \omega_X|$ and define

$$\mathcal{E}_f = \frac{\max_{\text{nine sides}} |p^f|}{D_0} = \frac{\max_{X=A,B,C} |\tau_X(\omega_X - \Omega)|}{D_0}. \quad (123)$$

This definition is meaningful only for $D_0 > 0$ and gives $\mathcal{E}_0 = 1$ exactly. It measures the largest absolute indicated port power, not a new net transfer and not an invariant amount of circulating energy. In particular,

$$\mathcal{E}_C = \frac{1}{|R - 1|} \quad (A \text{ held}), \quad \mathcal{E}_C = \frac{|R|}{|R - 1|} \quad (B \text{ held}). \quad (124)$$

Table 16. Five pitch-compatible compound configurations. Multiple-planet tooth phasing, clearance, and equal load sharing remain additional physical assumptions.

Configuration	$(Z_A, Z_{pA}, Z_{pB}, Z_B)$	(s_A, s_B)	R
I: simple planetary	(24, 18, 18, 60)	(−1, +1)	−5/2
II: stepped planet and ring	(20, 22, 20, 62)	(−1, +1)	−341/100
III: two external members	(30, 20, 19, 31)	(−1, −1)	62/57
IV: two rings	(62, 20, 21, 63)	(+1, +1)	30/31
V: two rings, close ratios	(100, 58, 59, 101)	(+1, +1)	2929/2950

For each choose $m_0 = 1/500$ m, $q = 3$, $w = 2\pi$ rad/s, $\tau_* = 47/4$ N m, and interval $[0, 3/2]$ s. All four frame ratios follow directly from (121)–(123):

Table 17. All held-member and frame choices for the five configurations. The smallest value is 100/441 and the largest is 2950/21. Neither is a continuous bound for other geometries or frames.

Configuration	Held	Ground	Carrier	Member A	Member B
I	A	1	2/7	1	2/5
I	B	1	5/7	5/2	1
II	A	1	100/441	1	100/341
II	B	1	341/441	341/100	1
III	A	1	57/5	1	57/62
III	B	1	62/5	62/57	1
IV	A	1	31	1	31/30
IV	B	1	30	30/31	1
V	A	1	2950/21	1	2950/2929
V	B	1	2929/21	2929/2950	1

Each signed work is separately $W_X^f = (3/2)\tau_X(\omega_X - \Omega)$; internal works carry the corresponding opposite signs. This ideal algebraic boundary has no constitutive storage coordinates, so both endpoint store sums are empty. The external works sum to zero and $r_E = 0$ without an impulse. Adding finite inertias would require independent kinetic endpoints and, for acceleration, different torques. Configuration I reduces exactly to the earlier simple train.

At $R = 1$, the constraints impose $\omega_A = \omega_B$, not a stopped carrier. With either member held, both are zero, $\tau_C = 0$, and $D_0 = 0$ even if $w \neq 0$. A nonzero relative circulation effort can still be imposed in the ideal network. Its ratios are then nonzero divided by zero, or zero divided by zero; both are undefined rather than finite observations. For any positive rational $R = p_0/q_0$, the two-external assembly $(Z_A, Z_{pA}, Z_{pB}, Z_B) = (2q_0, 2p_0, p_0 + q_0, p_0 + q_0)$ realizes that ratio with positive integer teeth. Thus approaches from either side of one are algebraically possible, subject again to physical tooth and clearance constraints.

7.2 Complete ports and mapped terminal products

Use two ideal transformers with primary branches A–C and B–C and secondary branches both P–C. For each transformer impose

$$V_X - V_C = h_X(V_P - V_C), \quad j_X = -h_X i_X, \quad j_A + j_B = 0, \quad i_C = -(i_A + i_B). \quad (125)$$

The parallel-secondary current law gives $i_B = -Ri_A$; the common return then gives $i_C = (R - 1)i_A$. With the power-preserving scale $\alpha = 1 \text{ V}/(\text{rad/s})$, these reproduce (121)–(122). The conversion from revolutions per second includes 2π in every power and work.

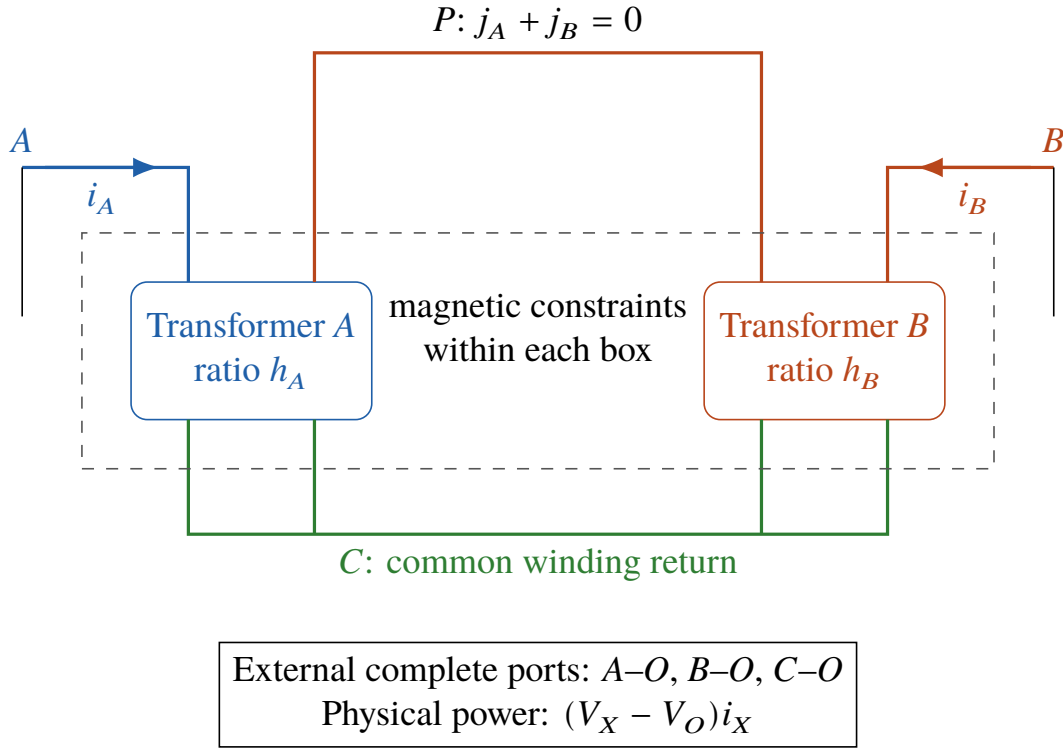


Figure 8. Schematic two-transformer compound correspondence. Both secondary windings share $P-C$; the complete external ports connect A, B, C to the physical return O . The four winding ports are internal to the complete network. A common shift of all node potentials, including O , preserves every complete port voltage.

On the whole electrical boundary the three external ports are $X-O$, with power $(V_X - V_O)i_X$. On individual transformer boundaries the four complete winding powers are $(V_X - V_C)i_X$ and $(V_P - V_C)j_X$; each pair sums to zero. The nine mechanical sides correspond instead to signed copies of the terminal products $(V_X - \alpha\Omega)i_X$. They therefore match exactly in every frame in the table, but they are not all complete electrical ports. For a coordinate shift, an individual external-port return contributes $-(V_O + g)i_X$, even when the sum of all return currents is zero.

The maximum winding power divided by D_0 equals $\mathcal{E}_C$. The ratio including all three external and four winding ports is

$$\mathcal{E}_{\text{phys}} = \max(1, \mathcal{E}_C), \quad (126)$$

which is invariant under a common coordinate shift and generally differs from $\mathcal{E}_f$. For the table, $\mathcal{E}_{\text{phys}} - \mathcal{E}_f$ ranges from $-241/100$ to $2929/21$, and differs from zero for 26 of the 40 frame choices. These are differences between defined observables, not failed matches of the corresponding terminal and mechanical products. A fixed physical-port set cannot realize all frame ratios through changing only the voltage zero. Constant ideal winding voltage here does not establish a sustainable finite-core DC transformer.

The same correspondence retains zero velocity with nonzero torque, zero current, sign reversals, and exact current zeros. Replace the constant applied torque by $\tau_A(t) = \tau_*[t/(1\text{ s}) - 3/4]$ on $[0, 3/2]$ s, retaining constant rates. Each port has $P_j(t) = p_{j*}[t/(1\text{ s}) - 3/4]$, with p_{j*} in watts. Its separate work integrals are

$$W_j(0, 3/4\text{ s}) = -\frac{9}{32}p_{j*}\text{ s}, \quad W_j(3/4\text{ s}, 3/2\text{ s}) = \frac{9}{32}p_{j*}\text{ s}, \quad (127)$$

$$W_j(0, 3/2 \text{ s}) = 0, \quad \int_0^{3/2 \text{ s}} |P_j(t)| dt = \frac{9}{16} |p_{j*}| \text{ s}. \quad (128)$$

Thus $|W_j| = 0$, whereas integrated absolute power is positive when $p_{j*} \neq 0$. Zero-power ports remain zero, and every instantaneous power vanishes at the reversal itself. Zeros and maximizing-port ties follow from the individual polynomial products; undefined zero-throughput ratios must be retained. Arbitrary constant and linear common offsets, including the earlier nine choices, change terminal products according to (79) while leaving the complete-port works fixed.

8 Finite correspondence, active references, and limiting models

8.1 Magnetic elasticity and capacitive inertia

A finite compound network can have real stored states without being a rigid gear train. Let the oriented branch-incidence matrix B have columns $e_A - e_C, e_P - e_C, e_B - e_C, e_P - e_C$, with the held node eliminated when necessary. Each coupled pair has

$$\mathbb{L}_X = L_0 \begin{pmatrix} h_X^2 & kh_X \\ kh_X & 1 \end{pmatrix}, \quad L_0 > 0, \quad 0 \leq k < 1. \quad (129)$$

Let C be the diagonal grounded node-capacitance matrix and R the positive branch-resistance matrix. The electrical model is

$$C\dot{v} = I_{\text{ext}} - Bi, \quad Li = B^T v - Ri. \quad (130)$$

Source, receiver, and probe conductances are included explicitly in I_{ext} , each with its own physical return. Independently specify a rotational Maxwell model with inertias $J = \alpha^2 C$, elastic deformations z , stiffness $K = \alpha^2 L^{-1}$, and mobility $M = R/\alpha^2$. Its equations are

$$J\dot{\omega} = \tau_{\text{ext}} - Bf, \quad f = Kz, \quad \dot{z} = B^T \omega - Mf. \quad (131)$$

For a single pair set $u_z = z_1/h_X$, $v_z = z_2$, and $q_{\pm} = (u_z \pm v_z)/\sqrt{2}$. The elastic energy is

$$U_X = \frac{\alpha^2 q_+^2}{2L_0(1+k)} + \frac{\alpha^2 q_-^2}{2L_0(1-k)}, \quad f = \frac{\partial U}{\partial z}. \quad (132)$$

Differentiation verifies the specified stiffness. The exact map

$$v = \alpha \omega, \quad i = f/\alpha, \quad z = Li/\alpha, \quad I_{\text{ext}} = \tau_{\text{ext}}/\alpha \quad (133)$$

substitutes (130) into (131). Compatible initial states and uniqueness of the regular linear equations establish correspondence throughout each interval. The magnetic store equals U , and capacitor energy equals rotational kinetic energy. Magnetic energy maps to elasticity, while inertia maps to capacitance; interchanging those assignments would invalidate the comparison. Residuals in amperes, volts, radians, and radians per second cannot be merged into a dimensionful physical norm without scales.

One illustrative coefficient choice is $L_0 = 3/25 \text{ H}$, $R_X = (2/25) \text{ diag}(h_X^2, 1) \Omega$, $R_s = 3/4 \Omega$, with member inertias $3/100, 7/200 \text{ kg m}^2$, carrier inertia $1/50 \text{ kg m}^2$, three planets each of mass $3/25 \text{ kg}$ and spin inertia $1/1250 \text{ kg m}^2$. Include their orbital inertia $3m_p a^2$ separately, using (120). Grounded receiver and driver capacitors $1/100$ and $1/200 \text{ F}$ parallel the free-member and carrier capacitors; a reset conductance is $1/2 \text{ S}$ and source amplitude $2\pi \text{ V}$ at $\alpha = 1$ in the stated units. The branch stores may be counted as

four shared/leakage modal stores, and the free-member, carrier, planet, receiver, and driver stores as five more. The held-member capacitor has identically zero voltage. All nine nontrivial terms are evaluated constitutively at both endpoints.

For a completely specified example select configuration I, member B held, $k = 19/20$, and $\alpha = 1 \text{ V}/(\text{rad/s})$. Use the reduced node order (A, C, P) . Order the windings as the primary and secondary of A , followed by the primary and secondary of B . The matrices are

$$B_r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad C_r = \text{diag} \left(\frac{1}{25}, \frac{160219}{6250000}, \frac{3}{1250} \right) \text{F}, \quad (134)$$

$$L_A = \begin{pmatrix} 27/400 & -171/2000 \\ -171/2000 & 3/25 \end{pmatrix} \text{H}, \quad L_B = \begin{pmatrix} 27/2500 & 171/5000 \\ 171/5000 & 3/25 \end{pmatrix} \text{H}, \quad (135)$$

$$R = \text{diag} \left(\frac{9}{200}, \frac{2}{25}, \frac{9}{1250}, \frac{2}{25} \right) \Omega. \quad (136)$$

Here $L = \text{diag}(L_A, L_B)$. The carrier capacitance is $1/50 + 1/200 + (9/25)(21/500)^2 \text{ F}$; the free-member capacitance includes $1/100 \text{ F}$, and the planet capacitance includes three spin inertias. Every voltage and winding current starts at zero. The source drives C – O through $R_s = 3/4 \Omega$, with the following piecewise constant commands. The probe is absent, $G_P = 0$.

Table 18. Fully specified reversed-drive preparation, receiver change, relaxation, and reset trial. Each endpoint uses the value from its own interval side; the states are continuous.

Interval, s	u , V	G_O , S	G_C , S	G_R , S
$[0, 1/10]$	-2π	$1/10$	0	0
$[1/10, 1/5]$	2π	$1/10$	0	0
$[1/5, 3/10]$	2π	0	$2/5$	0
$[3/10, 1/2]$	0	0	$2/5$	0
$[1/2, 1]$	0	0	$2/5$	$1/2$

Let $e_F = (1, 0, 0)^T$, $e_C = (0, 1, 0)^T$, and $d_{FC} = e_F - e_C$. On each interval define the nodal conductance matrix and state x by

$$G = R_s^{-1} e_C e_C^T + (G_O + G_P + G_R) e_F e_F^T + G_C d_{FC} d_{FC}^T, \quad (137)$$

$$x = (v_A, v_C, v_P, i^T)^T, \quad \dot{x} = A_7 x + b_7 u, \quad (138)$$

$$A_7 = \begin{pmatrix} -C_r^{-1} G & -C_r^{-1} B_r \\ L^{-1} B_r^T & -L^{-1} R \end{pmatrix}, \quad b_7 = \begin{pmatrix} C_r^{-1} e_C / R_s \\ 0_4 \end{pmatrix}. \quad (139)$$

Thus $y = (x, 1)$ obeys the constant augmented matrix $A = \begin{pmatrix} A_7 & b_7 u \\ 0 & 0 \end{pmatrix}$. For the carrier-return receiver put $\ell = (1, -1, 0, 0, 0, 0, 0, 0)^T$; its quadratic form is $Q_{L,C} = -G_C \ell \ell^T$. Its separate interval work is $\mathcal{W}(Q_{L,C}; A, y_0, T)$, while the source uses the two linear forms u and $(u - v_C)/R_s$. Every other power below supplies its own Q_j . Stores at each endpoint are evaluated as $v_r^T C_r v_r / 2 + i^T L i / 2$, with the constituent capacitances retained separately if their subboundaries are needed. This specifies a unique exact matrix-function answer for every port and stage without a decimal trajectory. The source-free reset interval cannot erase a nonzero state in finite time, by the invertibility argument following (98).

The same construction defines a parameter family with both held members, $k \in \{0, 19/20, 99/100\}$, initial receiver conductance $G_* \in \{0, 1/10, 1\}$ S, and $G_P \in \{0, 1/100\}$ S. Preparation uses either $u = 0$ or $u = -2\pi$ V from zero state; operation uses $+2\pi$ V. The receiver either changes from $G_O = G_*$ to $G_C = 4G_*$ or remains at ground with G_O raised to $4G_*$. These alternatives are distinct initial value problems, not results inferred from the particular example.

Finite elasticity introduces effort paths absent in the rigid reduction. For example $m_X = j_X + h_X f_X$, using mechanical secondary and primary efforts, is generally nonzero and supplies a couple between planet and carrier. The pin effort includes orbital acceleration,

$$T_{\text{pin}} = J_{\text{orb}} \dot{\omega}_C + (h_A - 1)f_A + (h_B - 1)f_B. \quad (140)$$

Here primary and secondary efforts are the components of f in (131), and $J_{\text{orb}} = 3m_p a^2$. Consequently opposite diagnostic mesh-side products need not cancel across an elastic element. Matching the original nine diagnostics is not proof that those alone exhaust a finite boundary.

8.2 Physical ports, event schedules, and ratios

The complete finite boundary contains all the above stores and source and branch resistors, with external source, receivers, probe, reset, and heat reservoirs. There are ten physical powers: source uI_s ; source heat $-(R_s I_s)I_s$; four winding heat exports $-(R_j i_j)i_j$; ground-return receiver $-(V_F - V_O)G_O(V_F - V_O)$; carrier-return receiver $-(V_F - V_C)G_C(V_F - V_C)$; probe $-(V_F - V_O)G_P(V_F - V_O)$; and reset $-(V_F - V_O)G_R(V_F - V_O)$. F names the free member. Each resistance also has its equivalent $-T\dot{S}$ export. The mechanical forms follow (77), with mobility dissipation $-f^T M \dot{f}$. Node and winding subboundary powers remain separate diagnostics. Multiplying the two network equations by v^T and i^T cancels $v^T B i$, deriving the energy balance and the corresponding mechanical one.

Preparation $[0, 1/10]$, operation $[1/10, 1/5]$, post-connection change $[1/5, 3/10]$, relaxation $[3/10, 1/2]$, and reset $[1/2, 1]$ s are distinct stages. An explicitly prescribed reversed-drive preparation is also a distinct path. A receiver can remain at $F-O$ or change to $F-C$ while every capacitor return stays at O . Ideal selectors have zero voltage when closed and zero current when open; a stipulated zero-current controller then has zero work within that selector model. This says nothing about a real actuator supply. With bounded resistive changes, all capacitor voltages, winding currents, and elastic deformations are continuous; no inductor is opened, capacitor reattached, or energized rotor clamped.

For constant connections and a forcing represented as in (97), each of the ten works is the separate closed form (98); endpoints use each model's own states and stores. A finite transition of width $\delta \in \{1/1000, 1/2000, 1/4000\}$ s can be specified by $s_e(t) = 0$ before t_e , $(t - t_e)/\delta$ during the transition, and 1 afterwards. At $t_e = 1/5$ s, set $G_O = G_*(1 - s_e)$ and $G_C = 4G_*s_e$ for commutation, or $G_O = G_*(1 + 3s_e)$ and $G_C = 0$ for a retained return. At $3/10$ s take $u = 2\pi(1 - s_e)$ V; at $1/2$ s take $G_R = s_e/2$ S. These histories define time-dependent matrices, for which the constant-matrix theorem alone supplies no evaluated work. Their powers and constitutive balance are specified, but no trajectory, ordering, or limiting switch work is inferred here from the instantaneous model.

Linearity gives an exact amplitude control: reversing the source and the compatible prepared state reverses all physical states and leaves their quadratic physical powers and works unchanged. Zero drive with zero state is the exact null. Common-offset cross terms gI_j do reverse with current and must be recomputed; they are not copied from positive drive. Neither amplitude homogeneity nor a regular finite model at $R = 1$ proves the singular rigid constraint or an energized locking law.

In ground coordinates, the actual trajectory has diagnostic denominator

$$D(t) = \max_{X=A,B,C} |V_X I_{\text{ext},X}|. \quad (141)$$

A receiver commutation does not authorize replacing this with the old circuit's counterfactual trajectory. The mapped nine-side ratio uses its own numerator; a complete-physical-port ratio uses another numerator. An interval ratio instead first integrates every signed work and then takes the prescribed maxima and quotient. It is neither the instantaneous ratio nor an integral of absolute power. Zeros, nonzero-over-zero, and zero-over-zero remain undefined; a small whole-boundary residual cannot certify a small or uncertain denominator.

An initially unenergized right-hand operation endpoint can have zero node throughput while the external source and its resistor already carry finite, opposite powers. Thus the mapped diagnostic can be zero-over-zero while the complete physical diagnostic is nonzero-over-zero. There is no finite ratio to report at that event. In the example (136), ratios at 3/20, 1/4, and 3/4 s are obtained by evaluating its exact state and individual powers there, not by reusing (124). Inertia, compliance, resistance, loading, and excitation all change them. No ordering or extremum over those stages follows from the rigid table.

The active-reference, finite-controller and prepared-limit constructions are proved in Appendix A. They retain the full sensor, actuator, converter and rail states, distinguish physical moving returns from coordinate changes, and include both passive and imposed-clamp limits. The finite actuator has an exact endpoint departure; the proof and its separate supplies remain with equation (174).

9 Physical reconnection and prepared states

The four-tap construction fixes an ideal terminal relation. Selecting another terminal for a charged receiving branch changes the physical equations. Murray's SERPS patent illustrates an isolated primary and a tapped secondary with two switched capacitor-resistor branches Murray (2017). The positive branch charges from the full secondary and returns through its lower section; the negative branch uses the opposite full and tapped connections. The companion develops that graph, both polarities, finite commutation, controller and material states. It is a separate apparatus from the conductive common-flux four-tap realization.

For one ideal positive interval, write $v_s = Ri + v_C$, $C\dot{v}_C = i$ during charging, and $v_C = Rj + f v_s$, $C\dot{v}_C = -j$ during return. The tap fraction is $0 < f < 1$. Thus $f v_s < v_C < v_s$ allows both positive charging current at the full winding and positive return current at the tap. The independent powers give

$$v_s i = Ri^2 + \frac{d}{dt} \frac{Cv_C^2}{2}, \quad -f v_s j = Rj^2 + \frac{d}{dt} \frac{Cv_C^2}{2}. \quad (142)$$

Integrate each source and resistor product on its own interval and evaluate the capacitor endpoints. The resulting source draw minus return equals receiver delivery plus store change. The intended quarter-cycle command does not establish the current direction throughout that interval.

For any complete source port with positive-inward power p_s , define

$$W_+ = \int \max(p_s, 0) dt, \quad W_- = - \int \min(p_s, 0) dt, \quad W_s = W_+ - W_-, \quad W_{\text{traffic}} = W_+ + W_-. \quad (143)$$

Gross draw, return, net input and absolute throughput have distinct meanings. When a generator is enclosed with the capacitor and switches, its electrical return is internal. Prime-mover, excitation and controller works remain external; the rotor and field stores join the endpoint inventory.

The optional series/parallel bank also retains each cell's state. Two ideal cells of capacitance C , both at voltage V , have store CV^2 . Their parallel terminals have $(V, 2C)$, their series-aiding terminals $(2V, C/2)$. Equivalent capacitance describes the connection, while the individual cell voltages determine the store. For unequal preparations, actual exchange paths determine receiver and switch work.

A terminal may also conceal a prepared state. For two equal series cells with an isolated floating midpoint and resistor R_L , put $s = V_1 + V_2$, $\delta = V_1 - V_2$. The cell laws give

$$\dot{s} = -\frac{2s}{R_L C}, \quad \dot{\delta} = 0, \quad E_C = \frac{C}{4}(s^2 + \delta^2), \quad W_L(T) = \frac{Cs(0)^2}{4}(1 - e^{-4T/(R_L C)}). \quad (144)$$

The last expression is the separately integrated positive receiver delivery. With $C = 1$ F, preparations $(1, 1)$ and $(3, -1)$ V produce identical entire load-terminal histories and cell stores differing by 4 J. Later isolated parallel equalization can expose that difference, but its division among useful receiver, switch and probe requires their actual paths. A finite observation can distinguish only the states covered by its observation map and calibrated error.

Conserved oriented state, absolute magnitude, stored energy, and work are also different objects. Isolated resistive exchange conserves $q_1 + q_2$ and makes $|q_1| + |q_2|$ nonincreasing. Inductive exchange can create opposite charge signs with equal initial and final total reactive energy. Changing the connection requires testing conservation again. For $\dot{\lambda} = B^T v + e_p U - Ri$, a putative winding invariant $h^T \lambda$ obeys $\dot{c} = (Bh)^T v + h^T e_p U - h^T Ri$. With independent states and strictly positive winding resistance, conservation for every state and drive forces $h = 0$. An isolated two-store invariant cannot be assigned to this driven lossy winding by analogy.

The companion gives the exact contraction and reversal controls, an all-time certificate through one finite shunt interval, and physical reconnection observation and work contracts. Those results clarify which states the selected terminal sees and which paths receive work. Preparation, operation and reset still require separate histories for a complete performance comparison.

10 Events, uncertainty, and limits of the conclusions

10.1 Discontinuities and impulses

A mesh separation or re-engagement with finite force can have continuous positions and velocities and hence zero store jump, while its force–velocity product jumps. The exact work is the sum of integrals on both event sides; no additional impulse is inserted. A compliant contact example is an overlap δ with spring force $k_m \max(\delta, 0)$ and stored energy $k_m \max(\delta, 0)^2/2$. Adding a dashpot requires its own non-negative relative-motion heat export. Switching a finite contact or damping force does not itself establish a jump in momentum.

Rigid impact with restitution is different: momenta and kinetic energy may jump. Its signed impulse work requires a declared impact law or finite regularization. Backlash during an accelerating carrier, unequal planet load sharing, nonzero pin couples, and radial centre motion cannot be certified by the simpler event or steady-state formulas. Likewise, physically opening an energized winding can produce clamp, arc, and displacement-current paths; changing a tap or moving a capacitor can involve charge sharing. Every switch, clamp, controller, and preparation path must then be explicit. A finite resistive receiver selector does not settle these reactive events.

10.2 Exact residuals and unresolved accuracy questions

All polynomial, trigonometric cycle, and matrix-function integrals reported here are exact in their declared models. The numerical integration residual is **exactly zero**, because no numerical integration is used. The instantaneous equation residual is zero by the independent constitutive substitution shown for each model, and the exact energy residual is zero only for the complete boundaries and effective-frame conventions specified. The physical model residual remains unquantified: it comprises omitted mesh, bearing, windage, churning, joint, magnetic, dielectric, switch, controller, and thermal physics. No apparatus measurements are reported.

It is useful to retain the mathematical limitations of approximate audits without turning their outcomes into physical theorems. Three distinct failure mechanisms illustrate why an apparently small balance is insufficient. First, a quadrature error estimate can vanish for a constant integrand while roundoff in a nonzero work sum remains. Second, an accumulation estimate proportional to the computed power can miss cancellation error when forming $\omega - \Omega$ near an exact zero. If the represented rates have errors $\delta\omega, \delta\Omega$, the port-power error at the exact held condition is exactly $\tau(\delta\omega - \delta\Omega)$ for exact τ ; its work can have either sign. Reducing an integration step cannot remove that input error. Zeroing the torque removes it for a different reason and is a useful but distinct null control.

For N contributions combined by a balanced addition tree of depth $d = \lceil \log_2 N \rceil$, the standard elementary relative-error model $|\delta| \leq u$, without underflow or overflow, gives the conservative exact bound $[(1+u)^d - 1] \sum_{j=1}^N |a_j|$ for summation error. This follows by bounding each leaf's product of at most d relative factors. It still bounds summation of the given a_j , not the error in constructing them. A first-order du replacement is unnecessary here. Completing a missing error component must be justified independently; changing a criterion only to admit a residual provides no validation.

Third, a boolean statement $|r_E| \leq b$ is unstable when its uncertainty spans the threshold. With certified residual uncertainty u_r , only $|r_E| + u_r \leq b$ establishes acceptance; when $|r_E| - u_r > b$ it establishes failure; the intervening case is unresolved. Comparing a small positive bound with a much larger absolute comparison tolerance can be vacuous even if nothing is known to disagree. These questions concern classification and detection, not necessarily energy transfer. The exact nulls in (42) and (73) settle those ideal equations but do not retrospectively certify any approximate calculation or unknown implementation.

Three unresolved accuracy questions must retain their own conditions. At a held or locked port with nonzero applied torque, a formation-error bound must cover the work error caused by subtracting nominally equal rates; a bound proportional only to the resulting power can collapse to zero. For the regular finite circuit, the uncertainty of each work integral must be established independently of cancellation in the summed residual. For the two separately loaded returns, the work uncertainties and the voltage-integral uncertainties require separate statements on their own prepared trajectories. Exact expressions supplied here determine the model quantities; they do not supply a missing uncertainty enclosure for another evaluation or measurement.

Complete power-zero information also leaves short transfers to be resolved. In (99), $V_0 \neq 0$ gives negative power throughout every finite interval, with no finite power zero. Nevertheless its signed work is $-CV_0^2(1 - e^{-2T/(RC)})/2$, which remains substantial when T exceeds the relaxation scale. A brief, same-sign startup transfer can therefore be missed even if every sign-changing root is known. Independently checking the port integrals and their fastest time dependence remains necessary.

Endpoint accuracy is a separate issue when a small change is obtained from two large stores. For a

capacitor with true endpoint voltage V and voltage error δV , its exact energy error is

$$\delta E_C = CV \delta V + \frac{1}{2} C (\delta V)^2. \quad (145)$$

The error in the interval change is the final value of this expression minus its initial value. It can change the signed residual even when every work integral barely changes. Both endpoint states and each port work require independent accuracy statements. Improving one calculation does not erase the earlier signed discrepancy or certify a physical model error.

For the illustrative $C_s = 1$ F rail, $V_s = 100$ V and a positive voltage error $\delta V_s = 1/100$ V give the single-endpoint error

$$\delta E_s = 100 \frac{1}{100} + \frac{1}{2} \left(\frac{1}{100} \right)^2 = \frac{20001}{20000} \text{ J}. \quad (146)$$

It exceeds the finite-actuator departure in (174). Cancellation between initial and final rail errors requires established correlation; it cannot be inferred from similar displayed voltages. The complete supply voltage–current product at an external port and the difference of two internal rail stores are different measurement choices. Each retains its actual boundary and all preparation work. The finite budgets in (51), (90), and (175) connect specific signals to required error scales; the general remainder uses (5).

Finite-circuit work uncertainty, finite voltage-integral uncertainty, and finite compound ratio uncertainty remain separate. As explicit illustrative engineering decision limits, one may declare

$$b_W = \frac{1}{5000000} \text{ J} + \frac{1}{500000} \sum_j |W_j|, \quad (147)$$

$$b_\Psi = \frac{1}{50000000} \text{ V s} + \frac{1}{500000} \sum_l |\Psi_l|. \quad (148)$$

The second sum covers the three readouts and the two separately defined copper-drop voltage integrals. These fixed criteria are not error enclosures. Uncertainty for a fixed-output circuit and for two separately loaded returns are separate questions; checking one cannot certify the other. A joule discrepancy must not be added to a volt-second discrepancy. Different integration estimates may disagree while an equation-based energy residual is small through cancellation. Power-zero locations and maximum-port switches require complete root reasoning for the continuous functions; a collection of sign brackets alone is not such a proof, and an identically zero interval is not a set of isolated crossings. Sensitivity evidence is not a rigorous enclosure or a hardware model-error estimate. An uncertainty interval for a denominator that contains zero leaves its ratio unresolved. None of these unresolved accuracy questions is closed merely by the exact identities of a different idealization.

10.3 Physical scope

The mechanical derivation assumes parallel-axis rigid pitch geometry, compatible assembly, constant centre radius, equal loading, and zero pin couple. Ravigneaux arrangements require further independent mesh equations; helical or nonparallel-axis motion requires additional degrees of freedom. Bodies with flexure, true restitution, combined backlash and Euler forcing, and rate reversals in frictional mechanisms require their own constitutive and event treatments. The kinematic count identities themselves remain valid through reversals, while absolute counts and friction signs need interval splits. Tooth counts are integers; irrational rate ratios are allowed by the linear constraints and are not a new energy law.

The lead-out model uses massless coupling members and a resolved axial torque path only. It omits

clearance, compliance, encoder quantization and direction ambiguity, bearing friction, joint-force and bending work, and the energetic cost of the ground-phase device. Its illustrative large articulation is not a buildability claim. A combined train and readout must recompute the forces and trajectory under the actual load. Conditions such as a zero pin flow follow from geometry; force and work magnitudes depend on the selected inertias and torques. Inertial contributions scale linearly with their mass coefficient for a fixed prescribed trajectory, and quasistatic terms scale linearly with applied torque; a loaded trajectory need not retain either scaling when those parameters change.

For the windings, physical ground paths, nonlinear core and thermal behavior, reactive switching supplies, and the complete preparation and reset cycle retain their local open energy questions. The finite-reference equations add explicit sensor and actuator stores, conversion loss, holding error, and supply cutoff. Their effective feedback and nonloading transduction remain assumptions; the signed physical model discrepancy is unmeasured. The passive limit and the clamped limit retain their distinct drives and preparation conditions. A calibrated comparison includes the source, receiver, probe, all returns, each energy store, and instrument loading on the same event-split interval. Any unexplained positive or negative remainder remains an open result with its magnitude, conditions, uncertainty, and completed checks. An expected balance does not determine the outcome of that investigation.

11 Conclusion

The two-U-joint planet connection gives a four-shaft apparatus with two independent speeds and two independent steady loads. Contact constraints and attached free bodies determine its complete operating family. In the ring-held one-receiver control, the four signed one-second works are $(7, -14/3, 0, -7/3)$ J; the ring carries a nonzero reaction and the receiver changes the carrier's work by $7/3$ J. The ideal two-transformer and four-tap constructions reproduce all four voltage/current products with the same loaded planet terminal.

The construction identifies what a correspondence must preserve. Correct joint phasing concerns an actual moving plane; a grounded phase actuator requires its own supply. A common electrical zero leaves complete port voltages unchanged; a moved receiver return changes the connected circuit. Finite magnetic storage maps to an elastic state and capacitance to inertia. The specified compliant models preserve the corresponding powers and stores, while rank and preparation conditions limit reduction to the rigid train.

Coin and spoke controls make fixed and relative counts accessible without assigning work from a count. The dimensioned support comparison and finite locking controls give bounded physical models. Switched tap and capacitor connections expose a further distinction: the same terminal history can hide different prepared stores, and equal endpoints need not fix individual work destinations. Receiver work follows the actual path and its source, switch, probe and controller products.

The appendices retain the compound ratios, supplied references, singular preparations and complete event arguments that establish the scope of these results. Their finite actuator control gives the stated exact departure from an ideal receiver endpoint; their prepared limits can retain energy while a terminal current tends to zero. Each is a result for its declared equations. *Finite Transfers and Open Energy Balances* develops the joint, winding, material and observation questions needed to test physical realizations. Independent work and endpoint uncertainty determine whether a measured signed residual is resolved; its physical magnitude and sign remain unassigned until those observations exist.

A Supplied references, prepared limits, and event proofs

A.1 An accelerating reference is not a voltage coordinate shift

For the finite rotational analogue, the relative kinetic-plus-elastic store is $K^f + U = K^0 - \Omega H + J\Omega^2/2 + U$, with $J = \sum J_j$ and $H = \sum J_j \omega_j$. Its additional Euler power is $-\dot{\Omega}(H - J\Omega)$, without the centrifugal potential convention used for E^f in (30). Either convention is valid when its terms are retained consistently. Relative deformations and mobility losses are unchanged under common rigid rotation.

A common electrical coordinate shift, however, also shifts V_O and leaves all actual capacitor voltage differences and physical energies unchanged. The difference between the mechanical relative store and that unchanged physical electrical store is exactly

$$\Delta E_{\text{ref}} = -\Omega H + \frac{1}{2}J\Omega^2. \quad (149)$$

An effective electrical expression built from shifted node values relative to an unshifted fictitious zero can imitate it, but then represents a virtual store rather than the actual grounded capacitors. Equation (149) therefore obstructs realization by a common coordinate shift alone. It does not exclude a different physical circuit with driven returns and separately supplied compensation currents.

A.2 An active capacitor cell and its reference driver

Place a capacitor between physical nodes A and R , and let $w = V_A - V_R$. Its boundary excludes two floating current sources, their controllers, and a separate driver setting V_R . Prescribe

$$C = J/\alpha^2, \quad V_R = \alpha\Omega, \quad I_d = \tau/\alpha, \quad I_e = -C\dot{V}_R, \quad C\dot{w} = I_d + I_e. \quad (150)$$

Both currents are positive from R into A . The independent mechanical law is $J\dot{\omega} = \tau$. Compatible initial data imply $V_A = \alpha\omega$ and $w = \alpha(\omega - \Omega)$ by differentiation of (150). The complete physical source powers and store are

$$P_d = wI_d = \tau(\omega - \Omega), \quad P_e = wI_e = -J\dot{\Omega}(\omega - \Omega), \quad E_C = \frac{1}{2}Cw^2 = \frac{1}{2}J(\omega - \Omega)^2. \quad (151)$$

The Euler counterpart is supplied or absorbed by its own active source. Summing KCL at the floating source and capacitor returns gives zero net bias current I_b from the reference driver. This cancellation does not set either floating source's complete power to zero.

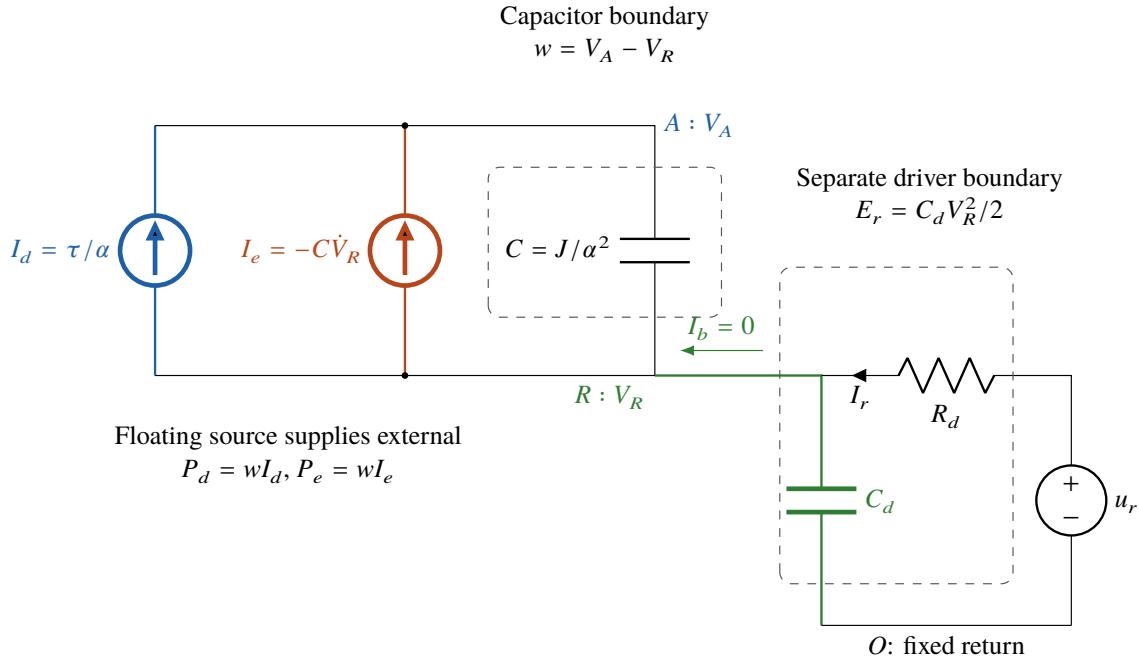


Figure 9. Schematic active reference cell. The two floating sources drive the capacitor between A and R ; their supplies lie outside the capacitor boundary. A separate R_d, C_d driver sets the physical reference potential. The net bias current is zero in this ideal floating cell, while the driver capacitor and resistor still require their own input work.

The driver boundary in Figure 9 contains a capacitor C_d from R to the fixed return O and a resistor R_d from its command node to R . For zero bias current, define

$$\begin{aligned}
 I_r &= C_d \dot{V}_R, & u_r &= V_R + R_d I_r, \\
 P_r &= u_r I_r, & P_{h,r} &= -(R_d I_r) I_r, \\
 P_b &= -V_R I_b = 0, & E_r &= \frac{1}{2} C_d V_R^2.
 \end{aligned} \tag{152}$$

Its command source and heat reservoir are external. The state law $C_d \dot{V}_R = (u_r - V_R)/R_d$ realizes the prescribed path when initialized compatibly. Each driver work is the integral of its own product in (152); driver energy is evaluated from V_R .

As an exact reference-only control choose $J = 1 \text{ kg m}^2$, $\alpha = 1 \text{ V}/(\text{rad/s})$, $C = 1 \text{ F}$, $\tau = 0$, and $\Omega = 2t \text{ rad/s}^2$ on $[0, 1] \text{ s}$, with zero initial physical and reference rates. Then $V_A = 0$, $V_R = 2t \text{ V/s}$, $w = -2t \text{ V/s}$, and $I_e = -2 \text{ A}$. Choose $C_d = 1/200 \text{ F}$ and $R_d = 3/4 \Omega$; consequently $I_r = 1/100 \text{ A}$ and $u_r = 2t \text{ V/s} + 3/400 \text{ V}$. Direct integration of the separate source and heat products yields

Table 19. Exact active control. Polynomial integrands in the table use numerical SI coordinates. Every boundary has zero energy residual. The combined capacitor and driver finish with 201/100 J, supplied by independently integrated works.

Boundary	Separate signed works on $[0, 1]$ s, J	Initial store, J	Final store, J
Capacitor	$W_d = 0, W_e = \int_0^1 4t \, dt = 2, W_b = 0$	0	2
Mechanical relative energy	$W_{\text{shaft}} = 0, W_{\text{Euler}} = 2$	0	2
Reference driver	$W_r = 403/40000, W_{h,r} = -3/40000, W_b = 0$	0	1/100

Disabling I_e is a distinct control. At zero torque the capacitor remains uncharged and V_A follows V_R ; its final energy is zero, whereas the specified mechanical relative energy is 2 J. Electrical minus mechanical store is -2 J. Both models separately balance, so this is a failed state and port correspondence, not an unexplained energy deficit. Finite changes of torque or reference acceleration preserve capacitor voltage and both mechanical rates; powers may jump but this control has no impulses. The current-source internals are outside the cell boundary. Their finite supply model is specified next rather than assigned zero work.

Whether a physical reference circuit reproduces these individual works and stores remains open [OP-TRF-12]. The reference-only control offers a direct observation: prescribe the finite ramp, record w, I_d, I_e , the driver voltage and current, and evaluate both capacitor endpoints. Its 2 J receiver change specifies the gross correspondence. The finite-actuator prediction and endpoint bound in (175) give a smaller comparison with explicit calibration requirements. The individual products must be integrated over the same ramp and its preparation. Supply endpoints are additionally needed when the supply lies inside the chosen boundary. The measured work and store differences, and any unexplained energy remainder, currently have unknown signs and magnitudes. A measured agreement or a surviving discrepancy would distinguish the alternatives; neither follows from the prescribed-source calculation. Finite actuation below gives an exact example of a signed departure that such an observation can resolve.

A.3 A compound reference supplied by a finite rail

Restore all four node coordinates, including the held member and its capacitance. Replace their grounded capacitor returns by a physical node R . Write $r = V_R - V_O$, $w = v - r\mathbf{1}$, and retain the complete incidence matrix B , for which $B^T \mathbf{1} = 0$. Four floating actuators inject $I_{\text{ext},X}(v, i, t)$ and four separate compensating sources inject $-C_X \dot{r}$ from R . The receiver boundary contains these capacitors, the two winding pairs, and their four resistors. Its state laws are

$$C\dot{w} = I_{\text{ext}} - Bi - C\mathbf{1}\dot{r}, \quad L\dot{i} = B^T w - Ri. \quad (153)$$

The actuators implement the original source and receiver current laws as commands. Their transfers are regenerative converter outputs; the original source-resistance and load coefficients no longer denote additional physical heat exports at this receiver boundary. The winding resistors remain physical. An actuator at the held member supplies the reaction needed to maintain its absolute potential. Omitting it after restoring that coordinate would change the model.

Substituting $v = w + r\mathbf{1}$ in (153) recovers (130). For selected node s , a physical feedback reference is $r = \sigma v_s$, initialized compatibly and driven by $\dot{r} = \sigma \dot{v}_s + \dot{\sigma} v_s$. Here $\dot{v}_s$ comes from the instantaneous KCL

state equation. Constant $\sigma = 1$ represents that member's frame; $\sigma = 0$ represents ground. A continuous change of σ changes the physical reference, including the $\dot{\sigma}v_s$ contribution.

The receiver's twelve signed powers and independent stores are

$$P_{a,X} = w_X I_{\text{ext},X}, \quad P_{e,X} = -w_X C_X \dot{r}, \quad P_{h,j} = -(R_j i_j) i_j, \quad (154)$$

$$E_{\text{rec}} = \frac{1}{2} w^T C w + \frac{1}{2} i^T L i, \quad X \in \{A, B, C, P\}, \quad j = 1, \dots, 4. \quad (155)$$

Multiplying (153) by w^T, i^T proves its balance. Under the finite map the first eight powers become the individual shaft and Euler products; the four heat powers become mobility losses. The store is relative kinetic plus elastic energy, not the centrifugal-effective store E^f of (30). Every signed receiver work is integrated on its own stage before internal supply transfers are combined.

Use the driver (152) with $C_d = 1/200$ F and $R_d = 3/4$ Ω ; it is additional to the capacitor already included at node C . Nine ideal isolated bidirectional converters supply the eight floating actuators and the driver's command branch. Let Π_j be each converter's signed output power: the eight $P_{a,X}, P_{e,X}$ and P_r . Their common rail is a capacitor $C_s = 1$ F with initial voltage $V_s = 100$ V. The constitutive converter law is $I_{\text{in},j} = \Pi_j/V_s$ while $V_s > 0$. A specified controller conductance $G_q = 1/1000$ S also loads the rail. Rail KCL and its exact solution for $\xi = V_s^2$ are

$$C_s \dot{V}_s = - \sum_{j=1}^9 \frac{\Pi_j}{V_s} - G_q V_s, \quad (156)$$

$$\xi(t) = e^{-\lambda_q(t-t_0)} \xi(t_0) - \frac{2}{C_s} \int_{t_0}^t e^{-\lambda_q(t-s)} \sum_j \Pi_j(s) ds, \quad \lambda_q = 2G_q/C_s. \quad (157)$$

This is a capacitor-current law, not an energy residual used to determine a trajectory. The nine rail works are $-\int \Pi_j dt$, its controller heat work is $-\int V_s(G_q V_s) dt$, and its endpoints are $C_s \xi/2$. Positivity of the exact expression for ξ is the supply-feasibility condition. No universal nondepletion claim follows from a balanced receiver.

On any constant-connection interval with constant σ , the underlying state is (97), and every Π_j is a quadratic form. Augmenting its outer-product coordinates with ξ makes (157) a constant linear system; its separate work integrals follow from $T\varphi_1$ just as in (98). Polynomial reference changes on a fixed underlying topology add finite polynomial coordinates to these products. Time-dependent receiver conductances still require their own path solution; the constant-matrix expression is not silently applied to them.

The rail has an explicit preparation on $[-1, 0]$ s with receiver and controller disconnected. A 100 A charging source through 1 Ω gives $V_s(t) = (100 \text{ V})(t + 1 \text{ s})/(1 \text{ s})$ and command $u_s = V_s + 100$ V. Its separately integrated source and heat works are 15000 J and -10000 J, while its endpoint stores are zero and 5000 J. Combining the prepared rail, driver, and receiver boundaries cancels each converter transfer only after its two signed integrals have been evaluated. The remaining operation exports are four winding heats, driver heat, and controller heat, and all three kinds of endpoint storage remain. An ideal converter and a chosen controller shunt do not quantify real conversion efficiency, bandwidth, sensor demand, or hardware discrepancy.

For a linear reference change from r_- to r_+ over $[t_e, t_e + \delta]$, let $\Delta r = r_+ - r_-$. Independent driver integrals give

$$W_{h,r} = - \int_{t_e}^{t_e + \delta} R_d (C_d \dot{r})^2 dt = - \frac{R_d C_d^2 (\Delta r)^2}{\delta}, \quad (158)$$

$$W_r = \int_{t_e}^{t_e + \delta} (r + R_d C_d \dot{r}) C_d \dot{r} dt = \frac{1}{2} C_d (r_+^2 - r_-^2) + \frac{R_d C_d^2 (\Delta r)^2}{\delta}. \quad (159)$$

The endpoint store change is evaluated from $C_d r^2/2$ and gives zero residual for every positive δ . For $r_- = 0$, $\Delta r = 1$ V and the stated driver values, durations 1/1000, 1/2000, 1/4000 s give heat works $-3/160$, $-3/80$, $-3/40$ J and command works 17/800, 1/25, 31/400 J; every store increase is 1/400 J. At fixed positive R_d , C_d and nonzero Δr , heat magnitude diverges as $\delta \rightarrow 0$. This driver supplied from a finite rail has no finite-energy instantaneous-reference limit. It is a precise model obstruction, compatible with the constructive finite-rate realization above.

A.4 Finite sensing, actuation, and conversion

The physical reference can have finite response while every constituent has its own specified energy law. Retain the four receiver capacitors, winding pairs, and winding resistors. Let $j_{a,X}$ and $j_{e,X}$ be the actual actuator and compensation currents from R into node X . The receiver obeys

$$C\dot{w} = j_a + j_e - Bi, \quad Li = B^T w - Ri, \quad v = w + r\mathbf{1}. \quad (160)$$

All four node voltages are now dynamic. In particular, a finite holding servo may leave a nonzero held-member voltage.

Nine RC buffers sense the four absolute voltages, four winding currents, and reference-driver current. Write $\kappa_i = 1/10$ V/A for current transduction. For each sensor, ξ is its supplied input voltage and z its capacitor voltage. With $C_z > 0$, $\tau_z > 0$, and $R_z = \tau_z/C_z$,

$$I_z = (\xi - z)/R_z, \quad C_z \dot{z} = I_z, \quad E_z = \frac{1}{2} C_z z^2. \quad (161)$$

The inputs are the respective v_X , $\kappa_i i_j$, and $\kappa_i I_r$, limited when necessary to a fraction κ_v of the positive rail voltage: $\mathcal{S}(x, V_s) = \max(-\kappa_v V_s, \min(x, \kappa_v V_s))$. Each buffer is powered by a converter. The transducer itself is stipulated nonloading; its physical loading error is still unquantified. Hats below denote the sensor voltages, divided by κ_i for a current observation.

For held member H and free member F , the four external-current commands are the explicit laws

$$J_{a,F} = -(G_O + G_P + G_R) \hat{v}_F - G_C (\hat{v}_F - \hat{v}_C), \quad (162)$$

$$J_{a,C} = (u - \hat{v}_C)/R_s + G_C (\hat{v}_F - \hat{v}_C), \quad (163)$$

$$J_{a,H} = \hat{i}_{H1} - G_H \hat{v}_H, \quad J_{a,P} = 0, \quad G_H = 1 \text{ S}. \quad (164)$$

Here i_{H1} is the held member's primary-winding current. For the selected sensor z_s , define $r_t = \sigma z_s$ and $\dot{r}_t = \sigma \dot{z}_s + \dot{\sigma} z_s$. Here σ is continuous and piecewise differentiable, with bounded derivative on each finite transition. An abrupt nonzero jump of r_t would require a separate event model. The four compensation commands are $J_{e,X} = -C_X \dot{r}_t$. These source and load coefficients determine regenerative commands; the corresponding passive resistors are not also present as receiver heat paths.

Each of the eight actual currents has a separate inductor $L_a > 0$ and resistor $R_a \geq 0$. Its source commands

$$e = \mathcal{S}\left(w_X + R_a j + \frac{L_a}{\tau_a}(J - j), V_s\right), \quad L_a \dot{j} = e - w_X - R_a j, \quad (165)$$

where J is the corresponding command. Without saturation this gives $\tau_a \dot{j} = J - j$ exactly. The local feedback is a declared constitutive law; detailed switching electronics remain unspecified.

The driver command has its own capacitor $C_u > 0$ and voltage u_r . Set $u_t = \mathcal{S}(r_t + R_d C_d \dot{r}_t, V_s)$ and

$$J_u = \hat{I}_r + \frac{C_u}{\tau_a}(u_t - u_r), \quad C_u \dot{u}_r = J_u - I_r, \quad I_r = \frac{u_r - r}{R_d}, \quad C_d \dot{r} = I_r. \quad (166)$$

The receiver and floating actuators have zero net bias current at R by their complete KCL, leaving the stated RC reference driver. Its capacitor remains additional to the carrier's receiver capacitance.

Each boundary's signed powers and independent store are now explicit:

Table 20. Finite reference boundaries. Every resistor power is a separate heat export $-T\dot{S}$. The voltage and current in each product belong to the same physical port.

Boundary	Individual inward powers	Store
Receiver	$w_X j_{a,X}, w_X j_{e,X}, -R_j i_j^2$	$w^T C w / 2 + i^T L i / 2$
Each actuator	$e j, -w_X j, -R_a j^2$	$L_a j^2 / 2$
Each sensor	$\xi I_z, -R_z I_z^2$	$C_z z^2 / 2$
Command stage	$u_r J_u, -u_r I_r$	$C_u u_r^2 / 2$
Reference driver	$u_r I_r, -R_d I_r^2$	$C_d r^2 / 2$

Eighteen converters supply the eight actuator sources, nine sensor buffers, and command stage. Let Π_j be one of their output powers from the table. For a chosen illustrative efficiency $0 < \eta_c \leq 1$, specify

$$P_{\text{in},j} = \begin{cases} \Pi_j / \eta_c, & \Pi_j \geq 0, \\ \eta_c \Pi_j, & \Pi_j < 0, \end{cases} \quad I_{\text{in},j} = P_{\text{in},j} / V_s, \quad P_{\text{heat},j} = -(P_{\text{in},j} - \Pi_j). \quad (167)$$

The converter has empty storage coordinates and the three separately integrated powers $V_s I_{\text{in},j}$, $-\Pi_j$, and $P_{\text{heat},j}$. Their constitutive products sum to zero in each sign interval. This law specifies conversion loss independently of an energy remainder. For the rail, retain $C_s = 1$ F and $G_q = 1/1000$ S, with

$$C_s \dot{V}_s = - \sum_j I_{\text{in},j} - G_q V_s, \quad E_s = \frac{1}{2} C_s V_s^2. \quad (168)$$

Each rail transfer is $-V_s I_{\text{in},j}$ and its controller heat is $-G_q V_s^2$. Preparation from zero uses the independently integrated charging path already given, with a general final voltage V_0 if desired. For duration $T_p = 1$ s and charging resistance $R_p = 1$ Ω , its current is $I_p = C_s V_0 / T_p$, source work is $C_s V_0^2 / 2 + R_p C_s^2 V_0^2 / T_p$, and heat work is $-R_p C_s^2 V_0^2 / T_p$. The endpoint store is $C_s V_0^2 / 2$.

The physical stages and finite receiver changes remain those specified earlier. Choose initial receiver, sensor, command, driver, and actuator states explicitly, zero for an unenergized preparation. For each stage $[t_m, t_{m+1}]$, integrate every product in the table and (167)–(168) separately, and evaluate every displayed store at both endpoints. Direct multiplication of the state laws proves each differential balance; cancella-

tion of already integrated internal transfers then leaves four winding, eight actuator, nine sensor, eighteen converter, one driver, and one controller heat export for the combined boundary. Its stores include all receiver, sensor, actuator, command, driver, and rail states. Bounded command or limiter changes preserve those states; powers can change discontinuously. A supply cutoff ends this operating model and does not specify a subsequent shutdown or reset.

For example, declare cutoff at $V_s = 1/100$ V. The remaining rail store is $1/20000$ J. With zero drive and every other state zero, the controller-only law gives $V_s = V_0 e^{-G_q t/C_s}$ and

$$t_{\text{cut}} = \frac{C_s}{G_q} \ln \frac{V_0}{1/100 \text{ V}}, \quad W_q(0, T) = -\frac{C_s V_0^2}{2} (1 - e^{-2G_q T/C_s}). \quad (169)$$

This independently integrated heat matches the constitutive endpoint change. One exact illustrative initial voltage is $V_0 = 2001/200000$ V. With active drive, receiver and actuator stores at cutoff generally remain as well; the cutoff state and its work require that drive's actual path.

Parameter distinctions include $\tau_z, \tau_a \in \{1/10000, 1/1000, 1/100\}$ s, $\eta_c \in \{1, 19/20, 4/5\}$, all four fixed references and a continuous reference change, both receiver connections, and edge durations $\{1/4000, 1/1000, 1/100\}$ s. The independent voltage-limit fractions $\{1/200, 1/20, 9/10\}$ and rail preparations $V_0 \in \{1/10, 1, 10, 100\}$ V change the actual commands and available operating interval. An illustrative joint approach uses $\tau_z = \tau_a = \epsilon^2/1000$ s, $L_a = \epsilon/10000$ H, $R_a = \epsilon/10$ Ω , $C_z = \epsilon^2/1000000$ F, $C_u = \epsilon/100000$ F, and $\eta_c = 1 - \epsilon/20$, with $0 < \epsilon \leq 1$. It keeps the receiver, rail, and RC reference-driver coefficients fixed. These definitions alone establish no unrestricted convergence through saturation or supply depletion. For arbitrary finite parameters, the nonlinear path integrals remain unevaluated here; no ordering or numerical error bound is assigned to them.

A.5 An exact energy departure caused by finite actuation

A single-cell control isolates actuation while prescribing the reference $r = at$, with constant $a > 0$ on $[0, T]$. Set shaft torque to zero and give the compensation actuator the command $J = -Ca$. Retain its positive L_a , nonnegative R_a , and response time $\tau_a = \tau > 0$ from (165), without voltage saturation. From zero current and capacitor voltage, the independent current and capacitor laws give

$$j = -Ca(1 - e^{-t/\tau}), \quad w = -a[t - \tau(1 - e^{-t/\tau})], \quad e = w + R_a j + L_a \dot{j}. \quad (170)$$

The reference driver is the prescribed-path RC driver above, outside the cell and actuator boundaries. This control does not presume that the full sensing model follows the same path.

Let $b = e^{-T/\tau}$ and $\mathcal{Q} = R_a C^2 a^2 [T - 2\tau(1 - b) + \tau(1 - b^2)/2]$. Direct integration of each physical product gives

$$W_C = \int_0^T w j dt = \frac{1}{2} C a^2 [T - \tau(1 - b)]^2, \quad (171)$$

$$W_{a,\text{out}} = \int_0^T (-w j) dt = -W_C, \quad W_{a,h} = \int_0^T (-R_a j^2) dt = -\mathcal{Q}, \quad (172)$$

$$W_{a,s} = \int_0^T e j dt = \frac{1}{2} C a^2 [T - \tau(1 - b)]^2 + \mathcal{Q} + \frac{1}{2} L_a C^2 a^2 (1 - b)^2. \quad (173)$$

The last line follows by expanding the specified e and integrating its three products, not by solving an energy remainder. Independent initial stores are zero, and final stores are $Cw(T)^2/2$ and $L_a j(T)^2/2$. Both

residuals are zero. The current and capacitor voltage remain continuous at command changes; no impulse appears. All expressions have zero numerical integration residual. Physical sensor loading, converter electronics, and component discrepancy remain unmeasured.

The ideal mechanical relative store under this zero-torque reference is $Ca^2T^2/2$. Consequently the signed electrical-minus-mechanical difference is

$$\delta E = \frac{Ca^2}{2} ([T - \tau(1 - e^{-T/\tau})]^2 - T^2) < 0. \quad (174)$$

The strict inequality follows from $0 < \tau(1 - e^{-T/\tau}) < T$ for $T > 0$. With illustrative $C = 1$ F, $a = 2$ V/s, $\tau = 1/10$ s, and $T = 1$ s, the final voltage is $w(1 \text{ s}) = -(9 + e^{-10})/5$ V, and the receiver finishes with $(9 + e^{-10})^2/50$ J instead of 2 J. Its signed departure from the ideal counterpart is $[(9 + e^{-10})^2 - 100]/50$ J (approximately -0.380 J). The separately supplied reference driver still has its own earlier works and store; it is not included a second time here.

Capacitor voltage and calibrated capacitance therefore provide a direct first observation of finite actuation. If $|C - \hat{C}| \leq u_C$ and $|w - \hat{w}| \leq u_w$, the exact single-endpoint bound is

$$u_{EC} \leq \frac{u_C}{2} (|\hat{w}| + u_w)^2 + \hat{C} \left(|\hat{w}|u_w + \frac{u_w^2}{2} \right). \quad (175)$$

For illustrative $\hat{C} = 1$ F, $u_C = 1/100$ F, $|\hat{w}| \leq 2$ V and $u_w = 1/100$ V, this gives 80501/2000000 J per endpoint. Two such observations of the finite and ideal final states have combined bound 80501/1000000 J. A proposed 1/10 J total comparison bound leaves 19499/1000000 J for reference setting, timing, and other errors; it separates the finite departure from zero if those requirements are met. The ideal comparator can alternatively be calculated from independently bounded C, a, T rather than observed as a second capacitor state. Initial states require their own bounds if the comparison uses changes instead of final stores.

For the complete energy observation, integrate each actuator source product, the separately supplied reference driver, and actual external heat, then evaluate actuator, capacitor, controller, and thermal endpoints. This full-boundary residual has its own uncertainty and may agree with zero or retain either sign. The finite-minus-ideal store difference above is a determined comparison, not that residual. Keeping the supply external and measuring its complete voltage–current product avoids subtracting two large rail stores to obtain its delivered work. Bringing the rail inside instead retains both endpoints and its preparation, as specified earlier.

For the complete finite-reference model, define on each matching interval

$$\delta W_j = W_{j,\text{rec}} - W_{j,\text{mech}}, \quad \delta E(t) = E_{\text{rec}}(t) - E_{\text{mech}}(t). \quad (176)$$

The comparator uses the same dimensional map and relative-store convention. Its mechanical reference is $\Omega = \sigma \omega_s$, where s denotes the selected mechanical member; the receiver uses its actual r . The difference $r - \alpha \Omega$ is a separately observable tracking error. Integrate the comparator's shaft, Euler, and mobility works separately. A zero balance residual within either model does not set these signed differences to zero. The exact lag control has a determined negative difference; general finite tracking, holding, and limiter paths may have either sign and need their own evaluation. Hardware discrepancies remain open with unknown signs and magnitudes until the local reference experiment is performed.

A.6 A joint finite-to-rigid limit with matched drives

Unity coupling at fixed inductance, resistance, and inertia is not the rigid algebraic network. A different, explicitly matched family does have that limit. For $0 < \epsilon \leq 1$, choose

$$L_0 = \frac{3}{25\epsilon} \text{ H}, \quad k = 1 - \epsilon^2, \quad r_\epsilon = \frac{2\epsilon^2}{25} \Omega, \quad R_X = r_\epsilon \text{ diag}(h_X^2, 1), \quad C_{X,\epsilon} = \epsilon C_{X,0}. \quad (177)$$

Scale every inertia counterpart, including the orbit and added receiver and carrier capacitances. The boundary contains all four node capacitors and both winding pairs. Four ideal voltage clamps prescribe the constant values $\bar{v}_X = \alpha \omega_X$ from (121); their currents are obtained from KCL. Four winding heat exports complete the boundary. The additional planet clamp is retained even though its limiting current is zero. These imposed drives are different from a passive load supplied through the original source resistor.

For either pair abbreviate $h = h_X$, and put $s = hi_1 + i_2$, $d = hi_1 - i_2$. With winding voltages u_1, u_2 , the two independent voltage laws become

$$\ell_+ \dot{s} = u_1/h + u_2 - r_\epsilon s, \quad \ell_- \dot{d} = u_1/h - u_2 - r_\epsilon d, \quad \ell_+ = \frac{3(2 - \epsilon^2)}{25\epsilon} \text{ H}, \quad \ell_- = \frac{3\epsilon}{25} \text{ H}. \quad (178)$$

If $s(0) = 0$ and $u_1/h + u_2$ is uniformly bounded on $[0, T]$, its exact convolution implies $|s(t)| \leq T \sup |u_1/h + u_2|/\ell_+$. If additionally $d, \dot{d}$ remain bounded, the second equation implies $|u_1/h - u_2| \leq \ell_- \sup |\dot{d}| + r_\epsilon \sup |d| \rightarrow 0$. These are conditional constitutive limits, not a claim that arbitrary initial data avoid a singular transient.

For the constant clamps, $u_1 = hu_2$ and $u_2 = \bar{v}_P - \bar{v}_C =: V_2$. Choose compatible initial ideal currents $i_1(0) = i_{1*}$, $i_2(0) = -hi_{1*}$, hence $s(0) = 0$ and $d(0) = d_* = 2hi_{1*}$. The exact solution on $[0, T]$ is

$$s(t) = A_s(1 - e^{-\beta_+ t}), \quad d(t) = d_* e^{-\beta_- t}, \quad A_s = 2V_2/r_\epsilon, \quad \beta_\pm = r_\epsilon/\ell_\pm. \quad (179)$$

No expansion in ϵ is used. The inequality $0 \leq 1 - e^{-x} \leq x$ for $x \geq 0$ proves the uniform bounds

$$|s(t)| \leq \frac{2|V_2|T}{\ell_+}, \quad |d(t) - d_*| \leq |d_*|\beta_- T, \quad 0 \leq t \leq T. \quad (180)$$

Both right sides tend to zero. Independently, the mechanical modal strains $a_z = z_1/h + z_2$ and $b_z = z_1/h - z_2$ obey $\dot{a}_z = 2\omega_{PC} - r_\epsilon a_z/\ell_+$ and $\dot{b}_z = -r_\epsilon b_z/\ell_-$ at $\alpha = 1$ in the stated units. Their compatible solutions satisfy $a_z = \ell_+ s$, $b_z = \ell_- d$, recovering the same forces from elastic constitutive laws.

Every signed operation work can be integrated before taking a limit. Define $F_\beta(T) = T\varphi_1(-\beta T)$, equal to $(1 - e^{-\beta T})/\beta$ for $\beta \neq 0$ and T at zero. The five separate primitives needed for one pair are

$$S_1 = \int_0^T s dt = A_s(T - F_{\beta_+}), \quad D_1 = \int_0^T d dt = d_* F_{\beta_-}, \quad (181)$$

$$S_2 = \int_0^T s^2 dt = A_s^2(T - 2F_{\beta_+} + F_{2\beta_+}), \quad D_2 = \int_0^T d^2 dt = d_*^2 F_{2\beta_-}, \quad (182)$$

$$S_D = \int_0^T s d dt = A_s d_* (F_{\beta_-} - F_{\beta_+ + \beta_-}). \quad (183)$$

All F terms in (183) have argument T . The primary and secondary current integrals are $(S_1 + D_1)/(2h)$ and $(S_1 - D_1)/2$. Their heat works are individually $-r_\epsilon (S_2 + D_2 + 2S_D)/4$ and $-r_\epsilon (S_2 + D_2 - 2S_D)/4$. For the two pairs assemble the four current integrals into $\mathcal{T}$. Each node clamp has work $W_X = \bar{v}_X(B\mathcal{T})_X$

since $\dot{v} = 0$. Thus the planet clamp's work is included explicitly. The nine diagnostic products have their own constant velocity factors times these same independently integrated effort functions; they are not extra inputs.

At each endpoint evaluate

$$E_{m,X}(t) = \frac{\ell_+ s(t)^2 + \ell_- d(t)^2}{4}, \quad U_X(t) = \frac{a_z(t)^2}{4\ell_+} + \frac{b_z(t)^2}{4\ell_-}, \quad E_C(t) = \frac{\epsilon}{2} \sum_X C_{X,0} \bar{v}_X^2. \quad (184)$$

For example the bound on s gives $\ell_+ s^2/4 \leq V_2^2 T^2/\ell_+$, while $\ell_- d^2/4 \leq \ell_- d_*^2/4$. Each store tends to zero, but is retained at every positive ϵ . Bounded currents and $r_\epsilon \rightarrow 0$ make each heat work tend to zero. The planet-clamp current tends to zero because the two ideal secondary currents cancel. Uniform effort convergence makes every signed diagnostic work converge to its own rigid value on $[0, 3/2]$ s, for every configuration, held member, and fixed frame in the compound table. This follows directly from (180) and the finite number of constant port factors.

Preparation is another explicit interval, $[-T_p, 0]$ with $T_p = 1/10$ s. Before reconnecting the network, independently prescribe linear current ramps $i = f i_*$ in each coupled pair and voltage ramps $v_X = f \bar{v}_X$ on each capacitor, where $f = (t + T_p)/T_p$. Each winding source has voltage $(L\dot{i} + Ri)_j$ and current i_j . Its work, its separate heat export, and each capacitor charging work are

$$W_{s,j} = \frac{1}{2} i_{j*} (Li_*)_j + \frac{T_p}{3} R_j i_{j*}^2, \quad W_{h,j} = -\frac{T_p}{3} R_j i_{j*}^2, \quad W_{C,X} = \frac{1}{2} C_{X,\epsilon} \bar{v}_X^2. \quad (185)$$

These follow separately from the products and $\int f \dot{f} dt = 1/2$, $\int f^2 dt = T_p/3$. Independent initial stores are zero; final stores are $i_*^T Li_*/2$ and the charged-capacitor values. Reconnection retains these voltages and fluxes, with compatible cancelling secondary currents, so this ideal event has zero state jump and zero impulse. Real reconnection transients require additional ports. A fixed nonzero initial s would instead give $\ell_+ s^2/4 \rightarrow \infty$ and falls outside this result.

A.7 Vanishing currents with finite prepared energy

The compatible-data restriction above leaves an important energy question: which stores survive when the observed currents or work differences approach zero? The following exact controls determine several alternatives within the finite constitutive family. Their physical realization retains the open magnetic and preparation questions already identified.

Write $a_L = 3/25$ H and $b_R = 2/25$ Ω , so $\ell_+ = a_L(2 - \epsilon^2)/\epsilon$, $\ell_- = a_L \epsilon$, and $r_\epsilon = b_R \epsilon^2$. For a fixed nonzero current s_0 , prepare one pair with $s_* = s_0 \epsilon^p$, $d_* = 0$, where $p \geq 0$. Its independent store is

$$E_{m,*} = \frac{a_L s_0^2}{4} (2 - \epsilon^2) \epsilon^{2p-1}. \quad (186)$$

It tends to zero for $p > 1/2$, to $a_L s_0^2/2$ for $p = 1/2$, and grows without bound for $p < 1/2$. In particular, $p = 1/2$ makes both winding currents tend to zero while the magnetic energy tends to a positive value. For $s_0 = 1$ A this is 3/50 J. The component inductances change along the family; a small current alone does not determine the energy of that family.

Enclose the isolated pair and its two resistors, with independently controlled winding sources and heat reservoirs outside. Prepare it on $[-T_p, 0]$, with $T_p = 1/10$ s, $f = (t + T_p)/T_p$ and $i = f(s_*/(2h), s_*/2)^T$. The source voltages are the independently specified $u = L\dot{i} + Ri$, with $R = r_\epsilon \text{diag}(h^2, 1)$. Each signed

source and heat integral is

$$W_{s,j} = \int_{-T_p}^0 u_j i_j dt = \frac{\ell_+ s_*^2}{8} + \frac{T_p r_\epsilon s_*^2}{12}, \quad W_{h,j} = - \int_{-T_p}^0 R_j i_j^2 dt = -\frac{T_p r_\epsilon s_*^2}{12}, \quad j = 1, 2. \quad (187)$$

The endpoint stores are independently 0 and $\ell_+ s_*^2/4$, giving zero residual after the four works are combined. The finite ramp changes current continuously and introduces no impulse. Subsequent reconnection needs its own compatible state and port treatment. The physical core discrepancy has unknown sign and magnitude; the exact integration residual here is zero.

For $p = 1/2$, the same preparation specifies the supply effort explicitly:

$$u_2 = \frac{a_L s_0 (2 - \epsilon^2)}{2T_p \sqrt{\epsilon}} + \frac{b_R s_0}{2} \epsilon^{5/2} f, \quad u_1 = h u_2. \quad (188)$$

At fixed T_p the required voltage increases without bound as $\epsilon \rightarrow 0$, even while the prepared currents vanish and their store tends to 3/50 J for $s_0 = 1$ A. The limit includes this supply requirement. It does not claim that a fixed voltage source prepares every member.

One finite member is a more direct electrical investigation. Choose $\epsilon = 1/4$, $s_0 = 1$ A, $T_p = 1/10$ s, and $h = -3/4$, the first winding ratio of configuration I. Then the constitutive parameters and prepared state are

$$\mathbf{L} = \begin{pmatrix} 27/100 & -27/80 \\ -27/80 & 12/25 \end{pmatrix} \text{H}, \quad \mathbf{R} = \text{diag}(9/3200, 1/200) \Omega, \quad (189)$$

$$\mathbf{i}_* = \begin{pmatrix} -1/3 \\ 1/4 \end{pmatrix} \text{A}, \quad E_{m,*} = \frac{93}{1600} \text{J}, \quad (190)$$

$$u_1 = -\frac{279}{160} - \frac{3f}{3200} \text{V}, \quad u_2 = \frac{93}{40} + \frac{f}{800} \text{V}. \quad (191)$$

For each winding $j = 1, 2$, independently integrating its source and heat products on $[-1/10, 0]$ s gives

$$W_{s,j} = \frac{2791}{96000} \text{J}, \quad W_{h,j} = -\frac{1}{96000} \text{J}, \quad E_m(-T_p) = 0, \quad E_m(0) = \frac{93}{1600} \text{J}. \quad (192)$$

The four works sum to the independently evaluated magnetic increase, with zero model and numerical integration residuals and continuous currents. These finite voltages and currents specify a two-channel preparation with a measurable store target. Conditional measured envelopes $|\hat{u}_j| \leq 5$ V and $|\hat{i}_j| \leq 1$ A, with voltage and current errors 1/100 V and 1/1000 A, give the same 1501/1000000 J per-source product bound as the commutation control. To compare the net input with the 93/1600 J target at uncertainty 1/100 J, the sum of the two source bounds leaves 3499/500000 J for timing, actual external heat, endpoint and calibration errors. Whether that budget holds is an independent measurement question. A physical winding pair needs calibrated finite coefficients; if they differ from the illustrative matrix, recompute the prediction. Thermal endpoints and the magnetic-state qualification in (85) remain part of that comparison. Later reconnection is another interval with its own ports and compatible states.

The corresponding changing-drive result follows without an expansion. Set $S(t) = u_1/h + u_2$, $D(t) = u_1/h - u_2$, and $\beta_\pm = r_\epsilon/\ell_\pm$. The exact modal solutions are

$$s(t) = s(0)e^{-\beta_+ t} + \frac{1}{\ell_+} \int_0^t e^{-\beta_+(t-u)} S(u) du, \quad (193)$$

$$d(t) = d(0)e^{-\beta_- t} + \frac{1}{\ell_-} \int_0^t e^{-\beta_-(t-u)} D(u) du. \quad (194)$$

On $[0, T]$, these give the exact bounds

$$|s(t) - s(0)| \leq |s(0)|\beta_+ T + T\|S\|_\infty/\ell_+, \quad (195)$$

$$|d(t) - d(0)| \leq |d(0)|\beta_- T + T\|D\|_\infty/\ell_-. \quad (196)$$

Thus a drive defect $D = \epsilon^\gamma D_0(t)$ contributes at most $T\|D_0\|_\infty \epsilon^{\gamma-1}/a_L$ to d . For $\gamma > 1$ this vanishes; for $\gamma = 1$ its limit is $a_L^{-1} \int_0^t D_0(u) du$, under a bounded continuous D_0 . An independently chosen differentiable effort $d_t(t)$ is realized exactly by $D = \ell_- \dot{d}_t + r_\epsilon d_t$, with compatible initial data. Cancellation in a particular drive can reduce its integral, so the actual signed history must be retained.

For an explicit noncancelling defect take zero initial currents and $u_1/h = D_\epsilon/2$, $u_2 = -D_\epsilon/2$ on $[0, T]$, where $D_\epsilon = D_0\sqrt{\epsilon}$. Then $s = 0$ and

$$d(T) = \frac{D_0 T}{a_L \sqrt{\epsilon}} \varphi_1(-b_R \epsilon T/a_L), \quad (197)$$

$$E_m(T) = \frac{D_0^2 T^2}{4a_L} \varphi_1(-b_R \epsilon T/a_L)^2. \quad (198)$$

The voltage mismatch tends to zero, while the current grows and the store tends to $D_0^2 T^2/(4a_L)$ for $D_0 T \neq 0$. With $D_0 = 1$ V and $T = 1$ s that limit is 25/12 J. With a fixed $D_\epsilon = D_0$ instead, both current and stored energy diverge.

For the same pair boundary, write $\beta = \beta_-$ and use the already defined $F_\beta(T)$. Each of the two source products is $D_\epsilon d/4$ and each heat product is $-r_\epsilon d^2/4$. Their individual works are

$$W_{s,j} = \frac{D_\epsilon^2}{4r_\epsilon} (T - F_\beta), \quad W_{h,j} = -\frac{D_\epsilon^2}{4r_\epsilon} (T - 2F_\beta + F_{2\beta}), \quad j = 1, 2. \quad (199)$$

The initial store is zero and the final store is (198). Their difference equals the separately integrated sum. For every fixed $\epsilon > 0$, the imposed voltage steps leave current continuous and introduce no impulse. The finite limiting energy has thus been reached along a specified drive, with all signed transfers retained.

A.8 Matching selected works while stores differ

The distinction also occurs in the full two-pair network. Keep the earlier constant voltage clamps and ideal initial currents, but add c_ϵ to the first secondary current and $-c_\epsilon$ to the other, leaving both primary currents initially unchanged. The added secondary currents cancel at the shared node, so the initial node-current constraint is compatible. Denote the two signs by $\sigma_A = 1$, $\sigma_B = -1$. The exact differences from the original clamped solution are

$$\delta s_X = \sigma_X c_\epsilon e^{-\beta_+ t}, \quad \delta d_X = -\sigma_X c_\epsilon e^{-\beta_- t}, \quad \delta i_{X1} = \frac{\sigma_X c_\epsilon}{2h_X} (e^{-\beta_+ t} - e^{-\beta_- t}). \quad (200)$$

For fixed $c_\epsilon = c \neq 0$, both primary-current differences tend uniformly to zero on a bounded interval, since their magnitudes are at most $|c|(\beta_+ + \beta_-)T/(2|h_X|)$. Their separately integrated differences are $\sigma_X c(F_{\beta_+} - F_{\beta_-})/(2h_X)$; the two secondary differences still cancel at their node. Every original diagnostic work has a constant velocity factor multiplying these primary-current or node-current integrals. Hence those work differences tend to zero, while the individual secondary currents retain opposite nonzero

differences.

The initial common-mode magnetic energy of the two pairs is independently $\ell_+ c_\epsilon^2/2$. The differential-mode contribution remains $\ell_- \sum_X (d_{X*} - \sigma_X c_\epsilon)^2/4$. For fixed c , the first term diverges. For $c_\epsilon = c\sqrt{\epsilon}$ it tends to $a_L c^2$, while the differential and node-capacitor stores tend to zero. Thus even convergence of every named rigid diagnostic work can coexist with a finite additional store.

No preparation or operation transfer is inferred from those endpoints. Equation (185) applies to the newly specified initial current vector, component by component. On operation, for each pair put $A_s = 2V_2/r_\epsilon$, $B_s = s(0) - A_s$, and $d_0 = d(0)$. Replace the common-mode primitives by their direct exponential integrals

$$S_1 = A_s T + B_s F_{\beta_+}, \quad (201)$$

$$S_2 = A_s^2 T + 2A_s B_s F_{\beta_+} + B_s^2 F_{2\beta_+}, \quad (202)$$

$$S_D = d_0(A_s F_{\beta_-} + B_s F_{\beta_+ + \beta_-}). \quad (203)$$

Retain $D_1 = d_0 F_{\beta_-}$ and $D_2 = d_0^2 F_{2\beta_-}$. Each winding heat and node-clamp work is then its earlier stated combination of these separately integrated products. Evaluate the full magnetic and capacitive stores at both endpoints. The zero residual belongs to that complete clamped boundary; it does not remove the difference in stores or winding states. This construction holds the voltages by actual clamp ports. A passive receiver imposes a different selection of efforts.

A.9 Passive receivers and preparation-dependent limits

Retain the source resistor and the original ground and carrier-return loads. Let $G_g = G_O + G_P + G_R$ and $G_n = G_C$, and set $a_v = 1 - h_F/h_H$. With the held voltage zero, the limiting voltage constraints give $v_F = a_v v_C$ and $v_P = (1 - 1/h_H)v_C$. Their KCL and ampere-turn constraints independently yield

$$\bar{v}_C = \frac{u}{1 + R_s[G_g a_v^2 + G_n(a_v - 1)^2]}, \quad \bar{v}_F = a_v \bar{v}_C, \quad \bar{I}_F = -[G_g a_v + G_n(a_v - 1)]\bar{v}_C. \quad (204)$$

The source and load select these efforts. They need not equal the fixed τ_* effort in the rigid comparison. A precise sufficient convergence condition can be derived from the finite equations. On a constant-connection interval, order reduced nodes as (F, C, P) . For each pair define the columns $B_{\pm, X} = B_{X1}/(2h_X) \pm B_{X2}/2$. Let C_0 be the positive diagonal capacitance before multiplication by ϵ , and let G include the source and receiver conductances. In the variables (v, d) the exact fast matrix is

$$A_\epsilon = \begin{pmatrix} -C_0^{-1}G & -C_0^{-1}B_- \\ (2/a_L)B_-^T & -(b_R/a_L)\epsilon^2 I \end{pmatrix}. \quad (205)$$

The time equation is $\epsilon(\dot{v}, \dot{d}) = A_\epsilon(v, d) + (C_0^{-1}(b_u u - B_+ s), 0)^T$. For $H = \text{diag}(C_0, a_L I/2)$, direct multiplication at $\epsilon = 0$ gives

$$HA_0 + A_0^T H = \text{diag}(-2G, 0). \quad (206)$$

This is a calculated matrix identity, with no energy-constancy premise.

Assume $R_s > 0$, $G_g + G_n > 0$, and $h_H \neq 1$. The null space of G then has $v_F = v_C = 0$. For an eigenvalue on the imaginary axis, (206) forces those two voltages to vanish. Their two nodal equations force both differential currents to vanish: the determinant of the corresponding two by two part of B_- is $(h_H - 1)/(4h_F h_H) \neq 0$. The remaining winding equation then forces $v_P = 0$. No nonzero such

eigenvector exists. Every eigenvalue of A_0 has negative real part. All five stated geometries meet the winding-ratio condition. A zero receiver and zero probe before reset need a separate damping check; they are not included by this sufficient condition.

For a dimensionally specified norm, choose fixed positive scales V_*, I_* and set

$$S = \text{diag}(V_* I_3, I_* I_2), \quad \tilde{x} = S^{-1}(v, d)^T, \quad \tilde{s} = s/I_*, \quad \tilde{A}_\epsilon = S^{-1} A_\epsilon S. \quad (207)$$

All following vector norms are Euclidean norms of these dimensionless coordinates and matrix norms are their induced norms. Similarity preserves the preceding spectral result. Consequently there are positive constants M, γ and a sufficiently small ϵ_0 such that $\|e^{\tilde{A}_\epsilon t/\epsilon}\| \leq M e^{-\gamma t/\epsilon}$ for $0 < \epsilon \leq \epsilon_0$. This follows by continuity of the finite matrices and their stable spectral separation. Let $x_* = (\bar{v}, \bar{d})$ solve the constrained source/load equations with zero common mode, on a smooth interval. Variation of constants gives the exact inequality for $\tilde{x}_* = S^{-1}x_*$, with $\tilde{d} = \bar{d}/I_*$:

$$\|\tilde{x}(t) - \tilde{x}_*(t)\| \leq M e^{-\gamma(t-t_0)/\epsilon} \|\tilde{x}(t_0) - \tilde{x}_*(t_0)\| \quad (208)$$

$$+ \frac{M}{\gamma} \left(\frac{I_*}{V_*} \|C_0^{-1} B_+\| \sup \|\tilde{s}\| + \frac{b_R}{a_L} \epsilon^2 \sup \|\tilde{d}\| + \epsilon \sup \|\dot{\tilde{x}}_*\| \right). \quad (209)$$

Here M is dimensionless and γ has units of inverse time. All suprema are over that interval. Writing $\tilde{v} = v/V_*$, the common-mode convolution separately gives $\sup \|\tilde{s}\| \leq \|\tilde{s}(t_0)\| + 2TV_* \|B_+^T\| \sup \|\tilde{v}\|/(\ell_+ I_*)$. For bounded drive, derivative, and initial fast states, substitution into (209) has a coefficient multiplying $\sup \|\tilde{x} - \tilde{x}_*\|$ that tends to zero. Taking it below one gives a finite uniform bound. If the initial common currents tend to zero, the right-hand side then tends to zero away from the initial endpoint. Each connection change begins another such decaying layer. Uniform convergence at a mismatched initial or switching state is not asserted. The modal-energy examples show why this state statement alone does not guarantee disappearance of every prepared store.

Fixed unbalanced common currents instead remain in the nodal forcing $-B_+ s$ and can change the limiting efforts. Zero data, compatible passive data, voltage or differential-current mismatch, opposite common currents, and unbalanced common currents of either sign are thus distinct preparations. For changing clamps, an exact reversal such as $v_C(t) = 2\pi[1 - 4t/(3\text{ s})]$ V on $[0, 3/2]$ s can be combined with the mesh voltage constraints, or with an independently prescribed d_t . Constant, square-root, linear, and quadratic powers of ϵ in a differential-drive defect have the respective behaviors determined by (194)–(198).

Every passive stage retains its ten physical powers and independent stores. Each linear preparation uses its actual initial currents and capacitor voltages in (185). For constant connections and these constant or polynomial commands, (98) independently gives each operation integral. Finite return ramps retain their stated paths and constitutive balance, with no unevaluated work promoted to a value. Source work can be negative: if a prepared state has $v_C > u > 0$, its source product $u(u - v_C)/R_s$ is negative, and continuity retains that sign on a following interval. All other transfers and the separately evaluated decrease of stored energy remain in the comparison. The preparation, returned work, and remaining store are three separate results.

A.10 Singular constitutive laws and prepared clamps

The regular inverse in (139) must not be used when a capacitance vanishes or $k = 1$. For prescribed linear external laws $I_{\text{ext}} = b_u u - Gv$, retain the descriptor equations

$$\underbrace{\begin{pmatrix} C & 0 \\ 0 & L \end{pmatrix}}_E \dot{x} = \underbrace{\begin{pmatrix} -G & -B \\ B^T & -R \end{pmatrix}}_{A_d} x + \begin{pmatrix} b_u u \\ 0 \end{pmatrix}. \quad (210)$$

Dynamic and algebraic coordinates are separated by the constraints, not by inverting E . For a regular pencil and compatible data the reduced dynamic system has its own matrix exponential and individual power integrals. Initial algebraic variables must satisfy the constraints and any necessary differentiated constraints. Across an event without a declared impulse, integrating (210) requires $E(x^+ - x^-) = 0$ for fixed constitutive matrices. An algebraic jump in the null space of this symmetric matrix changes no quadratic store; a required charge or flux jump cannot be hidden in an algebraic update.

For exactly $k = 1$ in one lossy pair, let $a_h = (h, 1)^T$, $L = L_0 a_h a_h^T$, $\phi = L_0 a_h^T i$, and assume its resistance matrix is positive definite. Its winding voltage vector u_w gives

$$\dot{\phi} = \frac{a_h^T R^{-1} u_w - \phi / L_0}{a_h^T R^{-1} a_h}, \quad i = R^{-1} (u_w - a_h \dot{\phi}), \quad E_m = \frac{\phi^2}{2L_0}. \quad (211)$$

The constraint $a_h^T i = \phi / L_0$ follows directly. Multiplication gives $u_w^T i - i^T R i = \phi \dot{\phi} / L_0$, so the magnetizing store remains. At the chosen scale $\alpha = 1$, the independent mechanical model has one spring coordinate ϕ , the same positive mobility matrix, and algebraic forces $f = R^{-1} (u_w - a_h \dot{\phi})$ when winding voltages are replaced by relative angular rates. Its spring energy is $\phi^2 / (2L_0)$. An arbitrary null-mode current is not an independent initial state; it is fixed by this algebraic law.

With only the free-member capacitance zero, its KCL equation becomes a force-balance counterpart without inertia. With all capacitances zero, $I_{\text{ext}} = Bi$ holds at every instant; a node with no external actuator also imposes its differentiated current constraint on $\dot{i}$. Zero source resistance is specified instead by $v_C = u$ and an unknown reaction current from KCL. Its source power uI_s is retained; no $1/R_s$ formula is evaluated at zero. These compound constrained models have their own compatible data and do not settle every singular limit of the earlier tapped-transformer experiment. A zero winding ratio is excluded by the positive tooth counts and nonzero h_X used here.

A prepared short or an energized lock changes stored energy. For a capacitor initially at v_- , connect an external constant voltage U through a positive resistance R_c . On the ensuing interval $[0, T]$, the single-capacitor event model has $v = U + (v_- - U)e^{-t/(R_c C)}$, $i = (U - v)/R_c$. With $a_T = e^{-T/(R_c C)}$, its separate signed port integrals are

$$W_s(0, T) = \int_0^T U i dt = CU(U - v_-)(1 - a_T), \quad (212)$$

$$W_h(0, T) = - \int_0^T R_c i^2 dt = -\frac{1}{2} C (U - v_-)^2 (1 - a_T^2). \quad (213)$$

Independently $E(0) = Cv_-^2/2$ and $E(T) = C[U + (v_- - U)a_T]^2/2$. Substitution gives zero residual for every $R_c, T > 0$. In a vanishing-duration approach with $T/(R_c C) \rightarrow \infty$, the event limits are

$$W_s^{\text{imp}} = CU(U - v_-), \quad W_h^{\text{imp}} = -\frac{1}{2} C (U - v_-)^2, \quad E^+ - E^- = \frac{1}{2} C (U^2 - v_-^2). \quad (214)$$

The source and heat terms have been integrated independently before taking the limit. A short sets $U = 0$ and exports $Cv_-^2/2$ as heat. The independent rotor law $J\dot{\omega} = (\Omega_t - \omega)/\rho_c$ gives motor power $\Omega_t(\Omega_t - \omega)/\rho_c$ and heat export $-(\Omega_t - \omega)^2/\rho_c$, with the same integrals after the dimensional map. In the vanishing-duration limit with other efforts bounded, applying this law to each rotor gives a full energized lock. For regular finite inductance and bounded winding voltages during the shrinking event, flux and current stay continuous; magnetic or elastic energy may then relax over the subsequent finite interval. Exactly singular pairs retain flux continuity and their algebraic current constraint.

A finite clamp conductance $20C/\delta$ over width $\delta \in \{1/1000, 1/2000, 1/4000\}$ s is a separate model. For the isolated capacitor its voltage error is exactly $(v_- - U)e^{-20}$; in the coupled network the other winding currents also enter KCL and must be retained. Neither finite transition is assigned the ideal clamp endpoint without its own path calculation. Prepared locked states may have zero ground shaft throughput while stored winding or elastic efforts give nonzero indicated powers in a rotating observer frame. Such a ratio remains undefined nonzero-over-zero; the clamp reaction belongs to the actual boundary, beyond the original nine diagnostics.

A.11 Finite reversible histories and complete polynomial event sets

A finite circuit can return to zero state after nonzero transfers without being a storage-free ideal transformer. For this control alone set the four winding resistances to zero and use externally prescribed node actuators; retain the positive capacitances and coupled inductances of the finite model. The receiver boundary is (155) with no heat export: four node-source ports and four compensation ports. Driver and rail are outside this particular boundary, with their costs given separately above.

Let $T_* = 1$ s, $z = t/T_*$, and prescribe $v(t) = \mathbf{u}f(z)$ on $[0, T_*]$, where $\mathbf{u}$ is a constant voltage vector and $F(z) = \int_0^z f(\zeta)d\zeta$. Three complete profiles are

$$f_1(z) = z(1-z)(1-2z), \quad F_1(z) = \frac{1}{2}z^2(1-z)^2, \quad (215)$$

$$f_2(z) = z(1-z)(z-1/3)^2(z-5/7), \quad F_2(z) = -\frac{z^2(1-z)^2(21z^2-18z+5)}{126}, \quad (216)$$

$$f_0(z) = 0, \quad F_0(z) = 0. \quad (217)$$

Differentiation checks every primitive. Both f and F vanish at 0 and 1. The second profile has an interior double zero at $z = 1/3$, with $F_2(1/3) = -8/15309 \neq 0$; thus products proportional to fF have a tangent zero there when their coefficient is nonzero.

The held member H and planet have $u_H = u_P = 0$. Let the free member be F and $U_* = 1$ V. For $k > 0$ choose

$$u_C = U_*, \quad u_F = U_* \left(1 + \frac{h_F}{h_H} - \frac{2h_F}{k} \right). \quad (218)$$

For $k = 0$ choose $u_C = 0$, $u_F = U_*$. These definitions apply to both held members of all five configurations and to the equal-ratio two-external assembly, with $k \in \{0, 19/20, 99/100\}$. They come from the no-external-planet-current constraint, not from a work requirement. Indeed, put $a_i = \mathbf{L}^{-1}B^T\mathbf{u}$. With zero initial flux the independent winding and KCL equations give

$$\lambda = T_*B^T\mathbf{u}F(z), \quad i = T_*a_iF(z), \quad I_{\text{ext}} = \mathbf{C}\mathbf{u}f'(z)/T_* + T_*Ba_iF(z). \quad (219)$$

For $k > 0$ the sum of secondary-current coefficients is proportional to $ku_C/h_H - k(u_F - u_C)/h_F - 2u_C$, which vanishes by (218). At $k = 0$ both secondary currents are zero. Since $v_P = 0$, the planet capacitive

current is also zero. Thus $I_{\text{ext},P} = 0$ throughout. Independent integration of the mechanical relative rates gives $z_m = T_* B^T \mathbf{u} F / \alpha$, and the prescribed elastic law gives $f_m = \alpha i$.

For any one of the ground, carrier, or two member frames write $r = u_f f(z)$, where $u_f = 0, u_C, u_A, u_B$, respectively. For an arbitrary subinterval $[T_* z_0, T_* z_1]$ write $[g] = g(z_1) - g(z_0)$. Each physical port integrates separately to

$$W_{a,X} = \frac{1}{2} C_X (u_X - u_f) u_X [f^2] + \frac{1}{2} T_*^2 (u_X - u_f) (B a_i)_X [F^2], \quad (220)$$

$$W_{e,X} = -\frac{1}{2} C_X (u_X - u_f) u_f [f^2], \quad (221)$$

$$E_C(z) = \frac{1}{2} \sum_X C_X (u_X - u_f)^2 f(z)^2, \quad E_m(z) = \frac{1}{2} T_*^2 a_i^T \mathbb{L} a_i F(z)^2. \quad (222)$$

The first line follows from the source product and $\int f f' dz = [f^2]/2$, $\int f F dz = [F^2]/2$; the second follows from the compensation product. The last line is evaluated from the state, not assigned from those works. Since $B^T \mathbf{1} = 0$, their independently summed difference equals the endpoint store change on every subinterval. At both endpoints of the complete interval every state and store is zero, and each of the eight signed works is zero. Intermediate stores are nonzero for the two nontrivial profiles. For instance $f_1(1/2) = 0$ and $F_1(1/2) = 1/32$, leaving positive magnetic energy when $a_i \neq 0$; the two half-interval transfers must therefore retain their opposite signs.

The original nine diagnostics also have separately integrated zero works. To see this without assuming cancellation, define their finite electrical counterparts in the chosen frame as three products $(v_X - r) I_{\text{ext},X}$ for $X = A, B, C$, the four mesh-side products

$$i_{X1} [(1 - h_X)(v_C - r) + h_X(v_P - r)], \quad -i_{X1}(v_X - r), \quad X = A, B, \quad (223)$$

and the two opposite pin products $\pm [J_{\text{orb}} \dot{v}_C / \alpha^2 + (h_A - 1)i_{A1} + (h_B - 1)i_{B1}](v_C - r)$. Every one is a linear combination of $f f' / T_*$ and $T_* f F$, so the same two primitives prove the separate full-interval zeros. This statement does not say that each finite mesh pair cancels instantaneously; its modal elastic path remains part of the physical model. All ratios of the zero full-interval signed works are undefined zero-over-zero. These histories also allow exact reasoning about all continuous power events. Each displayed power has the form $P_j = f(A_j f' + D_j F)$, with rational coefficients in the stated SI scale; its coefficients follow directly from its effort and flow. Thus the remaining factors for every power and every pairwise sum or difference are $A f' + D F$, using respectively (A_j, D_j) or $(A_j \pm A_l, D_j \pm D_l)$.

For the first profile put $q_z = z(1 - z)$. Then $f'_1 = 1 - 6q_z$ and $F_1 = q_z^2/2$, so every remaining factor is the quadratic

$$R_{A,D}(q_z) = A(1 - 6q_z) + \frac{1}{2} D q_z^2. \quad (224)$$

For $D \neq 0$ its candidate roots are $q_z = (6A \pm \sqrt{36A^2 - 2AD})/D$; only real values in $[0, 1/4]$ are retained. For $D = 0, A \neq 0$, the root is $q_z = 1/6$. Each retained value gives $z = (1 \pm \sqrt{1 - 4q_z})/2$. Together with $z = 0, 1/2, 1$, these are the complete events for the first profile, with multiplicities inherited from the displayed factors. The case $A = D = 0$ is an identity, not an isolated event.

For the second profile the explicit rational polynomials f_2 and $A f'_2 + D F_2$ similarly specify every factor. Their exact real roots on $[0, 1]$, isolated with multiplicities, include crossings and the double zero already identified. Between consecutive distinct roots no power sign or absolute-order relation can change, since $P_j^2 - P_l^2 = (P_j - P_l)(P_j + P_l)$. At any nonzero- f point the maximizing set is exactly the indices attaining $\max_j |A_j f' + D_j F|$; one interior point determines that set throughout each open partition interval. At a

root all equalities and inequalities are evaluated there; at a zero of f every power is zero. This includes tangencies that leave the neighboring maximizing sets unchanged. Coprime factors cannot share a root; distinct isolating intervals must be separated rather than merging nearby events. Endpoint roots and persistent ties are retained. The zero profile is identically tied throughout.

Every partition interval, including any isolating interval, retains its own works and stores from (222); their sum is the full-interval integral. Observational roots add no physical switch or impulse. This algebraic completeness argument applies to the displayed polynomials. Initial all-zero states are boundary ties with undefined zero-over-zero ratios. Hardware realization and arbitrary parameter interiors require further determination. Dissipative histories admit a different, bounded certificate.

A.12 What certifies a dissipative event set

For a fixed constitutive history with constant or linearly changing receiver conductances, the augmented state equation has the exact form $\dot{y} = [A_0 + A_1(t - t_e)]y$ on each smooth interval. Its state is analytic there. Each power is an analytic product of its actual effort and flow; maximizing-port events additionally satisfy $P_j - P_k = 0$ or $P_j + P_k = 0$. An identity remains a persistent equality, rather than a collection of isolated events. Negative amplitude with compatibly reversed preparation preserves the quadratic powers and event times; zero amplitude gives the identically zero control.

Theorem 2 (Completeness from a bounded event certificate). *For these analytic functions on a declared compact interval, suppose a finite cover establishes every nonidentically-zero factor as nonzero on each complementary interval and isolates all its remaining roots with their multiplicities. Suppose also that common roots or their separation are proved, endpoint roots are retained, and every maximizing set is determined both at the roots and on the complements. Then the resulting event partition is complete for that constitutive history.*

Proof. The cover leaves no time unclassified. A nonzero factor has fixed sign on each connected root-free complement. Including both pairwise sums and differences fixes the sign of $P_j^2 - P_k^2$ there, hence fixes the absolute ordering. The separately established root multiplicities retain tangencies, and the root equalities determine the maximizing sets at the events. An additional event would contradict one of these classifications. $\square$

Analytic enclosures can supply the needed sign and uniqueness statements. For a complex disk of radius ρ about c , the integral equation along each radius gives the exact bound

$$\|y(c + z)\| \leq \|y(c)\| \exp\left(\|A(c)\|\rho + \frac{1}{2}\|A_1\|\rho^2\right), \quad |z| \leq \rho, \quad (225)$$

using a consistent induced norm. Bound the analytic observable f on that disk by M_f , including its explicit time-dependent coefficients. Cauchy's coefficient formula gives $|a_n| \leq M_f / \rho^n$ for its exact local series. For $0 < r < \rho$, $q_r = r / \rho$, and its degree- N polynomial p_N , the complete remainders satisfy

$$|f - p_N| \leq \frac{M_f q_r^{N+1}}{1 - q_r}, \quad |f' - p'_N| \leq \frac{M_f q_r^N [(N + 1) - N q_r]}{\rho (1 - q_r)^2}. \quad (226)$$

Both inequalities follow by summing the corresponding convergent geometric tails. They are bounds on the exact function; no remainder is discarded and no truncated series is substituted as the physical model. Initial-state uncertainty must propagate into M_f and the coefficient enclosures.

A value enclosure excluding zero proves absence of a root. A strictly signed derivative and opposite endpoint signs prove a unique simple root. Squared loss factors preserve double zeros. Exact relations

$f(t) = c(t)g(t)$ with c finite and nonzero on the root interval establish coincident factors and their orders; overlapping intervals alone do not. Endpoint factors and identities require their exact algebraic treatment. Any undecided sign, root coincidence, or maximizing set leaves that part of the certificate unresolved.

A complete certificate therefore establishes a bounded dissipative event result for its specified linear or affine history, independently of earlier finite root brackets. Such a result does not settle the clipped, state-dependent reference model or other parameter histories. Here the polynomial controls have explicit algebraic event sets, and the theorem states the sufficient conditions for the dissipative case; no unevaluated nonlinear event set is claimed complete. Observational roots introduce no physical switch, impulse, or omitted interval of work. Each interval retains its full signed integrals and independent stores.

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